RM’S: AN AI-INTEGRATED INVENTORY AND E-COMMERCE MANAGEMENT SYSTEM FOR REAL-TIME DEMAND FORECASTING AND ENHANCED DISTRIBUTION AT REMHART

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This Research/ Capstone Project Study entitled **"RM'S: AN AI-INTEGRATED INVENTORY AND E-COMMERCE MANAGEMENT SYSTEM FOR REAL-TIME DEMAND FORECASTING AND ENHANCED DISTRIBUTION AT REMHART"** prepared and submitted by **RHAJ S. MADRONA, MARIA HANAH G. MENDOZA,** and **GWEN M. BASCONCILLO** has been successfully presented and recommended for approval by the Advisory Committee in partial fulfillment of the requirements for the degree, Bachelor of Science in Information Technology.

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EXECUTIVE SUMMARY

This research paper presents RM'S: An AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Enhanced Distribution at REMHART — A Unilever Products Distributor. The primary aim of this study is to address the operational inefficiencies experienced by REMHART, particularly in manual stock monitoring, inconsistent demand forecasting, and fragmented e-commerce operations. By replacing traditional, paper-based workflows with an intelligent digital platform, RM'S seeks to streamline inventory processes, automate replenishment, and expand REMHART's business accessibility through an integrated online storefront.

The system serves multiple types of users, including administrators, warehouse managers, sales personnel, and customers. Its core features include an AI-powered demand forecasting module using Facebook Prophet for time-series analysis, real-time inventory monitoring with automated replenishment alerts, a supplier and purchase order management module, a built-in e-commerce platform with secure GCash payment integration, and a geographic sales performance analysis dashboard. Together, these functionalities enable REMHART to minimize overstocking and stockouts, reduce manual

workloads, and make data-driven decisions with greater confidence.

The researchers employed the Agile Model within the System Development Life Cycle (SDLC) framework to guide the development process. This approach allowed for iterative development and continuous refinement based on user feedback and stakeholder requirements. The system was built through a descriptive-developmental research design, involving consultation with REMHART management, prototype testing, system deployment, and post-implementation evaluation.

The system underwent a comprehensive evaluation involving 65 respondents composed of business owners, staff, and customers. Two assessment frameworks were applied: the ISO 25010 Software Quality Metrics and the Unified Theory of Acceptance and Use of Technology (UTAUT) model. Under the ISO 25010 evaluation, RM'S achieved an overall mean of 3.36, interpreted as "Agree," reflecting satisfactory quality across all dimensions. Functional Suitability received the highest rating at 3.43, affirming that the system effectively covers all specified tasks and user objectives. Performance Efficiency followed at 3.36, Usability at 3.35, and Reliability at 3.30. Under the UTAUT model, the system earned an overall mean of 3.32, also interpreted as "Agree."

Performance Expectancy scored the highest at 3.38, indicating that users perceive RM'S as a tool that enhances productivity and operational efficiency. Facilitating Conditions scored 3.31, Behavioral Intention scored 3.30, and Effort Expectancy scored 3.29 — all affirming strong user acceptance and willingness to adopt the system.

The successful deployment of RM'S resulted in notable improvements across REMHART's core business processes. Inventory management accuracy was significantly enhanced through real-time tracking and AI-driven forecasting, reducing instances of overstocking and stockouts. The integrated e-commerce module expanded REMHART's market reach, allowing clients to place orders online with greater convenience. The analytics dashboard further empowered management with actionable insights for strategic planning and distribution coordination.

Based on the findings and conclusions of this study, the researchers recommend the development of a mobile application version of RM'S to enable on-the-go access for administrators and field staff. Regular user training sessions are also advised to ensure that all personnel remain proficient in utilizing the system's advanced features. Continuous collection of user feedback will be essential for iterative improvements and sustained system relevance. Additionally, expanding integrations with Unilever's supplier databases and third-party logistics providers is proposed to further automate procurement and shipment coordination. Strengthening data security measures, including encryption and role-based access control, is likewise encouraged to safeguard sensitive business and customer information.

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Chapter I

INTRODUCTION

Project Context

The rapid growth of e-commerce has reshaped how businesses operate, pushing companies to adopt smarter, faster, and more customer-centered systems. Online platforms today go beyond simple product listings—they integrate secure payments, order tracking, and real-time stock visibility to meet the expectations of modern consumers. With this shift, businesses must adapt to stay competitive by streamlining both their digital storefronts and backend operations.

At the core of these operations lies inventory management. Traditional methods, which rely on manual stock tracking and static historical records, are increasingly unable to keep pace with fast-changing customer demands. Overstocking leads to wasted resources, while stockouts create missed sales opportunities and customer dissatisfaction. To address these challenges, businesses are turning to automation, predictive analytics, and AI-driven forecasting to maintain efficiency and accuracy in their inventory and distribution processes.

This need is particularly relevant for REMHART, a trusted distributor of Unilever products. REMHART currently encounters several operational challenges: fluctuating demand, manual stock monitoring, and inefficient order processing. These issues often result in excess inventory, shortages, and delays in order fulfillment. Without an intelligent system to connect their e-commerce platform with inventory operations, REMHART risks inefficiencies that hinder both profitability and customer satisfaction.

To respond to these challenges, this project introduces RM'S—an AI-Integrated Inventory and E-Commerce Management System tailored for REMHART. The system provides a centralized solution that bridges digital storefronts with backend inventory processes. Through features such as AI-powered demand forecasting, real-time stock monitoring, automated replenishment, supplier and order management, and integrated e-commerce with secure checkout, RM'S enhances operational efficiency and customer experience. It also incorporates robust data security measures such as role-based access control and encryption to ensure safe transactions and system reliability.

By unifying e-commerce capabilities with intelligent inventory management, RM'S enables REMHART to align stock levels with market demand, automate order fulfillment, and support data-driven decision-making. This innovation not only minimizes waste and reduces operational costs but also positions REMHART to stay competitive in today’s evolving distribution sector. Furthermore, RM’S supports several Sustainable Development Goals (SDGs), including promoting sustainable consumption, driving innovation in industry, and fostering stronger partnerships across the supply chain.

In essence, RM’S is more than just a system—it is a strategic solution that transforms REMHART’s operations into a smarter, faster, and more resilient model. By addressing the pressing pain points of inventory inefficiency and disconnected e-commerce processes, RM’S empowers REMHART to achieve sustainable growth and deliver enhanced value to customers.

Objectives of the Study

The general objective of this study is to design and develop RM’s: An Ai-Integrated Inventory And E-Commerce Management System For Real-Time Demand Forecasting And Enhanced Distribution At REMHART: A Unilever Products Distributor.

Specifically, this study aimed to:

1. Incorporate an e-commerce feature where customers can browse, order, and purchase products online, with automatic inventory updates;

2. Create an inventory management module that allows users to add, update, and organize product information with essential details;
3. Adopt AI-driven demand forecasting to predict future product requirements based on historical sales data collected from REMHART's covered stores;
4. Employ intelligent replenishment suggestions by analyzing sales trends; and
5. Apply location-based analysis to evaluate sales performance based on geographic locations.

Scope and Limitation of the Study

The scope of "R RM'S: An AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Enhanced Distribution at REMHART" centers on developing a centralized digital platform tailored for REMHART's operations. The system is designed to streamline inventory management, automate stock tracking, coordinate suppliers, and integrate AI-driven demand forecasting with real-time data. Core features include a dashboard for real-time inventory updates, automated replenishment suggestions, AI-powered demand prediction, geographic sales performance analysis, supplier management tools, and a built-in e-commerce module for product browsing, ordering, and purchasing. This unified platform aims to enhance operational efficiency, minimize overstocking or stockouts, and enable data-driven decision-making across REMHART's distribution processes.

However, the system is developed specifically for REMHART and may not fully fit other distributors with different workflows or product types. The accuracy of AI forecasting depends on the quality and completeness of historical data, which could affect prediction reliability. The e-commerce feature focuses on basic online transactions and may lack advanced personalization or third-party integrations. While security measures like encryption and access control are in place, risks such as cyber threats, internet dependency, and system downtimes remain. Future market or technology shifts may also affect performance.

Significance of the Study

This study aims to develop RM'S: An AI-Integrated Inventory and E-Commerce Management System tailored for REMHART, a Unilever products distributor. It addresses critical inventory and distribution challenges through real-time demand forecasting, intelligent stock monitoring, and streamlined e-commerce integration. The significance of this study extends to the following stakeholders:

REMHART. As the primary beneficiary, REMHART will gain a smarter inventory system that tackles issues such as fluctuating demand, overstocking, and stockouts. The integration of AI and automation will enhance operational efficiency, reduce unnecessary costs, and ensure timely product availability. The e-commerce feature will further expand market reach and improve customer satisfaction by offering a seamless ordering experience.

Supply Chain Professionals. This study highlights the practical application of AI and predictive analytics in modern supply chain environments. By leveraging real-time monitoring and time series forecasting, professionals can adopt a more agile, responsive, and data-driven approach to inventory and distribution management. It serves as a blueprint for improving supply chain resilience and competitiveness.

IT Experts. The system's development involves cutting-edge technologies, including secure web-based platforms, role-based access controls, and data encryption protocols. It provides IT professionals with hands-on experience in designing AI-powered systems that are both scalable and secure—an increasingly essential skill set in digital supply chain environments.

Business Managers. Managers will benefit from actionable insights generated by AI dashboards, sales trend analytics, and inventory tracking. These features enable data-informed decision-making, better demand alignment, and more efficient resource allocation, ultimately leading to smarter business operations and increased profitability.

Researchers and Future Developers. This study offers a foundational framework for further innovation in AI-driven inventory and e-commerce systems. It serves as a reference for applying predictive analytics and automation in distribution businesses, offering inspiration and guidance for future projects across various sectors.

Conceptual Framework

The conceptual framework for this study outlines the integration of inputs, processes, and outputs to develop an advanced AI-driven inventory management system for RM'S. The framework serves as the foundation for understanding how data is collected, processed, and transformed into actionable insights to optimize inventory management and supply chain operations. It provides a structured visual representation of the relationships between the various components of the system.

At its core, the framework is built around three interconnected stages that mirror the natural flow of information within the organization. The first stage focuses on data inputs, which encompass all relevant information that the system requires to generate meaningful outputs. This includes historical sales transactions, current stock levels, supplier lead times, customer order records, and external factors.

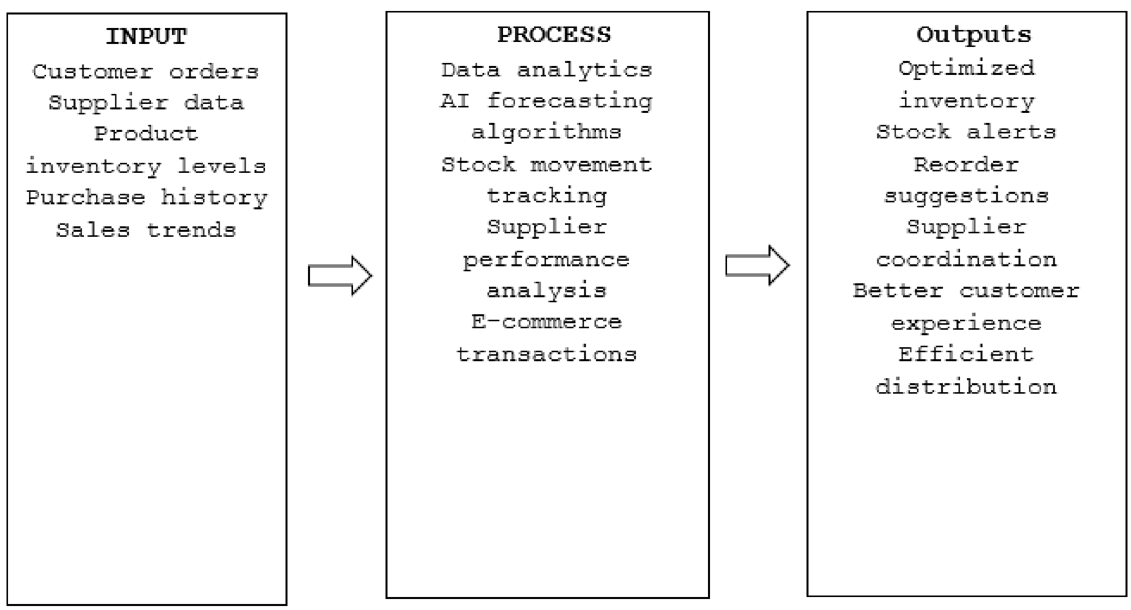


Figure 1. RM'S Conceptual Framework

Definition of Terms

To facilitate clarity, researchers break down the following terms:

AI-Driven Inventory Management System. A technology-enabled solution using artificial intelligence and predictive analytics to optimize inventory control, automate processes, and provide real-time data insights for RM’S's distribution operations.

Automated Reordering. A system functionality that triggers restocking processes based on demand forecasts, minimizing manual intervention and reducing stockouts.

Centralized Web-Based Platform. An integrated online tool that enables RM’S’s team to monitor inventory, access reports, and manage distribution processes from a single interface.

Comprehensive Reporting. A tool that provides visual insights into inventory levels, sales trends, and forecast accuracy to support data-driven decision-making.

Customer Satisfaction. The improved experience retailers and customers gain from reliable product availability and timely deliveries enabled by the system.

Data Encryption Protocols. Security measures implemented to protect sensitive inventory and sales data from unauthorized access and breaches.

Demand Forecasting. The use of historical sales data and predictive models to estimate future product demand, helping RM’S align stock levels with market needs.

Market Fluctuations. Unpredictable changes in customer demand or supply chain disruptions that the system seeks to mitigate through real-time adjustments.

Operational Efficiency. The ability to manage resources effectively, reducing costs and improving service delivery through optimized inventory and distribution processes.

Predictive Analytics. The application of AI techniques to analyze historical and current data, generating insights for proactive decision-making in inventory management.

Real-Time Inventory Monitoring. A feature that provides up-to-date tracking of stock levels to ensure efficient inventory management and timely decision-making.

Supply Chain Management. The process of managing product flow from Unilever to RM'S and onward to retailers, streamlined through AI-driven inventory tools.

Time Series Analysis. A statistical method applied to three years of sales data to identify patterns and trends, improving the accuracy of inventory predictions.

Chapter II

REVIEW OF RELATED LITERATURE/SYSTEM

This chapter presents the review of related literature and system that provides the researchers a strong foundation for the study.

Related Literature

According to Smith (2024), predictive analytics helps retailers forecast demand with greater accuracy by analyzing historical data and real-time market trends. The research demonstrates that AI-driven systems reduce stockouts and overstocking, minimizing costs and enhancing operational efficiency. By integrating machine learning algorithms, retailers can achieve dynamic inventory replenishment based on predicted demand patterns. The study highlights the manual stock tracking and improving customer satisfaction. The research suggests that predictive analytics offers a competitive edge in inventory management, particularly in industries with fluctuating demand, such as retail and fast-moving consumer goods (FMCG). However, the study also notes challenges, including data quality issues and the need for proper training among staff to effectively use these technologies.

Taylor (2024) provided a comprehensive review of AI applications in inventory control, highlighting how machine learning and automation are reshaping supply chain management. The study identifies several key benefits of AI, including real-time stock monitoring, predictive demand forecasting, and automated restocking. AI models, especially those based on deep learning algorithms, help businesses predict market fluctuations, optimize storage space, and minimize excess inventory. Taylor (2024) emphasizes the scalability of AI solutions, making them suitable for small and large businesses alike. However, the research also addresses limitations such as the reliance on accurate historical data and the initial setup costs. The study concludes that AI in inventory control not only increases operational efficiency but also strengthens the decision-making process, providing businesses with data-driven insights into consumer behavior and market trends.

Martinez (2023) presented a case study on the implementation of AI in inventory management in the fast-moving consumer goods (FMCG) sector. The study investigates how AI-driven demand forecasting and stock optimization have led to significant improvements in inventory management practices for FMCG companies. Martinez (2023) found that using AI algorithms for predictive analysis helped companies reduce wastage and lower storage costs by accurately predicting customer demand. The research emphasizes the importance of leveraging AI to automate reordering processes, allowing for faster restocking and better alignment with market demand. The study concludes that AI applications in FMCG inventory systems improve supply chain transparency, streamline operations, and increase profitability.

Zhang and Li (2024) investigated the application of AI and machine learning in improving the accuracy of demand forecasting for supply chain management. The researchers developed an AI model that incorporates both historical sales data and real-time market signals to predict demand with higher accuracy than traditional methods. Their study focused on the fast-moving consumer goods (FMCG) industry, where managing inventory levels can be particularly challenging due to volatile demand patterns. The authors found that machine learning algorithms, especially those based on neural networks, outperformed traditional statistical methods by identifying complex patterns and trends in data that were not apparent through basic analysis. Additionally, the study demonstrated that real-time data feeds, such as weather patterns and social media trends, could further enhance forecasting accuracy. Zhang and Li concluded that the integration of AI into demand forecasting offers significant benefits for FMCG companies, particularly in reducing stockouts and improving customer satisfaction. However, they also noted that successful AI implementation requires careful calibration and ongoing monitoring to maintain forecast accuracy.

Brown (2023) explored the integration of real-time inventory systems and their effect on supply chain efficiency. The research highlights that real-time data tracking enables businesses to maintain optimal stock levels, preventing both stockouts and excess inventory. According to the study, AI-powered systems provide detailed insights into product movements, sales trends, and inventory needs. By incorporating machine learning, businesses can predict stock requirements with high accuracy, thereby reducing manual interventions and human errors. The study further examines the challenges businesses face when implementing real-time systems, such as integrating new technologies into legacy systems and training staff. Despite these challenges, Brown (2023) argues that the benefits—improved decision-making, reduced costs, and better customer satisfaction—outweigh the initial hurdles.

Singh and Verma (2023) explored the integration of AI-powered inventory management systems in the pharmaceutical industry. Their research focused on how AI could address specific challenges in pharmaceutical distribution, such as shelf-life management, demand fluctuations, and stockouts. The study found that predictive analytics could forecast the demand for drugs based on various factors, including seasonality, market trends, and historical data. The researchers also highlighted the potential of AI in minimizing waste and reducing costs by ensuring that inventory levels are aligned with actual demand. One of the key findings was the efficiency gained from AI-based automated reordering systems, which proactively restocked items based on predicted shortages. However, the study also raised concerns about data privacy, particularly regarding patient information, which could be inadvertently shared during the demand forecasting process. The authors recommended stricter cybersecurity measures for companies adopting AI in inventory management. This study is significant for its contribution to the specialized field of pharmaceutical distribution and its insights into AI's potential to enhance efficiency and reduce waste in sensitive industries.

White and Brown (2023) explored the use of AI in optimizing warehouse management systems (WMS) to improve inventory tracking and distribution processes. Their study found that AI could streamline warehouse operations by automating tasks such as order picking, stock sorting, and shelf management. By integrating AI with robotic process automation (RPA), the system could process orders more quickly and accurately, reducing human errors and operational bottlenecks. The authors also highlighted how AI-powered WMS could monitor real-time stock levels and adjust picking and packing workflows based on current inventory levels. Additionally, they pointed out that AI algorithms could optimize the use of warehouse space by predicting which items would be in high demand, ensuring that frequently ordered products were stored in more accessible locations. While the study showed that AI systems led to significant improvements in warehouse efficiency, White and Brown also noted the need for substantial upfront investment in infrastructure and the potential challenge of integrating AI systems with legacy WMS platforms. The study concluded that AI's ability to enhance warehouse productivity could be a game-changer in industries like e-commerce and manufacturing.

Related System

In recent years, the application of artificial intelligence (AI) in inventory management and demand forecasting has revolutionized the way businesses operate, enhancing operational efficiency, reducing costs, and improving customer satisfaction. AI-driven systems are increasingly being adopted across various sectors such as retail, e-commerce, manufacturing, healthcare, and food distribution to streamline inventory control, optimize stock levels, and predict demand with unprecedented accuracy. By analyzing vast amounts of historical sales data, market trends, and real-time customer behavior, AI tools can forecast product demand, automate stock replenishment, and minimize the risk of overstocking or stockouts.

A study by Zhang and Tan (2024) developed an AI-powered supply chain optimization tool for food distributors to streamline inventory management and distribution. By analyzing historical sales data and real-time demand trends, the system was able to forecast product demand accurately, minimizing waste and ensuring on-time deliveries. The platform also incorporated predictive analytics for automated reorder suggestions, leading to a 10% reduction in operational costs and a 25% improvement in customer satisfaction.

Brown and Williams (2024) developed an AI-based stock replenishment system designed for electronics retailers to enhance demand forecasting accuracy and automate inventory restocking.The system used predictive analytics and machine learning to adjust inventory levels dynamically, optimizing the supply chain process. As a result, the retailer experienced a 40% reduction in stockouts and a 25% increase in sales, demonstrating the benefits of AI in managing high-demand electronics products.

In their 2024 study, Cheng and Zhao implemented a machine learning algorithm to provide real-time inventory monitoring and analytical insights for e-commerce enterprises.By integrating AI with sales data, the system predicted stockouts, allowing businesses to take proactive measures. The study found that businesses could reduce lost sales by 18% while improving customer satisfaction through faster restocking processes.

In their 2024 study, Smith and Thompson examined the implementation of an AI-driven order management system to enhance inventory management and streamline customer order fulfillment. By leveraging AI to predict demand and automate restocking processes, the system reduced manual interventions and increased order fulfillment accuracy. The system contributed to a 30% increase in customer satisfaction by ensuring timely deliveries.

A study conducted by Fernandez and Liu (2023) developed a predictive supply chain optimization system for manufacturing companies. Using AI and machine learning, the system forecasted demand and adjusted inventory levels accordingly, leading to a 30% reduction in overproduction and storage costs. The system also automated the restocking process, ensuring optimal material availability for production without excessive lead times.

Green and Miller (2023) developed a smart AI-based reordering system designed to forecast demand for medications and medical supplies within pharmaceutical supply chains. By leveraging historical sales data and seasonal trends, the system minimized stockouts and waste, resulting in a 22% reduction in pharmaceutical supply chain costs. The project showed how AI could enhance demand forecasting in highly regulated industries.

In their 2023 study, Thompson et al. proposed a predictive analytics tool aimed at improving demand forecasting and management within healthcare supply chains.

The AI system utilized machine learning models to forecast demand for medical supplies based on patient data and historical trends. This led to a more accurate inventory management process, with a reduction in both overstocking and stockouts. The system also automated inventory updates, improving operational workflows in hospitals and clinics.

Sanchez and Tan (2023) developed an AI-based system for data-driven inventory optimization in distribution centers. The system predicted demand using both historical sales data and external factors, allowing for precise inventory planning. The study showed that the AI solution reduced storage costs by 25% and ensured that inventory levels were always aligned with actual demand, improving distribution efficiency.

An investigation by O'Brien and Williams (2023) presented an AI-powered inventory management system that uses real-time data to improve supply chain efficiency. By employing machine learning for demand forecasting, the system optimized stock levels and automated reordering, leading to a 15% improvement in operational efficiency. The AI solution provided valuable insights for timely decision-making, thus streamlining supply chain operations.

A project by Rodriguez and Suarez (2023) focused on developing a cloud-based inventory management system for consumer goods distributors. The system utilized AI and machine learning to predict demand and automate the replenishment process. The implementation resulted in a 30% improvement in stock turnover and a reduction in product shortages. By providing real-time insights into inventory status, it enabled faster decision-making, reducing delays and increasing operational efficiency across distribution centers.

Patel and Singh (2023) introduced an AI-based inventory control system aimed at enhancing stock management efficiency for online grocery stores. The system analyzed real-time sales data and weather patterns to predict demand for perishable items. By automating reordering and optimizing inventory levels, it reduced waste by 12% and improved delivery accuracy. Customers benefitted from improved product availability and faster deliveries, reinforcing the importance of AI in e-commerce logistics.

Lee et al. (2023) implemented a smart demand forecasting system in retail chains to predict sales by analyzing real-time customer behavior and external market factors. The system utilized time series analysis combined with AI algorithms to enhance forecasting accuracy, thereby optimizing stock levels. It helped reduce inventory holding costs by 18% while improving stock availability. The research showed that AI's predictive capabilities could significantly improve the agility of supply chains, especially in the fast-paced retail environment.

In their 2022 study, Lee and Chen developed an AI-based inventory forecasting model for fashion retailers that leverages historical sales and fashion trends to enhance demand prediction and reduce surplus inventory. The research highlighted a 20% increase in sales efficiency and a 15% improvement in stock availability. The AI-driven system enabled retailers to adapt quickly to market fluctuations, ensuring they remained competitive in a dynamic industry.

An investigation by Kumar and Gupta (2022) explored the integration of AI in warehouse management systems. The AI-driven platform helped manage inventory levels in real-time by forecasting demand based on previous sales and market trends. It also optimized storage space and automated the reordering process. The study found a 35% increase in warehouse operational efficiency and a reduction in manual errors, showcasing AI’s potential to transform traditional warehouse operations.

Rivera and Martin (2022) investigated real-time inventory tracking in the automotive sector using an AI-based system. The platform employed predictive analytics to anticipate part demand and streamline restocking operations, which reduced supply chain delays. Implementation of the system enhanced production continuity and decreased downtime by 20%, thereby improving overall operational efficiency and profitability.

Smith and Johnson (2022) designed an AI-powered inventory management solution tailored for e-commerce platforms. By integrating machine learning algorithms for demand prediction, the system automated stock replenishment and optimized inventory levels. Utilizing historical sales data, it successfully cut stockouts by 20% and lowered overstocking by 15%. Furthermore, the solution provided a centralized dashboard for live monitoring and reporting, supporting more informed decision-making and operational improvements.

In conclusion, AI and machine learning are revolutionizing inventory management by providing businesses with the tools to forecast demand more accurately, automate reordering, and optimize stock levels. While the challenges of data quality, system integration, and skilled labor remain, the overall impact of AI on improving efficiency and reducing costs is undeniable. Continued advancements in these technologies, along with a focus on overcoming existing barriers, will likely further enhance their adoption and effectiveness in inventory management.

Chapter III

METHODOLOGY

The purpose of this chapter is to introduce the methodology implemented in development process which will include wide coverage of the components of the process.

Development Method

The development of "RM’S: An AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Intelligent Distribution at REMHART – A Unilever Products Distributor" will follow the System Development Life Cycle (SDLC) framework to ensure the delivery of a functional, scalable, and user-centered solution. To address the system's complexity, which includes AI-driven forecasting, inventory automation, real-time monitoring, supplier management, and e-commerce integration, the Agile methodology is adopted.

Agile supports incremental development, allowing the team to deliver key features in short, manageable sprints while remaining responsive to stakeholder feedback and evolving requirements. This iterative approach is ideal for handling the dynamic needs of an AI-powered and data-intensive system like RM’S.

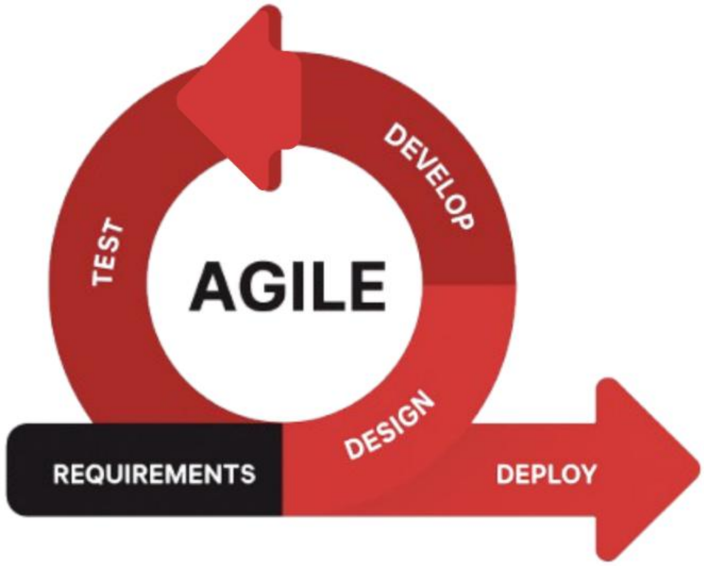


Figure 2. Agile Model

This study will employ the Agile Model, an adaptive project management approach that breaks the project into manageable phases and emphasizes continuous collaboration and iterative improvement throughout the system development process.

The Agile model consists of the following sequential phases:

1. **Requirements.** Initial consultations were conducted with REMHART management and staff to identify operational challenges and system needs. Data collection focused on existing inventory workflows, sales processes, supplier coordination, and demand forecasting practices. Historical sales and inventory records from REMHART's covered stores were analyzed to determine data availability for AI forecasting. The gathered requirements emphasized real-time inventory

tracking, automated replenishment alerts, AI-based demand forecasting, supplier management, geographic sales analysis, and an integrated e-commerce module for digital ordering.

2. Design. Based on the validated requirements, system architecture, data flow diagrams, and user interface wireframes were created. The design stage outlined the structure of the centralized dashboard, inventory management module, AI forecasting engine, and e-commerce components. Special focus was given to usability, ensuring a clean and intuitive interface suitable for both administrative and operational users. Database schemas and API integrations were also designed to support real-time data synchronization and secure transactions.

3. Development. The development phase followed an iterative Agile process. The system was developed using Node.js for backend processing and MySQL for database management. Core modules—including Inventory Management, E-Commerce Ordering, Supplier Management, and AI-Driven Demand Forecasting—were coded and integrated. The AI forecasting model was trained using historical sales data from REMHART's covered stores to predict future product demand and optimize stock levels. Continuous updates were implemented based on stakeholder feedback during development sprints.

4. Testing. Comprehensive testing was performed throughout the development cycle. Functional testing verified that all system features operated according to specifications, while performance testing assessed system responsiveness and stability under multiple user scenarios. Usability testing was conducted with REMHART staff to ensure ease of use and clarity of system functions. Data accuracy testing was also applied to validate inventory records, forecasting outputs, and e-commerce transactions.

5. Deployment. After successful testing and approval, RM’S was deployed at REMHART’s operational environment. The system was configured using real business data, and staff accounts were created based on access roles. System monitoring was conducted to ensure stability, accuracy of forecasts, and proper synchronization of inventory and sales data. User training sessions were provided to ensure smooth adoption and effective system utilization.

AI Demand Forecasting Process

The figure illustrates the AI-based demand forecasting workflow implemented in the study entitled "RM'S: An AI-Integrated Inventory and E-Commerce Management System with Real-Time Demand Forecasting and Enhanced Distribution for REMHART." Unlike traditional machine learning approaches that require extensive model training, RM'S utilizes Facebook Prophet, a time-series forecasting model designed to generate accurate forecasts using historical data with minimal manual training and configuration.

The primary objective of integrating AI forecasting into RM'S is to assist REMHART in predicting future product demand, minimizing stock shortages and overstock situations, and supporting data-driven inventory and distribution decisions. The forecasting process followed a structured workflow, consisting of the following stages:

1. Data Collection

The first stage involved the collection of historical sales and inventory data from REMHART's covered stores. These records included past transaction volumes, product demand trends, and time-based sales patterns (daily, weekly, and monthly). The historical data served as the primary input for the Prophet model, ensuring that forecasts were based on real operational data rather than simulated or synthetic datasets.

2. Data Preparation

After data collection, preprocessing was conducted to ensure compatibility with Prophet’s requirements. This included organizing the data into a time-series format, handling missing values, and standardizing date and quantity fields. The prepared dataset focused on identifying demand trends, seasonality, and recurring sales patterns across REMHART’s stores.

3. Forecast Generation Using Prophet

Instead of traditional model training, RM’S employs Prophet’s built-in forecasting mechanism, which automatically detects trends, seasonal effects, and holiday patterns from historical data. The system generates demand forecasts by analyzing past sales behavior and projecting future demand values over a specified time horizon.

Prophet’s strength lies in its ability to produce reliable forecasts even with limited data and without the need for manual hyperparameter tuning. This approach ensures faster deployment and consistent forecasting performance suited for business environments.

4. Forecast Integration into the System

The generated demand forecasts are seamlessly integrated into RM'S dashboard, where they are displayed alongside inventory levels and sales data. These forecasts support decision-making by providing insights into expected product demand, enabling timely restocking, improved distribution planning, and proactive inventory control.

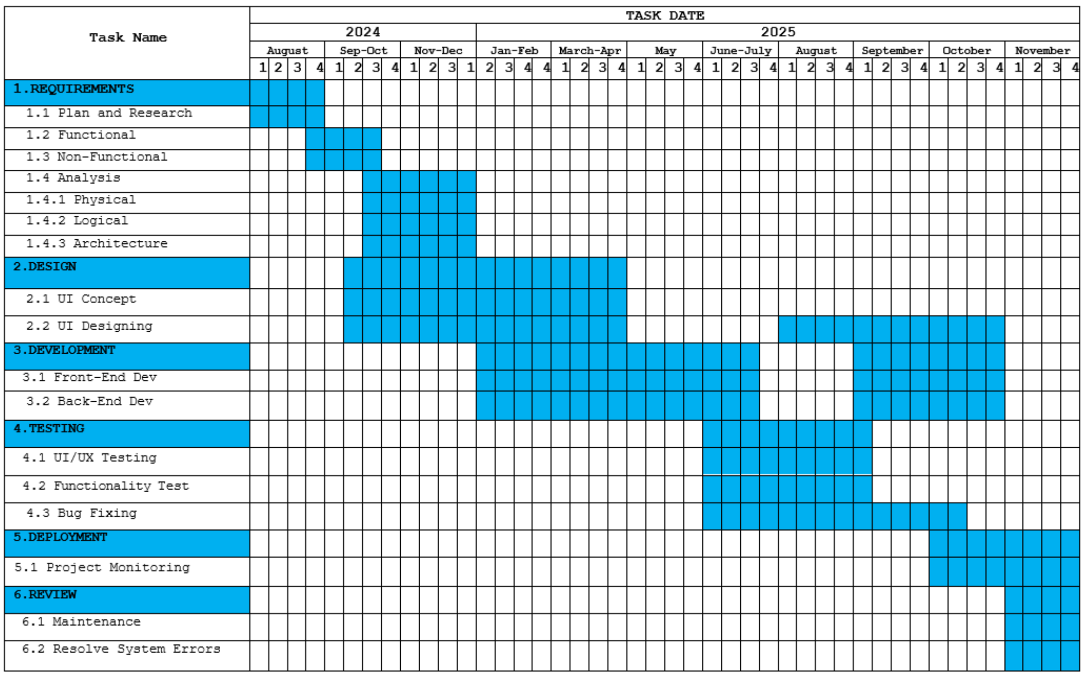
5. Monitoring and Continuous Update

As new sales data becomes available, the system continuously updates the forecasting inputs. Prophet recalculates forecasts based on the latest historical data, allowing RM'S to adapt to changing demand patterns without retraining or redeploying complex models. This ensures that the forecasting results remain relevant and aligned with REMHART's current operations.

Gantt Chart

In this section, a Gantt Chart is presented to outline the project timeline for "RM'S: An AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Enhanced Distribution at REMHART – A Unilever Products Distributor". The chart documents all stages of the development process, ensuring that tasks are well-organized and tracked against deadlines.

Table 1. Gantt Chart



Legend: ■ - Completed/ Done

Requirements Specification

The user must meet the required functions of the system to fully use the system.

Functional Requirements

Functional Requirements define how the system works and how it should be worked to function properly to avoid unnecessary events happening.

Table 2. Functional Requirements

Features	Description
Dashboard Overview	Provides a centralized dashboard showing current stock levels, low-stock alerts, and recent movements.
Product Inventory Management	Allows users to add, edit, and organize product information entries.
AI Demand Forecasting	Uses AI to analyze historical sales data and predict future product demand.
Replenishment Suggestions	Automatically generates reorder suggestions based on sales data, lead times, and demand forecasts.
Stock Movement Tracking	Logs all inventory movements
E-Commerce Management System	Built-in e-commerce module for browsing, cart management, and online purchases.

Table 2 shows that the RM'S features consist of predictive analytics for demand forecasting, time series analysis for demand prediction, real-time inventory monitoring, automated reordering system, comprehensive reporting and data visualization, secure data encryption for sensitive information, centralized web-based access for monitoring and reporting, role-based access control, forecast accuracy monitoring and feedback, e-commerce product management and order sync, online shopping cart and checkout system, and supplier and purchase order management.

User Interface

The user interface for the created system is depicted in the figure below.

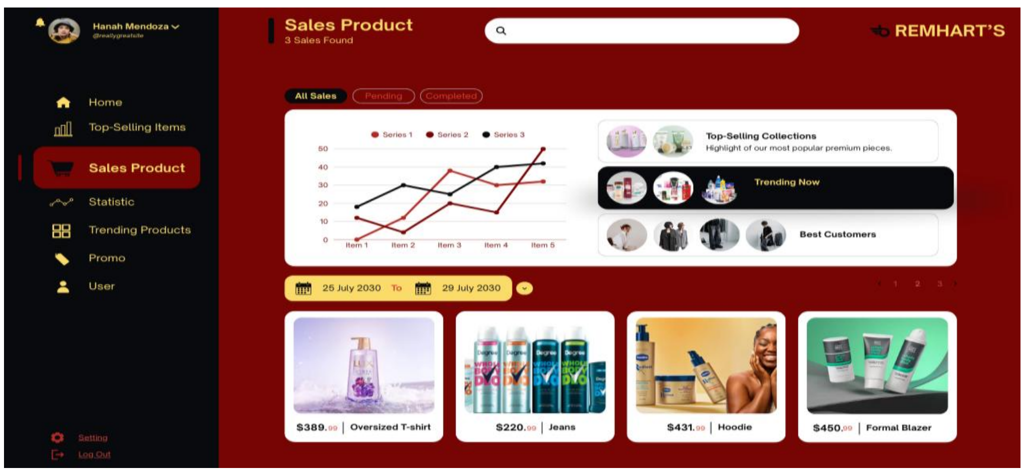


Figure 3. User Interface

The design presents a dark-mode aesthetic dominated by a deep, rich red and black color palette, creating a bold and sophisticated interface. The primary background is a dark red, contrasted with a stark black sidebar, which gives the dashboard a powerful, high-contrast, and somewhat luxurious feel.

The main content area utilizes white cards and sections against the dark red background to highlight key information, making the data pop and ensuring readability. The navigation is clearly delineated in the black sidebar, where the active selection ("Sales Product") is emphasized with a bright red background and a small graphic icon.

The layout is well-structured, featuring a left-hand navigation panel, a top search bar, and a main display area for metrics (the line graph) and product listings. The use of simple line icons and minimalist typography contributes to a clean, modern user experience. The combination of the dark, intense colors and the clean, white data cards successfully balances an energetic, professional look with optimal user focus on the core sales data. This deliberate contrast in design not only enhances visual clarity but also reduces eye strain during extended use.

Hardware Interface

The hardware interface defines how the physical hardware components interact and communicate to support the operation of the web-based sales and inventory management system. This includes the connectivity ports, communication interfaces, and networking mechanisms required for proper system functionality.

The system is designed to run on standard desktop or laptop computers equipped with essential hardware interfaces such as USB ports, network interface cards (NIC), and wireless adapters. USB ports are utilized for connecting peripheral devices including keyboards, mice, and external storage devices used for system operation, data input, and maintenance.

For network communication, the system relies on an 802.11ac dual-band (2.4 GHz and 5 GHz) wireless network interface to ensure stable and reliable internet connectivity. This interface enables real-time communication between the client devices and the web server, allowing users to access the system dashboard, sales reports, inventory data, and analytics modules without interruption.

Communication between the client-side interface, server-side application, and database is facilitated through standard networking protocols. The Transmission Control Protocol/Internet Protocol (TCP/IP) is used for reliable data transmission, while Hypertext Transfer Protocol Secure (HTTPS) ensures secure access to the system. These interfaces allow seamless interaction between system components, supporting real-time data processing, inventory updates, and sales monitoring.

Software Interface

The software interface defines how the different software components of the system interact, communicate, and exchange data to ensure seamless system operation. It focuses on the integration between the client-side interface, server-side application, and database management system used in the web-based sales and inventory management system.

The client-side interface is accessed through a web browser and is developed using modern web technologies and frameworks. It provides users with interactive dashboards, sales reports, inventory monitoring, and analytics visualizations. User inputs such as product updates, sales transactions, and inventory adjustments are captured through the graphical user interface and sent to the server for processing.

The server-side application is built using Node.js with the Express.js framework, which handles business logic, request routing, authentication, and data validation. The server processes incoming requests from the client-side interface and communicates with the database to retrieve, update, and store system data. This layer ensures that system rules, access control, and data consistency are properly enforced.

For data storage and management, the system utilizes Firebase and relational database technologies such as MySQL. Firebase is used for real-time data synchronization and fast access to frequently updated records, while MySQL is utilized for structured data storage such as product information, sales records, and inventory logs. Communication between the server and database is secured and optimized to maintain data integrity and system reliability.

Standard communication protocols are employed to facilitate software interaction. Hypertext Transfer Protocol Secure (HTTPS) is used to ensure secure data transmission between the client and server, while Application Programming Interfaces (APIs) enable structured communication between the system modules.

Security Requirements

The system implements security measures to protect data integrity, confidentiality, and user access within the web-based sales and inventory management system.

These security requirements ensure that sensitive information such as sales records, inventory data, and user credentials are safeguarded from unauthorized access.

User authentication is enforced through a secure login mechanism, allowing only authorized users to access the system. Role-based access control is implemented to restrict system functionalities based on user roles, such as administrators and staff, ensuring that critical operations are accessible only to permitted users.

These security measures collectively ensure reliable system operation, protect sensitive business data, and maintain user trust in the proposed system.

Technical Background

The technical background gives important information regarding technical aspects of the project which makes it easier to define what is required in easy-to-understand words for developers. The next sections go over hardware and software specifications.

Hardware Specifications

Hardware Specifications refer to the technical descriptions of the hardware items and their components.

Table 3. Hardware Specifications

Hardware	Function	Specifications		Unit
		Minimum	Recommended	
Processor	The component that oversees all the arithmetic, logical, and input/output tasks of a computer system by executing its instructions is referred to as the computer's CPU.	4, cores, 2.5 GHz	8 cores, 3.0 GHz	1
Memory	Has the ability to retain information indefinitely, allowing the user to access or retrieve the data as needed.	8GB	16GB or Higher	1
Network	Enables data communication between the computer and network.	802.11ac 2.4/5 GHz Wireless Adapter	802.11ac 2.4/5 GHz Wireless Adapter	1

Table 3 outlines the minimum and recommended hardware requirements for the system. It is advised to use a processor with at least 4 cores running at 2.5 GHz, while an 8-core CPU with a clock speed of 3.0 GHz or higher is ideal for optimal performance. The system should have a minimum of 8GB of RAM, with 16GB or more recommended for smoother multitasking. For network connectivity, an 802.11ac dual-band (2.4/5 GHz) wireless adapter is required, both for minimum and recommended setups.

Software Specifications

Software Specifications refer to the representation of the software used by the system, detailing the tools, frameworks, libraries, and platforms that form the technical foundation of the RM'S application. Table 4 outlines the minimum and recommended versions for each component, ensuring the application runs smoothly with the latest features and security updates. These specifications were carefully selected to align with the system's functional and performance requirements, providing a stable and scalable environment for development, deployment, and ongoing maintenance.

The software components identified in this section cover all critical layers of the application, from the operating system and web server to the database management system, programming frameworks, and AI/ML libraries.

Table 4. Software Specifications

Component	Minimum Specifications	Recommended Specifications
Operating System	Windows 10 (64-bit) / Ubuntu 20.04 LTS	Windows 11 Pro (64-bit) / Ubuntu 22.04 LTS
Web Server	Apache 2.4 or Nginx 1.14	Apache 2.4 or Nginx 1.14 (latest stable)
Database	Firebase Realtime DB / MySQL 5.7 / PostgreSQL 10	Firebase Firestore / MySQL 8.0 / PostgreSQL 13
Web Browser	Google Chrome / Mozilla Firefox (latest)	Google Chrome (latest stable)
AI/ML Libraries	TensorFlow 2.0 / Scikit-learn 0.24	TensorFlow 2.12+ / PyTorch 1.13+
IDE	Visual Studio Code, Python 3.8+, Node.js 16+	JetBrains PyCharm Professional, Node.js 18+
Vue JS Framework	Vue.js 2.6.x	Vue.js 3.x (latest)
Bootstrap / Tailwind	Bootstrap 5 / Tailwind 2.x	Bootstrap 5 / Tailwind 3.x (latest)

As shown in Table 4, it outlines the minimum and recommended software specifications required for the system. It includes components such as the operating system, web server, database, web browser, AI/ML libraries, IDEs, and development frameworks like Node.js, Express.js, Vue.js, Bootstrap, and Tailwind. These specifications ensure that the system runs smoothly, supports modern tools, and performs its intended functions efficiently.

System Analysis and Design

System Analysis and Design are concerned with the planning and development of information systems by understanding and specifying in detail what a system should perform as well as how the system's components should be implemented and work together.

System Overview

The system is a comprehensive inventory management solution tailored for RM'S, a Unilever products distributor. It integrates advanced technologies such as predictive analytics, time-series analysis, real-time inventory tracking, and automated reordering to streamline inventory control and enhance distribution efficiency.

This web-based platform leverages AI algorithms and firebase's robust data-handling capabilities to deliver precise demand forecasts, optimize stock levels, and provide actionable insights.

System Architecture

A system architecture shows the representation and structure of the system.

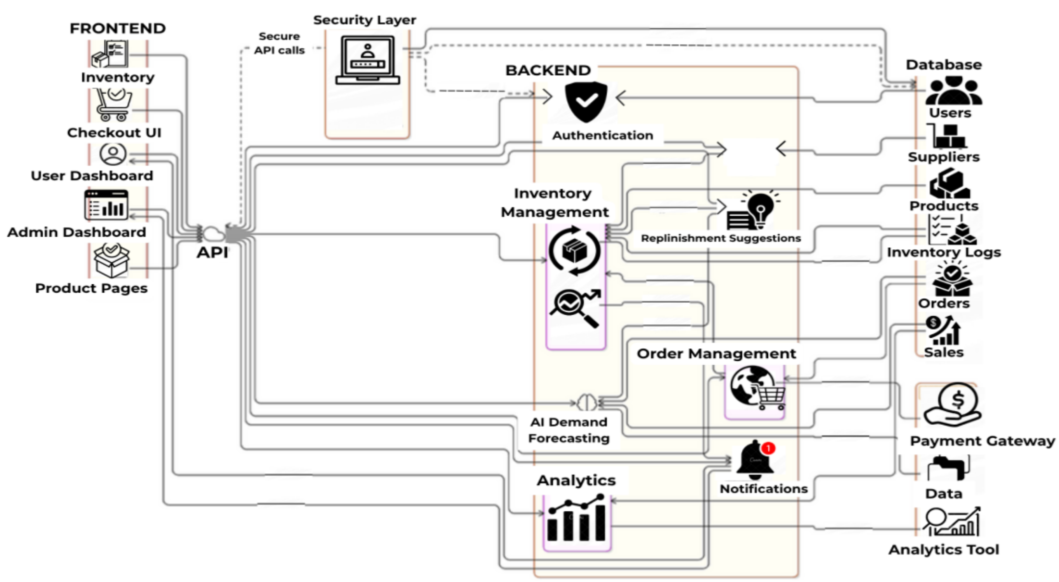


Figure 4. System Architecture

Figure 4 shows the representation and structure of the system. Figure 3 below shows the workflow of the system. Users access the platform via the frontend, which is built with Vue.js. They interact with real-time data and dashboards to track inventory and sales. The client sends requests to the backend (Node.js with

Express.js) for inventory data or to generate demand forecasts. The backend fetches data from the Firebase database and, for demand forecasting, sends the data to the AI/ML layer for analysis.

The AI/ML model processes the data and returns a forecast. This forecast is then passed back to the frontend for display. If the system detects that stock levels fall below the threshold based on forecasted demand, it automatically triggers the reordering process. Once the reordering process is initiated, the system generates a replenishment alert and logs the transaction for record-keeping and future reference.

The entire workflow is designed to operate seamlessly and with minimal manual intervention, ensuring that inventory levels are consistently maintained at optimal thresholds. Each layer of the architecture communicates through secure and well-defined API endpoints, guaranteeing data integrity and response efficiency throughout the process. The separation of concerns between the frontend, backend, and AI/ML layer also allows for independent scaling and maintenance of each component, making the system highly adaptable to REMHART's growing operational demands.

Use Case Diagram

The primary interactions between the users and the system are depicted in the UML Use-Case Diagram for "AI-Driven Inventory Insight for Real-Time Demand Forecasting and Enhanced Distribution for RM’S: A Unilever Products Distributor".

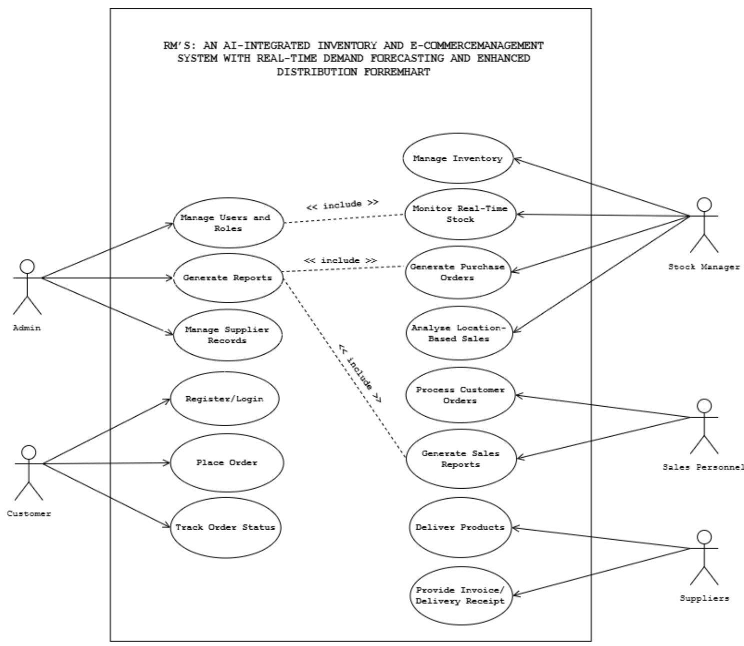


Figure 5. Use Case Diagram

Figure 5 presents the Use Case Diagram for *RM’S: An AI-Integrated Inventory and E-Commerce Management System*. The diagram illustrates the primary interactions between system actors and the core functionalities within the system. These use cases collectively represent the essential operations that support real-time inventory control, intelligent demand prediction, and enhanced distribution processes.

Activity Diagram

This activity diagram represents the workflow of the "RM'S System" by outlining the key interactions.

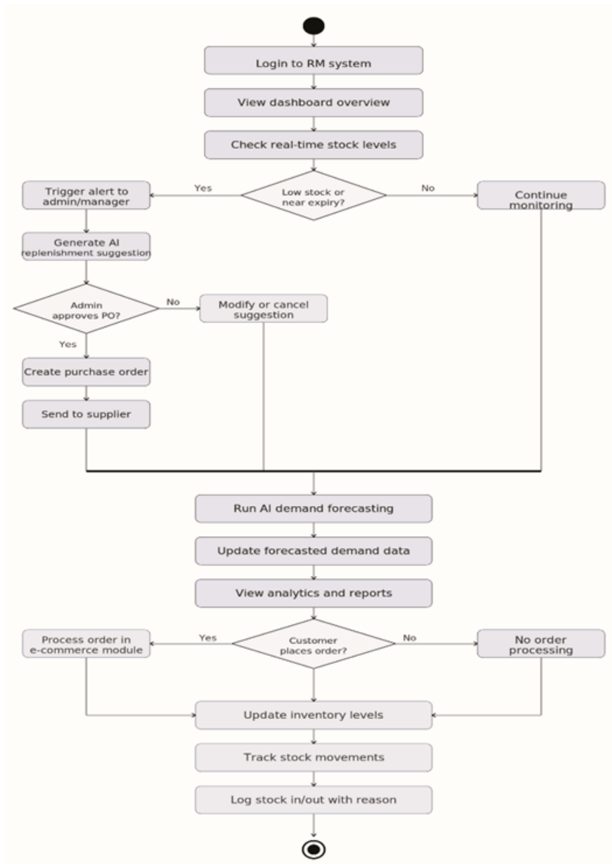


Figure 6. Activity Diagram

Figure 6 shows the main operational workflow of the system. The process begins when a user logs into the platform. Once authenticated, the user accesses the dashboard to view an overview of inventory status, including real-time stock levels and analytics. The system continuously monitors stock quantities, and if it detects that certain items are low in stock or nearing expiry, it automatically triggers an alert to the admin or stock manager. At this point, the system generates AI-powered replenishment suggestions. The admin then has the option to approve the suggested purchase order, which leads to the creation and sending of a purchase order to the appropriate supplier. If the admin chooses not to approve, they may modify or cancel the suggestion.

The system runs AI demand forecasting algorithms, updating demand predictions based on historical data, sales trends, and external factors. These forecasts feed into analytics dashboards and reports, helping managers make data-driven decisions. If a customer places an order through the e-commerce module, the system processes the order and automatically updates inventory levels to reflect the transaction. Throughout these activities, the system logs all stock movements — whether incoming, outgoing, or internal transfers — including the reason for movement to ensure traceability. The activity ends once the necessary actions are completed, keeping inventory, demand forecasting, and e-commerce processes tightly integrated and aligned.

Context Diagram

The Context Diagram for RM'S: AI-Integrated Inventory & E-Commerce System illustrates how the system interacts with external entities and supporting components. while the Customer places online orders directly through the integrated e-commerce module.

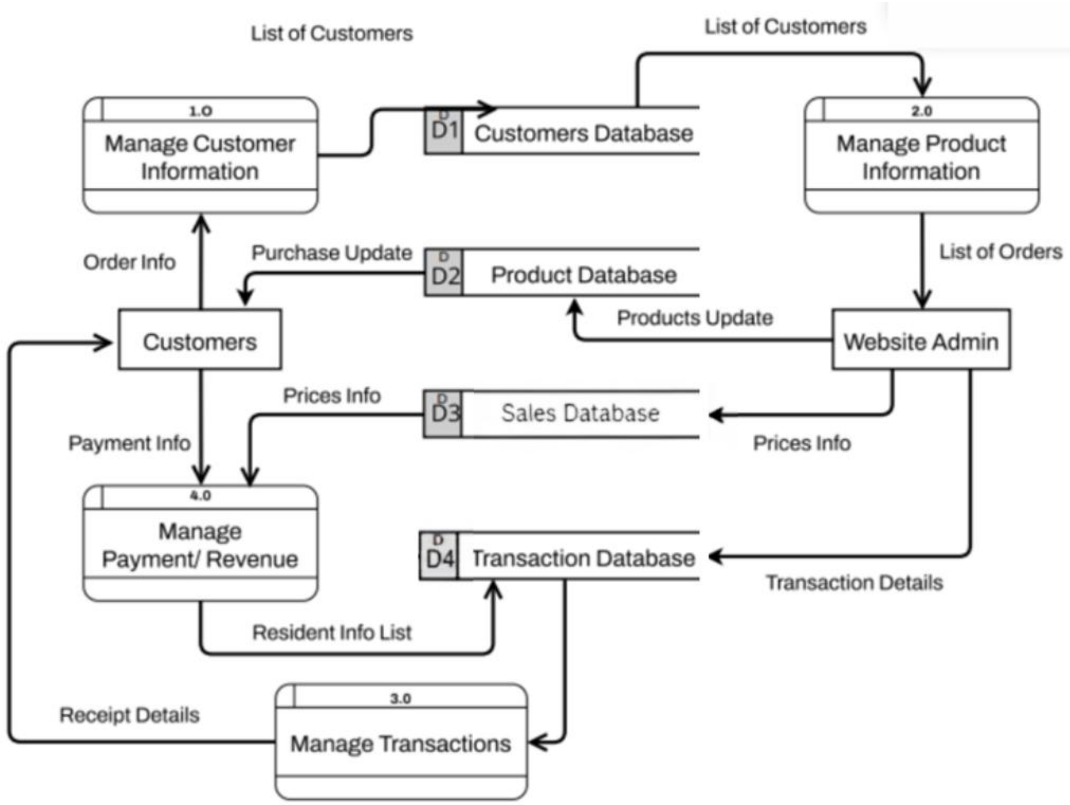
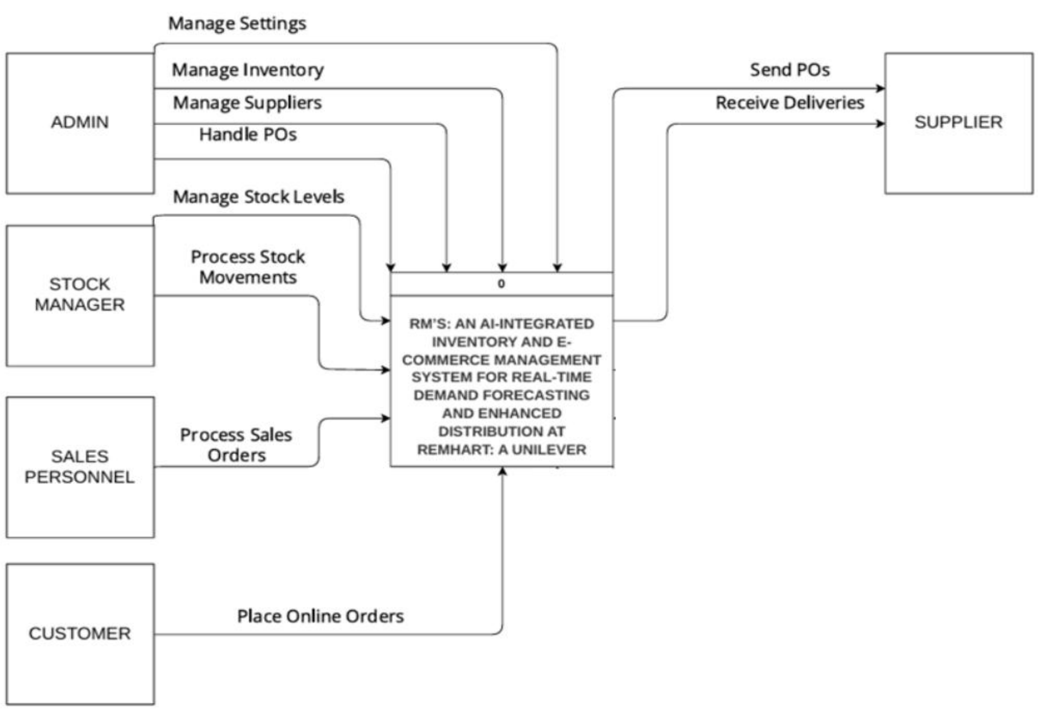


Figure 7. Context Diagram

The figure 7 shows how the system will be developed concerning the different fields in which they are part for specification into the users that will make use of it.

Data Flow Diagram (DFD)

The Level 0 Data Flow Diagram illustrates the main entities interacting with the "RM’S System," which serves as the central process for inventory management and demand forecasting. This diagram highlights the flow of information between external entities and the system, demonstrating how data is exchanged and processed to fulfill operational



requirements.

Figure 8. Diagram 0

Figure 8 DFD Level 0 illustrates the interactions and data flows between the Admin, Suppliers, and Inventory System.

Database Schema

The database schema for the RM’S Inventory System is a structured framework designed to facilitate efficient management.

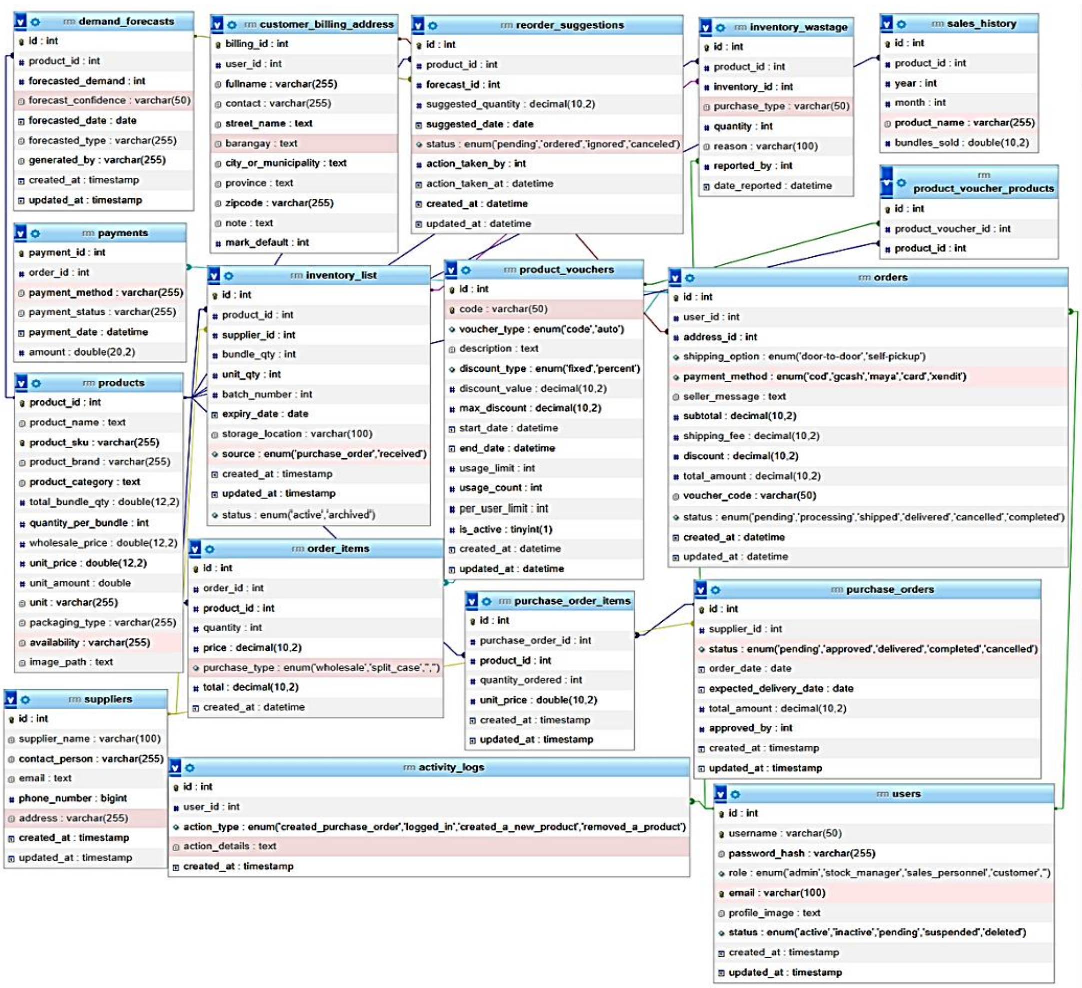


Figure 9. Database Schema

Legend:

One to One   Many to Many   One to Many

Figure 9 shows the structure of the E-recruitment database, where all the basic tables are connected using primary and foreign keys. Primary keys are unique identifiers for each record in a table, ensuring that each entry is distinct.

Testing and Evaluation

The testing and evaluation methodology for the RM'S Inventory System is designed to ensure the system meets its intended functionality, usability, and performance objectives. This methodology incorporates a structured approach involving participants, respondents, data gathering instruments, and a scaling system to systematically assess the system's effectiveness.

Table 5. Respondents of the Study

Category	Number of Respondents
Admin	1
Warehouse Managers	7
Sales Representative	10
Technical Experts	1
End Users	15
Total	**40**

Table 5 shows the respondents of the study, including the number of each category respondents

Data Gathering Instruments

Observation Checklists: Used to monitor system usability during live operations, focusing on tasks such as inventory updates, demand forecasting, and automated reordering.

Survey Questionnaires: Distributed to respondents to gather feedback on system performance, user interface design, and overall satisfaction. The survey is designed with both open-ended and closed-ended questions.

Performance Logs: Collected directly from the system to evaluate response times, accuracy of predictions, and overall system uptime during testing.

Focus Group Discussions (FGDs): Conducted with stakeholders and end users to gather qualitative insights into their experience with the system.

Table 6. Likert Scale

Scale	Range	Verbal Interpretation
4	3.50 - 4.00	Strongly Agree
3	2.50 - 3.49	Agree
2	1.50 - 2.29	Disagree
1	1.00 - 1.49	Strongly Disagree

Testing Phases

Unit Testing: Each module is tested independently to verify functional correctness

Integration Testing: Modules are combined to evaluate interactions between components such as inventory updates triggering reorder notifications.

System Testing: End-to-end testing ensures the entire system works cohesively, focusing on both functional and non-functional requirements.

User Acceptance Testing (UAT): Respondents test the system in a simulated real-world environment, providing feedback on usability and overall effectiveness.

Table 7. Evaluation Metrics

Metric	Method of Evaluation
System Usability	Feedback through survey questionnaires and focus group discussions.
Forecast Accuracy	Analysis of system-generated forecasts against real-world data.
Performance Efficiency	Monitoring response time, system uptime, and performance logs.
Impact on Workflow	User feedback on operational efficiency and observed workflow changes.
Security and Reliability	Penetration testing results and system uptime monitoring.

Implementation Plan

The implementation plan for the RM’S Inventory System outlines the steps required to successfully deploy the system and ensure it operates effectively in a live environment. The plan is divided into distinct phases, each focusing on critical activities to facilitate a seamless transition from development to full operational use.

Preparation Phase: During this phase, all resources required for implementation are procured and prepared. This includes setting up the necessary hardware, installing software components, and configuring development and production environments. Stakeholders are informed of the implementation timeline, and roles are assigned to ensure smooth execution.

Deployment Phase: The system is deployed on the production servers, and initial data, such as inventory records and user accounts, is migrated to the live database. Configuration settings, including access controls and automated features, are customized to meet the operational needs of the organization. Rigorous system-wide testing is conducted in the live environment to identify and resolve any potential issues before the system goes live.

Training Phase: Key users, including admin staff, warehouse managers, and sales representatives, are provided with comprehensive training sessions. These sessions cover all system functionalities, including inventory management, demand forecasting, real-time updates, and reporting tools. User manuals and training materials are distributed to aid understanding and ensure users are confident in operating the system.

Go-Live Phase: The system is officially launched for live operations, with all users granted access. A dedicated support team is on standby to address any immediate issues and assist users as they navigate the system during the initial weeks of operation. Performance metrics are closely monitored to ensure the system functions as intended, and any feedback from users is promptly addressed.

Post-Implementation Review: After the system has been operational for a designated period, a review is conducted to evaluate its performance and overall impact. This phase involves collecting feedback from users, analyzing performance data, and identifying areas for improvement.

Chapter IV

RESULTS AND DISCUSSION

This chapter presents, analyzes, and interprets the results gathered from the evaluation of the developed system. It provides an overview of the data collected from respondents, analyzed using appropriate statistical tools to determine the system's performance and level of acceptability.

Presentation of System Output

This section highlights how the RM'S: AI-Integrated Inventory and E-Commerce Management System delivers and displays its output, focusing on clarity, efficiency, and user accessibility.

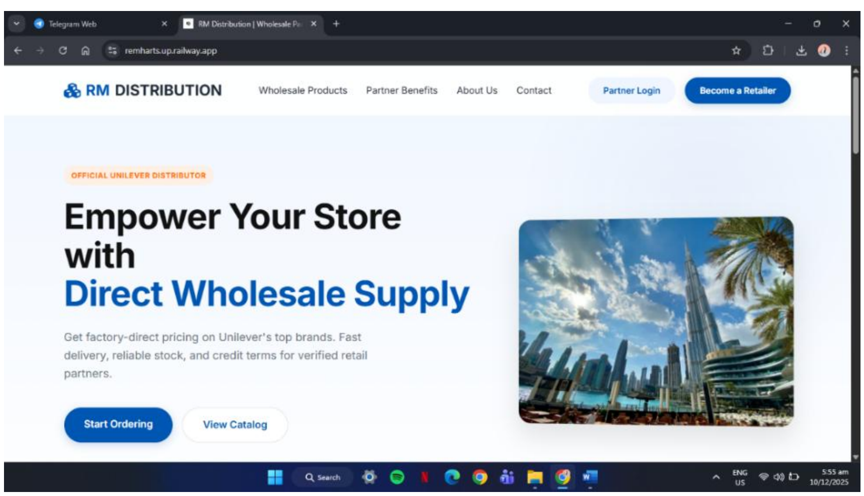


Figure 10. Home Page

Admin Side

The Admin Side showcases the interface built for system administrators, providing comprehensive tools for oversight and configuration. Through this panel, admins can manage users, monitor transactions, adjust system settings, and ensure that every module functions as intended. This centralized access point enables administrators to maintain order, enforce policies, and oversee the system's overall performance.

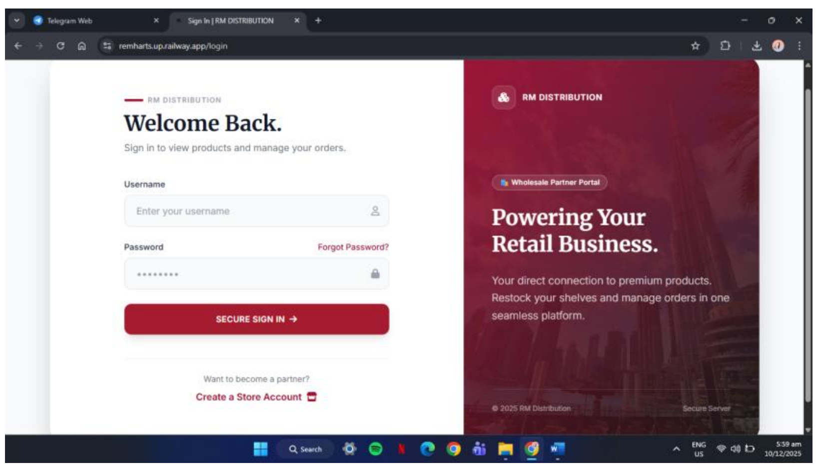


Figure 11. Login Form

Figure 11 shows the Login form that provides secure access to the system for authorized users. It uses role-based authentication, allowing specific access to modules based on user roles such as admin, warehouse manager, or sales representative.

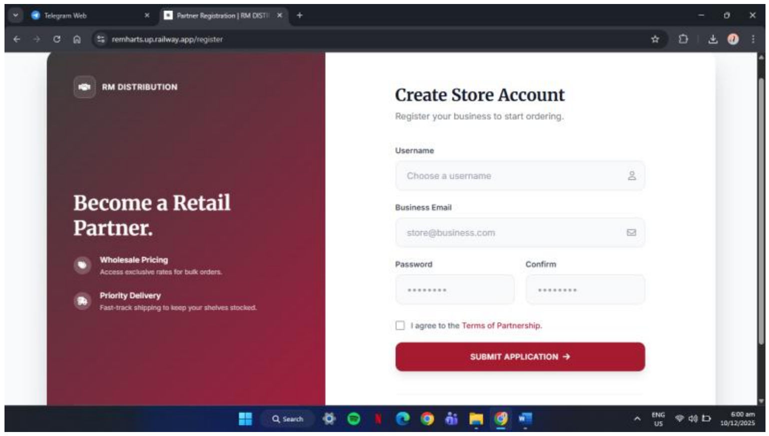


Figure 12. Sign-Up Form

Figure 12 shows the Sign In form where it is the system's gateway for user authentication. Users enter their registered email and password to gain access. It features input validation, error messaging for incorrect credentials, and redirects to the appropriate dashboard based on the user's role. This ensures that only authorized personnel, such as sales reps, admins, or warehouse staff, can interact with the system's modules.

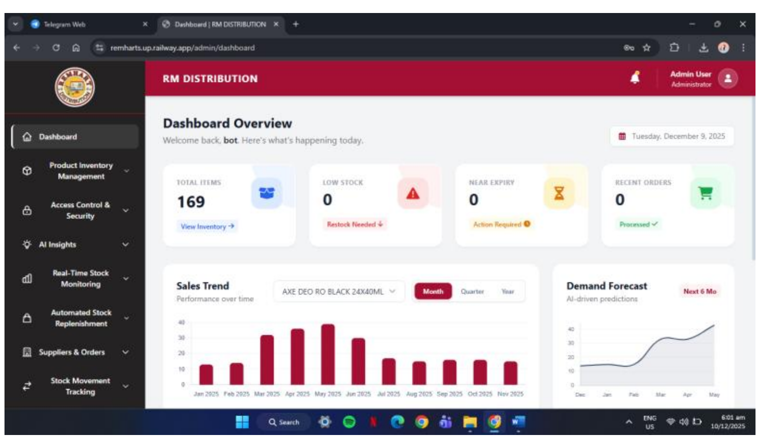


Figure 13. Dashboard

Figure 13 shows The central hub of the system. Displays a snapshot of key metrics such as current inventory levels, recent sales, low stock alerts, and demand forecasts. Graphs and charts provide quick insights, allowing users to monitor performance and trends in real time.

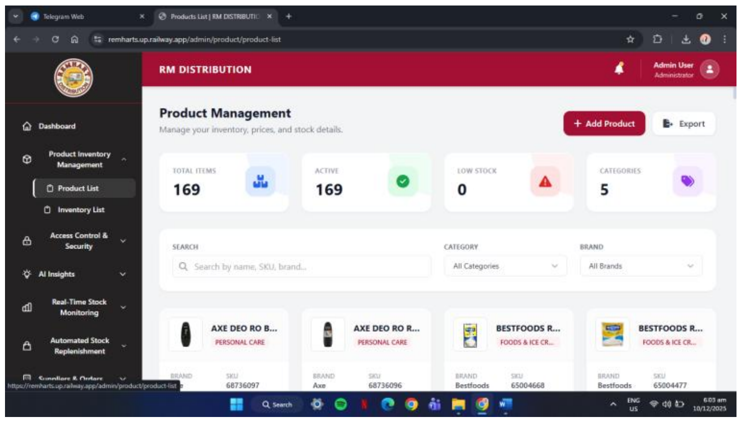


Figure 14. Product List

Figure 14 presents a comprehensive list of all available products in the system. Each entry includes the product name, description, product SKU, and price.

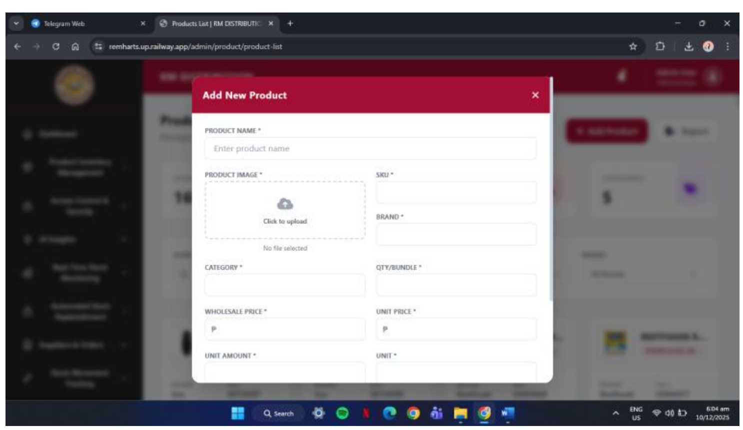


Figure 15. Add Product

Figure 15 presents the add product form that allows users to register new products into the system. The form includes fields for product name, description, price, unit, and product SKU. Once submitted, the product becomes visible in both the inventory and the product catalog.

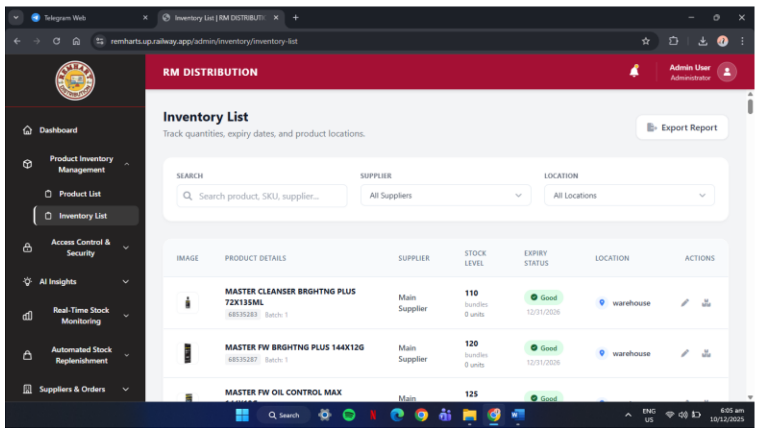


Figure 16. Inventory List

Figure 16 shows the inventors list that displays all stock currently in inventory. Users can see product names, available quantity, last updated date, and storage location.

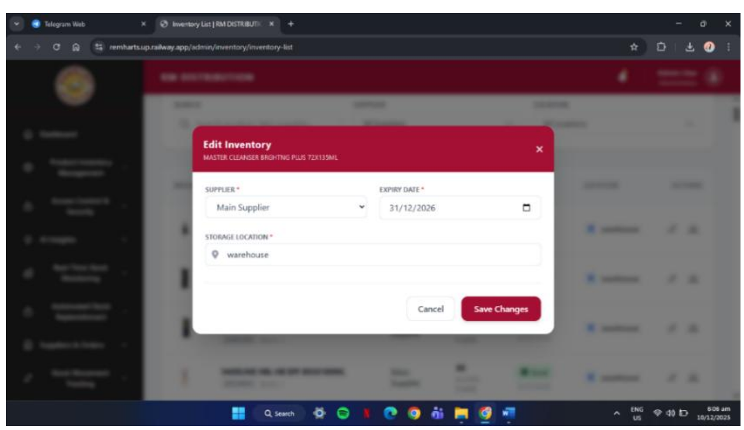


Figure 17. Edit Inventory Product

Figure 17 shows the edit inventory product that allows users to modify existing product details within the inventory.

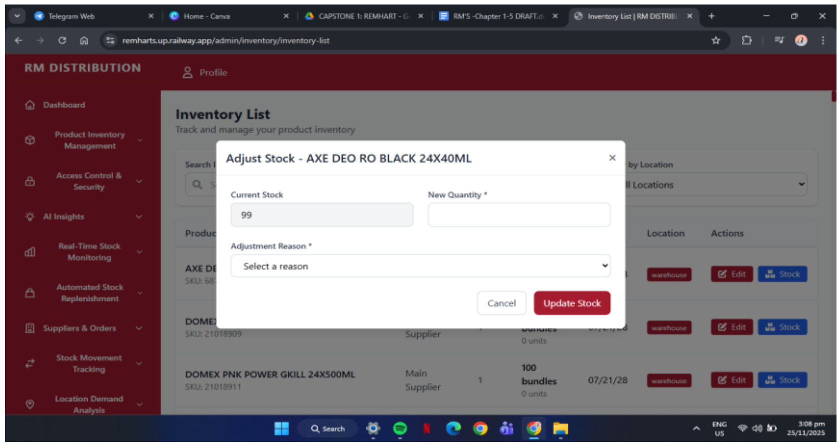


Figure 18. Edit Inventory Stock

Figure 18 shows the edit inventory stock that enables adjustments to stock quantity for cases like inventory audits, loss, or damage.

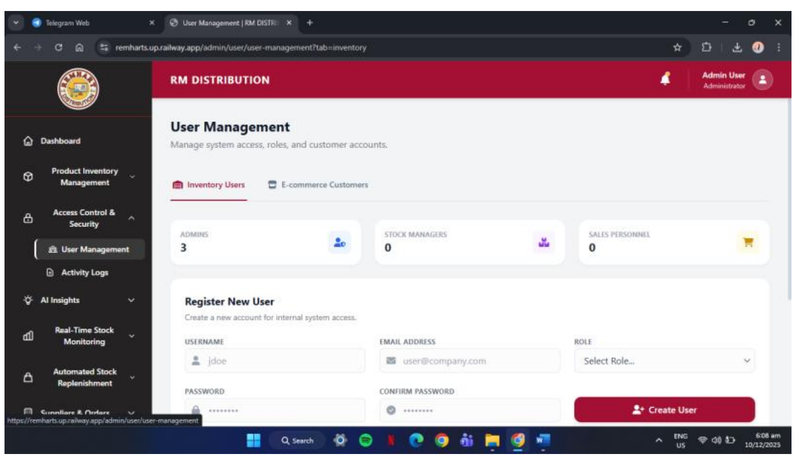


Figure 19. User Management

Figure 19 shows the User Management module that handles all tasks related to adding, editing, and organizing system users. Admins can assign roles, update permissions, and deactivate accounts as needed. It ensures that each user has the correct level of access, improving security and maintaining proper workflow separation across departments such as sales, warehousing, and admin operations.

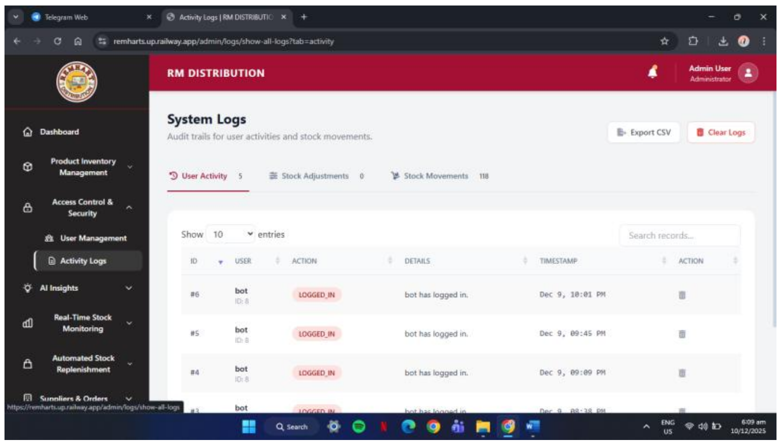


Figure 20. Activity Logs

Figure 20 shows the Activity Logs module that records every action performed by users within the system. It captures details such as timestamps, accessed modules, and the type of activity executed. This feature supports auditing, accountability, and security monitoring by allowing administrators to trace user behavior and verify system integrity.

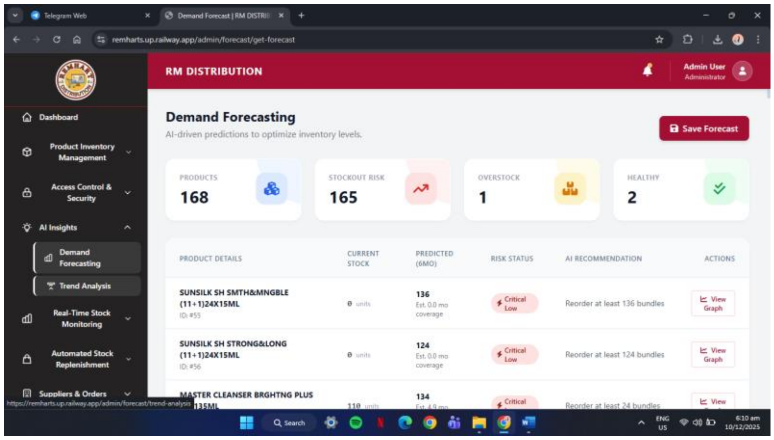


Figure 21. Demand Forecasting

Figure 21 shows The Demand Forecasting feature that analyzes past sales data and inventory movement to predict future product needs. It helps the company prepare for seasonal spikes, fast-moving items, and potential shortages. This module supports smarter purchasing decisions by minimizing overstocking and preventing stockouts.

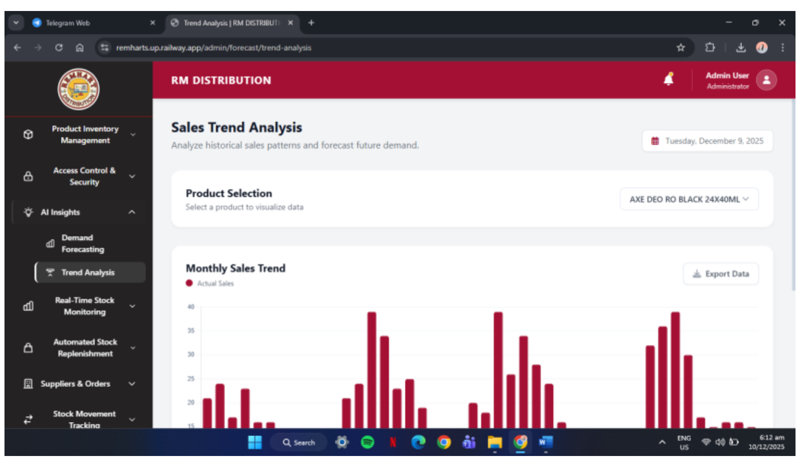


Figure 22. Trend Analytics

Figure 22 shows the Trend Analytics that provides insights into long-term sales and product performance. It visualizes shifts, growth patterns, and declining trends, allowing managers to understand what's working and what needs attention. This module supports data-driven planning for promotions, pricing, and inventory strategies.

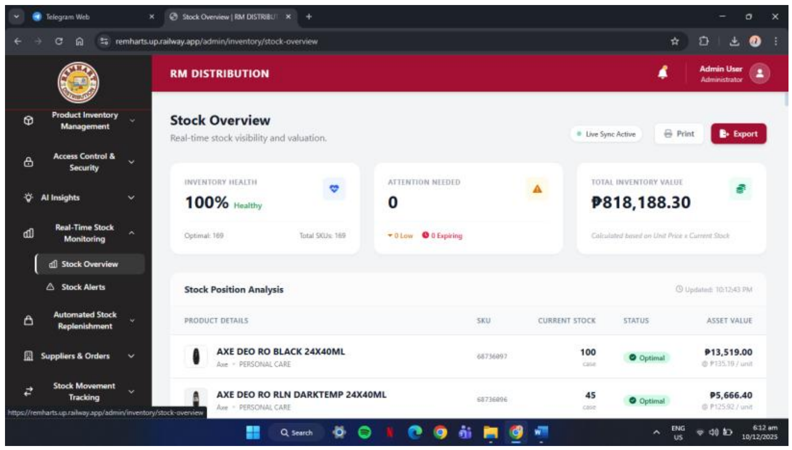


Figure 23. Stock Overview

Figure 23 shows the Stock Overview page that serves as the central dashboard for current inventory status. It displays available quantities, stock conditions, critical levels, and item summaries. Users can quickly gauge product availability without navigating multiple sections, improving decision-making and operational efficiency.

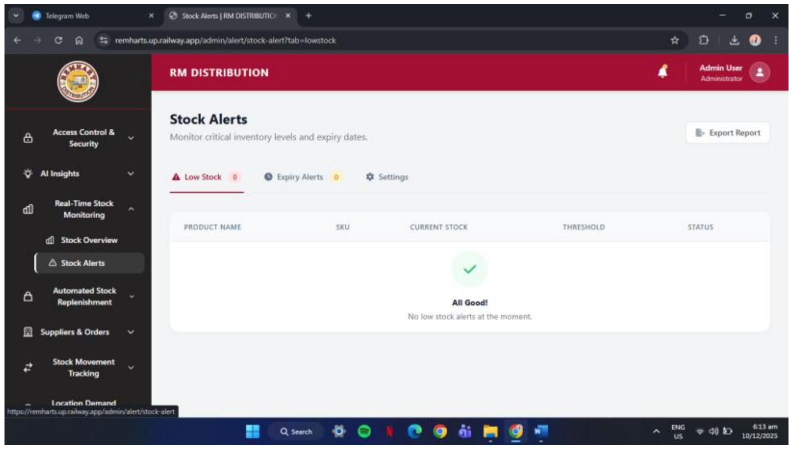


Figure 24. Stock Alerts

Figure 24 shows the Stock Alerts that notify users when items reach their minimum quantity threshold. This module ensures that staff are immediately informed about low stock, preventing delays in replenishment. It plays a crucial role in maintaining healthy inventory levels, especially for high-demand products.

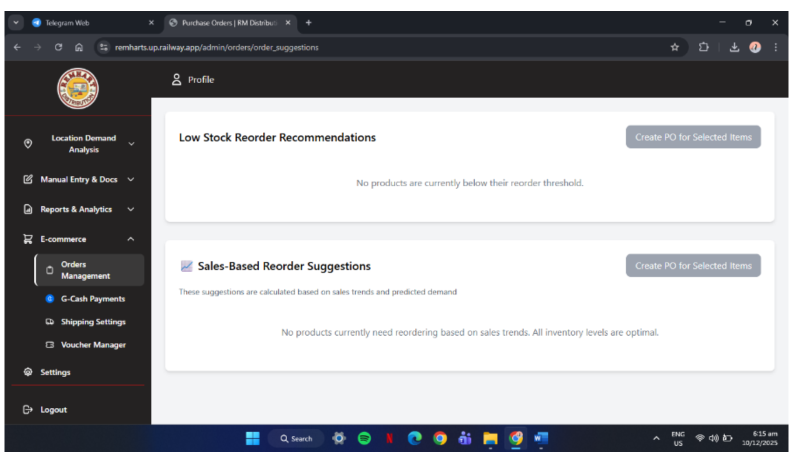


Figure 25. Reorder Suggestions

Figure 25 shows the Reorder Suggestions feature that provides automated recommendations for items that need restocking. Based on sales velocity, demand patterns, and set thresholds, it guides purchasing staff on what to order and when. This reduces manual monitoring and supports a seamless replenishment process.

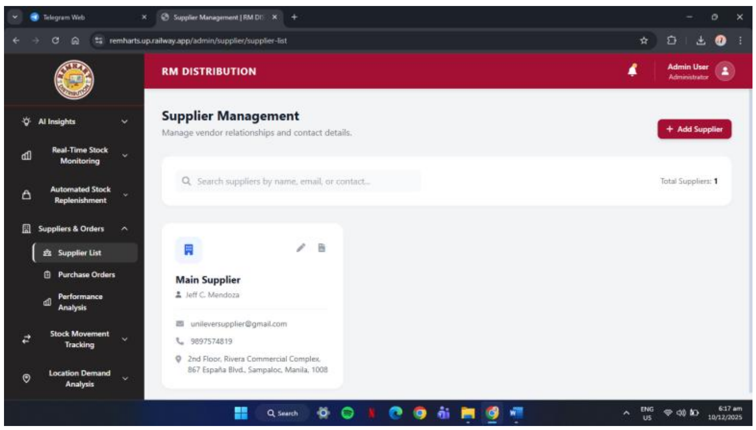


Figure 26. Supplier Management

Figure 26 shows the Supplier Management module that stores complete profiles of all suppliers, including contact details, product scopes, and transaction history. It centralizes vendor information to make communication and purchasing more efficient. This ensures smoother procurement operations and better supplier relationship management.

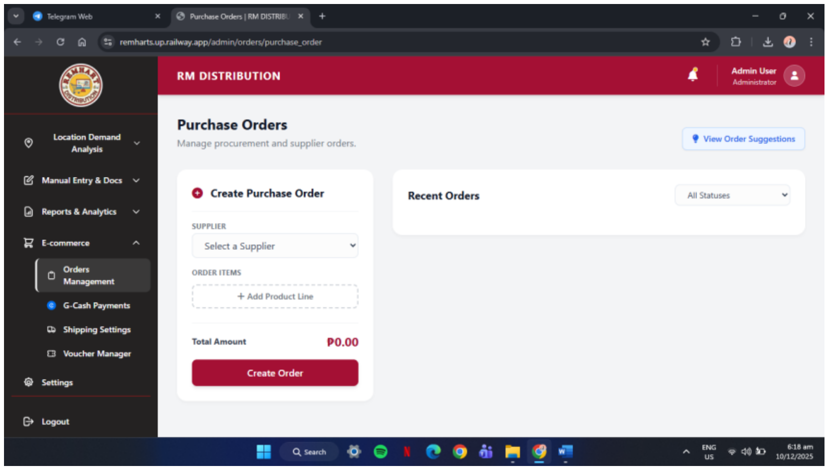


Figure 27. Purchase Orders

Figure 28 shows the Purchase Orders module that facilitates the creation, submission, and tracking of orders sent to suppliers. Users can monitor status updates such as pending, approved, delivered, or canceled. It supports accurate documentation and maintains a complete history of all procurement activities.

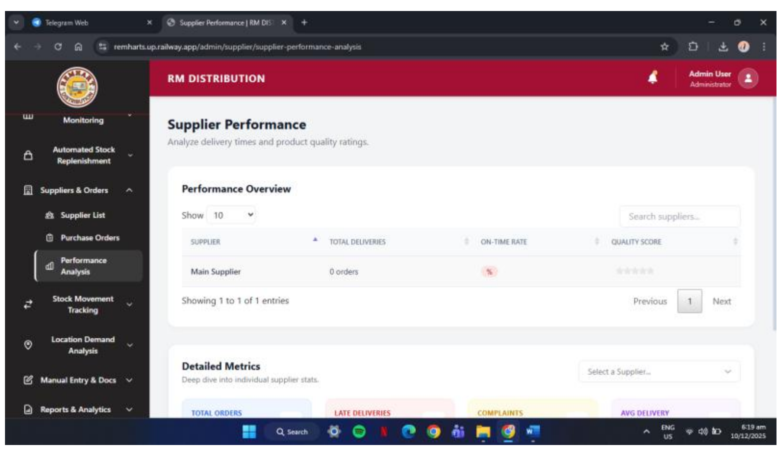


Figure 28. Performance Analysis

Figure 28 shows the Performance Analysis that offers a detailed breakdown of sales, product movement, and operational efficiency. It highlights best-selling items, underperforming SKUs, and overall market behavior. This module helps managers evaluate business performance and create strategies for improvement.

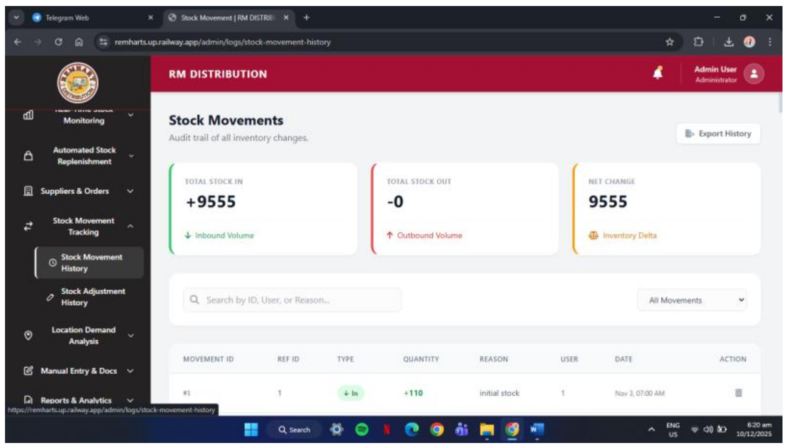


Figure 29. Stock Movement History

Figure 29 Stock Movement History records every inbound and outbound movement of inventory. It includes transfers, deliveries, returns, and withdrawals, along with timestamps and responsible users. It ensures traceability and helps identify discrepancies in stock counts.

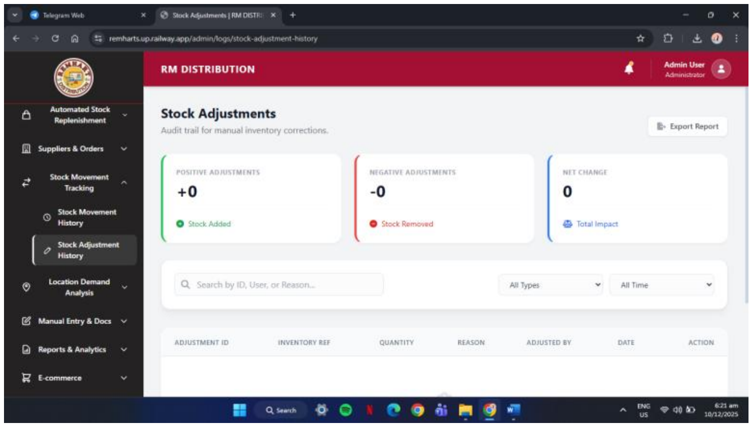


Figure 30. Stock Adjustments History

Figure 30 the Stock Adjustments History module logs all manual changes made to inventory quantities. It captures reasons such as damage, loss, corrections, or returns.

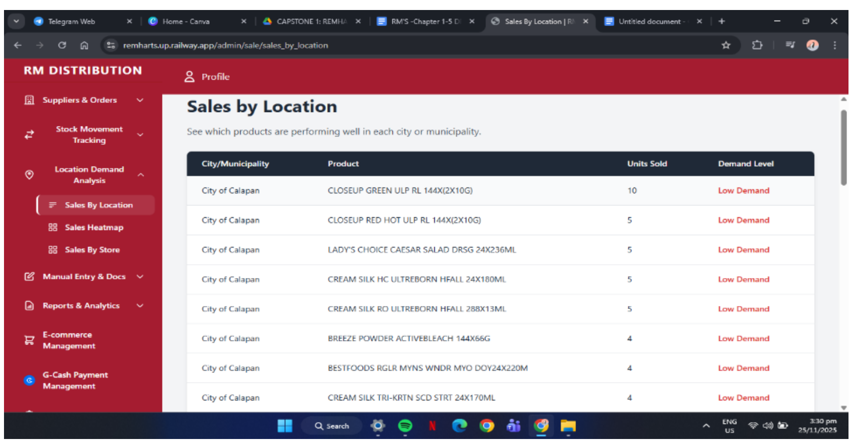


Figure 31. Sale by Location

Figure 31 shows the Sale by Location that provides a geographic breakdown of sales performance. It shows which areas or routes generate the highest revenue, helping managers optimize distribution efforts.

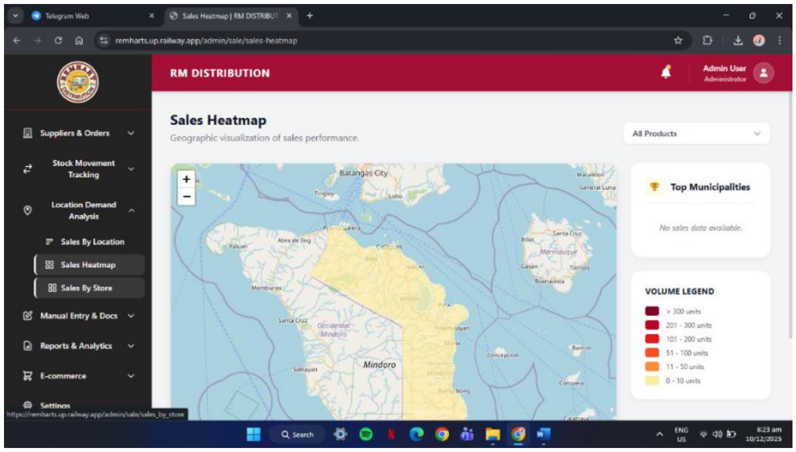


Figure 32. Sales Heatmap

Figure 32 the Sales Heatmap visualizes sales intensity using color coding to highlight strong and weak periods. It allows users to quickly identify peak hours, days, or locations.

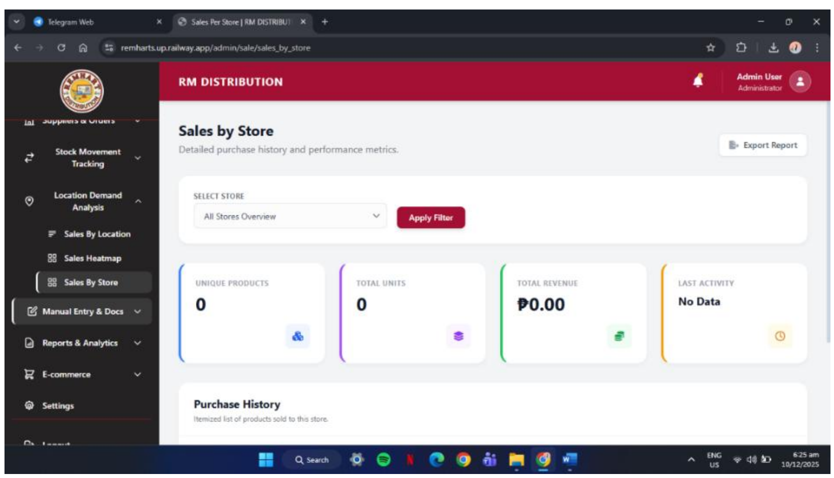


Figure 33. Sales By Store

Figure 33 shows this module that compares sales results across different branches or stores. It highlights high-performing locations and identifies those needing improvement. It helps management evaluate branch productivity and allocate resources more effectively.

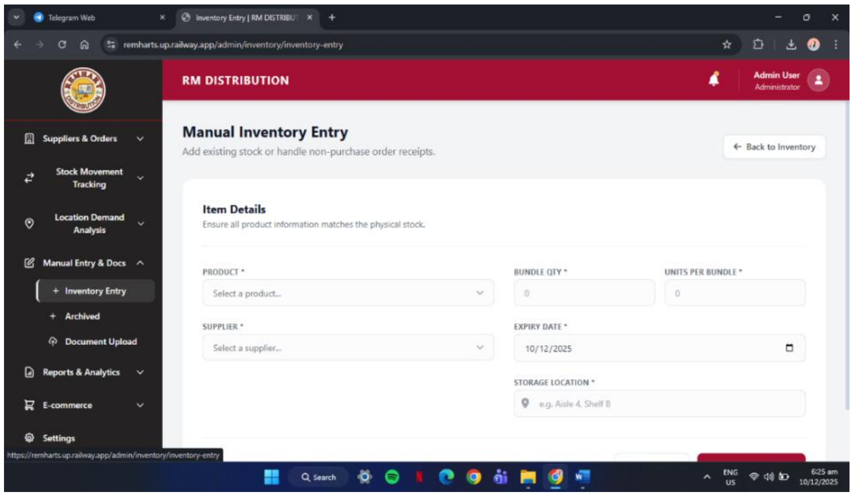


Figure 34. Inventory Entry

Figure 34 shows the Inventory Entry that allows users to manually input new stock arrivals, updates, or corrections. This module ensures that all changes in inventory are accurately recorded and immediately reflected in the system.

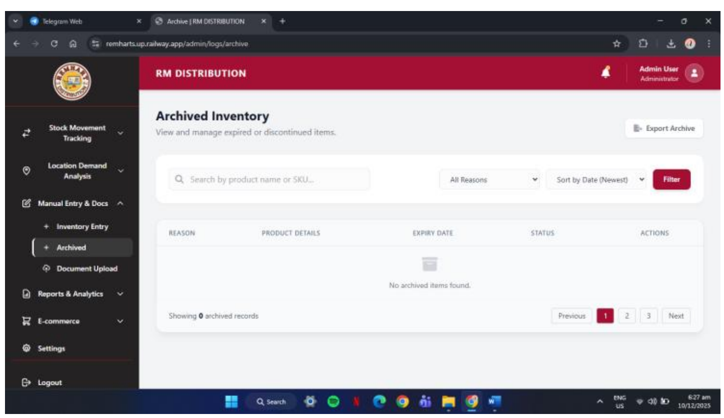


Figure 35. Archived

Figure 35 shows the Archived module that stores old or inactive records that are no longer used but must still be kept for reference. It helps declutter active modules while preserving important historical data. Users can retrieve archived information anytime when needed.

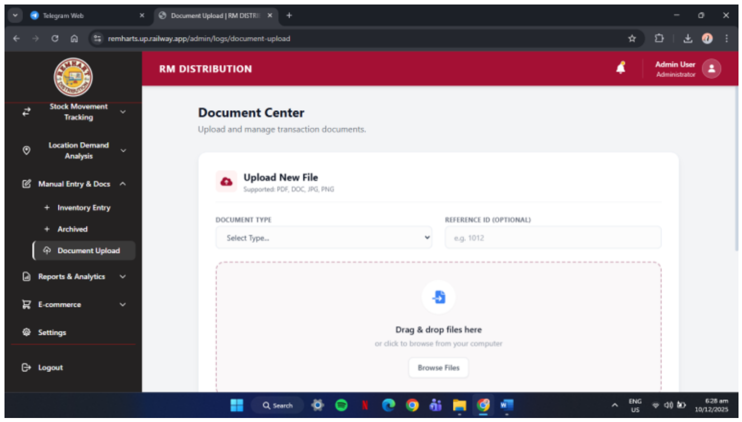


Figure 36. Document Upload

Figure 36 shows the Document Upload that allows users to attach files such as invoices, receipts, product documents, and transaction proof.

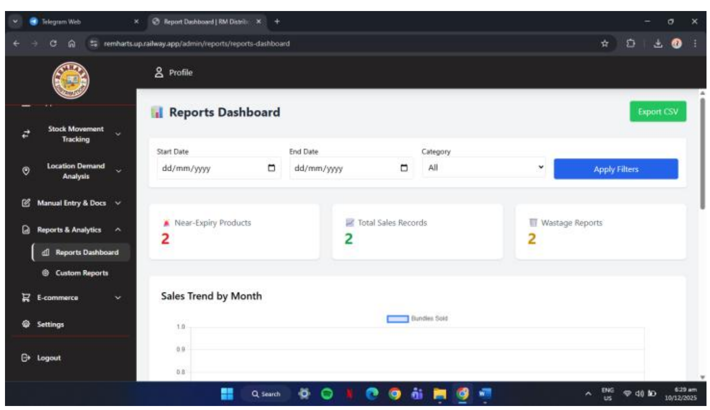


Figure 37. Reports Dashboard

Figure 37 shows the Reports Dashboard that consolidates key business metrics into visual charts and summaries. It gives users a quick snapshot of performance across sales, inventory, and operations.

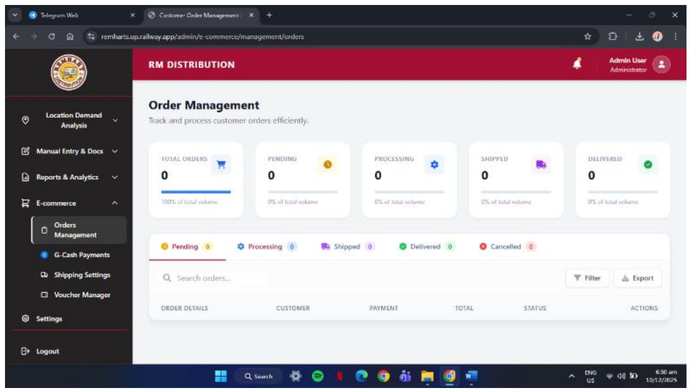


Figure 37. E-commerce Management

Figure 37 shows the E-commerce Management that oversees online orders, customer transactions, and product listings. It syncs digital sales with physical inventory to prevent discrepancies.

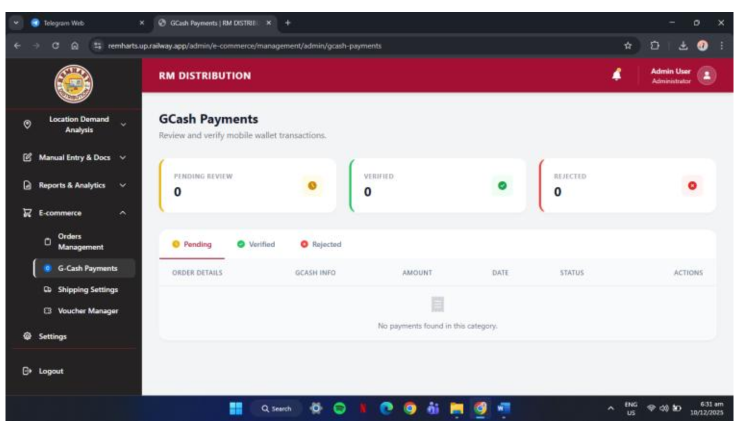


Figure 38. Gcash Payment Management

Figure 38 shows this module that handles all payments made through GCash. It records transactions, verifies payment status, and supports reconciliation. It ensures smooth digital payments and helps maintain accurate financial records across all customer purchases.

User Side

The User Side provides the main interface designed for customers or end-users, offering a smooth and intuitive experience for browsing, purchasing, and interacting with the system. Through this panel, users can explore products, manage their carts, track orders, and access relevant features with ease. This streamlined environment ensures convenient navigation, supports efficient transactions, and enhances overall user satisfaction by making every essential action readily accessible.

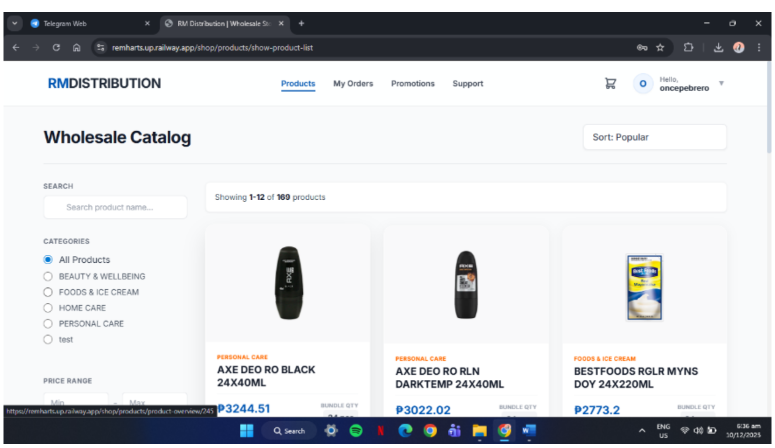


Figure 39. Home Page - All Products

Figure 39 shows the Home Page that displays all available products in one view, giving users quick access to the full catalog.

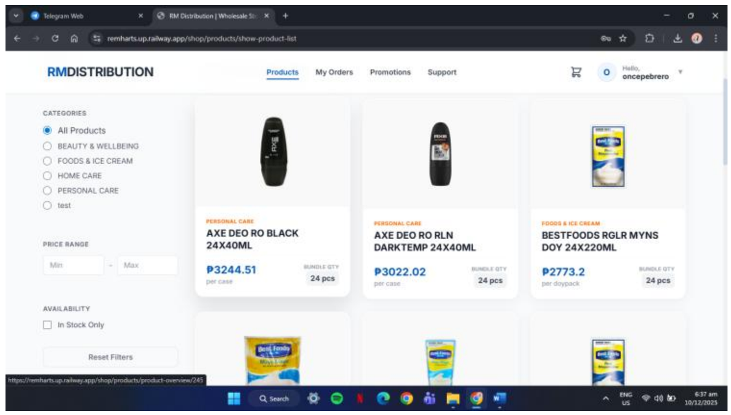


Figure 40. All Categories

Figure 40 shows the All-Categories section that organizes products into groups for easier navigation. It allows users to quickly filter items based on their interests or needs. This module improves product discovery and helps customers find what they’re looking for without hassle.

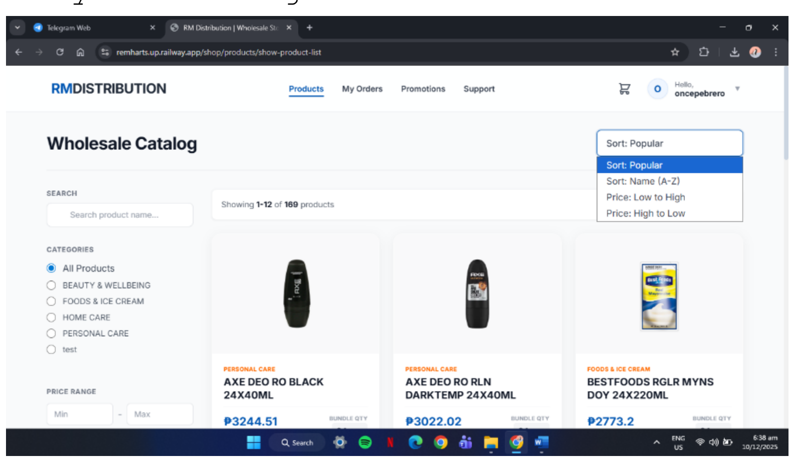


Figure 41. Sort Popular

Figure 41 shows the Sort Popular that lets users arrange products based on popularity or high engagement.

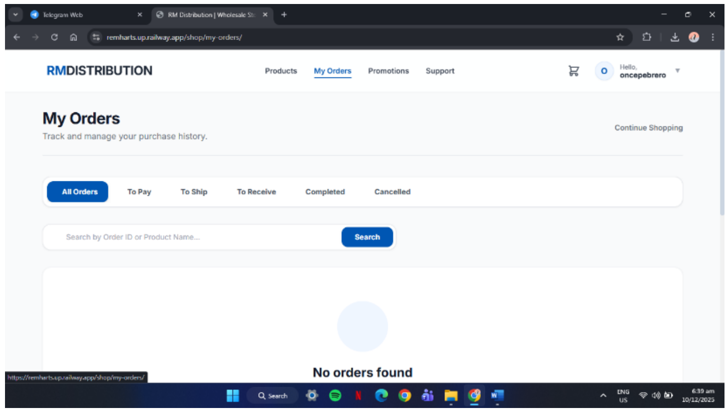


Figure 42. My Orders

Figure 42 shows the My Orders that provides users with a convenient overview of their past and ongoing purchases. It displays order details, statuses, and timestamps to keep customers updated throughout the fulfillment process.

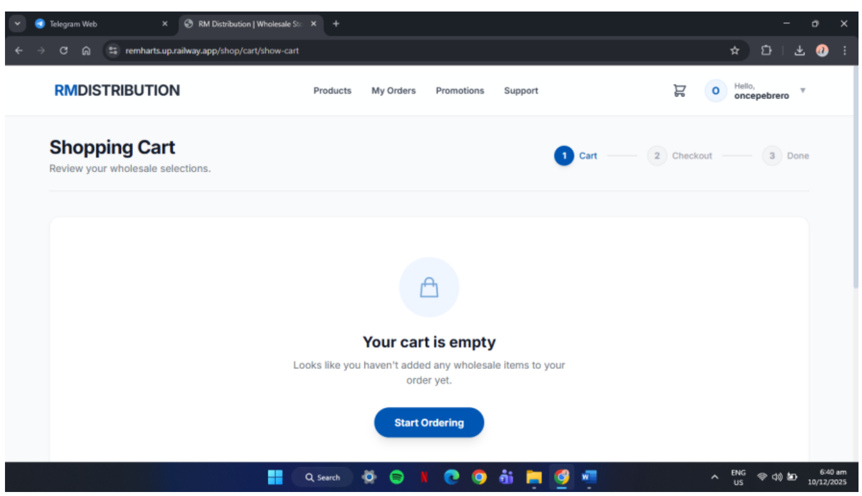


Figure 43. Shopping Cart

Figure 43 shows the Shopping Cart items that users have selected for purchase, along with quantities and pricing details. It allows customers to review, update, or remove products before checking out. This module ensures accuracy, convenience, and a seamless transition to the payment process.

Evaluation of the System

After designing, developing, and testing RM'S: An AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Enhanced Distribution at REMHART, the system underwent a comprehensive evaluation to determine its quality, usability, and acceptance based on ISO 25010 Software Quality Metrics and the Unified Theory of Acceptance and Use of Technology (UTAUT) model.A total of 65 respondents participated in the evaluation, composed of business owners, staff, and customers, who assessed the system's performance across multiple dimensions: Functional Suitability, Performance Efficiency, Usability, Reliability, Performance Expectancy, Effort Expectancy, Facilitating Conditions, and Behavioral Intention. Furthermore, the evaluation process involved administering a structured survey and guided system walkthroughs to ensure that respondents could accurately assess each metric based on their actual interaction with the platform.

1. ISO 25010 Metrics Tables

Table 8. Functional Suitability of the System

Functional Suitability	**Mean**	**Rank**	**Verbal Interpretation**
1.1 The functions of the system cover all specified tasks and user objectives.	3.45	2	Agree
1.2 The system produces accurate and reliable results.	3.43	4	Agree
1.3 The system facilitates the accomplishment of defined objectives effectively.	3.47	1	Agree
1.4 The system provides complete features necessary for business operations.	3.42	5	Agree
1.5 The system meets all intended business and user requirements.	3.40	6	Agree
1.6 The system functions correctly under different operational conditions.	3.38	7	Agree
1.7 The system integrates inventory and e-commerce processes smoothly.	3.44	3	Agree
Overall Mean	**3.43**		**Agree**

The overall mean ofu **3.43**, interpreted as *"Agree,"* demonstrates that the respondents recognized RM'S as functionally suitable for business use. The highest-rated indicators — "facilitates the accomplishment of defined objectives effectively" (3.47) and "covers all specified tasks and user objectives" (3.45) — emphasize that the system effectively addresses business needs and supports operational workflows.

Table 9. Performance Efficiency of the System

Performance Efficiency	Mean	Rank	Verbal Interpretation
2.1 The system responds quickly and meets user requirements.	3.41	1	Agree
2.2 The system operates efficiently using available resources.	3.36	4	Agree
2.3 The system maintains consistent performance during heavy usage.	3.33	7	Agree
2.4 Loading times are reasonable and suitable for real-time operations.	3.38	3	Agree
2.5 The AI features operate without delays.	3.35	5	Agree
2.6 The system handles multiple users simultaneously without lag.	3.39	2	Agree
2.7 Data synchronization between modules is seamless.	3.34	6	Agree
Overall Mean	**3.36**		**Agree**

The overall mean of **3.36** reflects an *"Agree"* interpretation, indicating that the system delivers a satisfactory level of performance efficiency. The top-rated indicator, "The system responds quickly and meets user requirements" (3.41), implies that users experience minimal delays while interacting with the system. Respondents appreciated its consistent responsiveness even under heavy usage, as well as its ability to synchronize data in real time between modules. Although AI features occasionally experienced slight delays, users still found them dependable for forecasting and decision-making. Overall, the findings highlight RM'S capability to operate efficiently with stable response times and reliable multi-user performance.

Table 10. Usability of the System

Usability	Mean	Rank	Verbal Interpretation
3.1 The system is easy to use and understand.	3.40	1	Agree
3.2 The interface is organized and user-friendly.	3.38	3	Agree
3.3 The system minimizes user errors.	3.32	7	Agree
3.4 The system can be used by users of varying experience levels.	3.33	6	Agree
3.5 Icons and labels are clear and self-explanatory.	3.36	5	Agree
3.6 The layout is visually appealing and consistent.	3.39	2	Agree
3.7 Navigation between features is smooth.	3.37	4	Agree
3.8 The system provides tool tips or help prompts when needed.	3.28	8	Agree
Overall Mean	**3.35**		**Agree**

The overall mean of 3.35 signifies that respondents "Agree" that RM'S demonstrates high usability. The items "easy to use and understand" (3.40) and "visually appealing and consistent layout" (3.39) earned the highest ratings, suggesting that users appreciate its intuitive and organized design. The system's user interface promotes effortless navigation and minimizes errors, making it suitable for both technical and non-technical users. Although a few respondents suggested adding more help prompts, the overall feedback highlights that the platform's simplicity and clarity significantly enhance user satisfaction and accessibility.

Table 11. Reliability of the System

Reliability	Mean	Rank	Verbal Interpretation
4.1 The system performs reliably under normal operation.	3.33	2	Agree
4.2 The system recovers easily from interruptions.	3.29	4	Agree
4.3 The system ensures stability during continuous use.	3.34	1	Agree
4.4 Backup and recovery functions work properly.	3.28	5	Agree
4.5 The system ensures data consistency after crashes.	3.31	3	Agree
4.6 The system operates reliably in different environments.	3.25	7	Agree
4.7 The system avoids duplication or loss of data.	3.27	6	Agree
Overall Mean	**3.30**		**Agree**

The overall mean of 3.30 indicates that users "Agree" with the system's reliability. The highest-rated indicator, "ensures stability during continuous use" (3.34), demonstrates confidence in the system's consistent performance during extended operation. Respondents confirmed that the backup and recovery mechanisms perform well, reducing data loss risks during interruptions. While a few users noted occasional slow recovery after errors, the majority affirmed that RM’S maintains operational integrity, reliability, and consistent functionality across different conditions. This establishes its dependability for daily business activities.

Table 12. ISO 25010 Software Metrics Summary Results of the Evaluation

Items	Mean	Rank	Verbal Interpretation
1. Functional Suitability	3.43	1	Agree
2. Performance Efficiency	3.36	2	Agree
3. Usability	3.35	3	Agree
4. Reliability	3.30	4	Agree
Overall Mean	**3.36**		**Agree**

The overall mean of **3.36** across ISO 25010 dimensions indicates that the RM’S system achieved a satisfactory level of software quality. The highest rating in *functional suitability* shows that the system effectively meets business requirements and user objectives, while the consistent “Agree” ratings in other categories demonstrate that it performs efficiently, is easy to use, and operates reliably. This overall evaluation confirms that RM’S fulfills its intended purpose of automating REMHART's inventory and e-commerce operations with accuracy and efficiency.

2. UTAUT Model Metrics Table

Table 13. Performance Expectancy

Performance Expectancy	**Mean**	**Rank**	**Verbal Interpretation**
1.1 The system improves performance in inventory and sales.	3.41	2	Agree
1.2 The system helps complete work efficiently.	3.39	3	Agree
1.3 The system makes tasks easier and faster.	3.43	1	Agree
1.4 The system improves work quality.	3.37	5	Agree
1.5 The system is useful for daily operations.	3.38	4	Agree
1.6 The AI forecasting optimizes restocking.	3.35	7	Agree
1.7 The system enhances decision-making.	3.34	8	Agree
1.8 The system supports business growth and efficiency.	3.36	6	Agree
Overall Mean	**3.38**		**Agree**

The overall mean of 3.38 represents an "Agree" interpretation, indicating that RM'S effectively enhances productivity and decision-making. The top-rated item, "The system makes tasks easier and faster" (3.43), shows that users recognize its ability to simplify complex tasks. Respondents noted that AI-driven forecasting significantly improved restocking accuracy and reduced manual workload. Additionally, they appreciated the system's ability to support daily operations and promote efficient work completion. These findings suggest that RM'S contributes to operational excellence and improved organizational performance.

Table 14. Effort Expectancy

Effort Expectancy	Mean	Rank	Verbal Interpretation
2.1 Learning to use the system is simple.	3.32	2	Agree
2.2 The system is easy to operate.	3.33	1	Agree
2.3 Interaction with the system is straightforward.	3.30	3	Agree
2.4 Using the system saves time.	3.26	7	Agree
2.5 The system is flexible to use.	3.29	4	Agree
2.6 Interface design supports quick understanding.	3.28	5	Agree
2.7 Minimal training is required.	3.27	6	Agree
2.8 Experience remains smooth for new users.	3.25	8	Agree
Overall Mean	**3.29**		**Agree**

The overall mean of **3.29** reflects that respondents *"Agree"* that RM'S is easy to use and understand. The highest-rated indicators — "easy to operate" (3.33) and "learning to use the system is simple" (3.32) — show that the system's design effectively supports usability. Users found it intuitive, requiring minimal training even for beginners. They also noted that the interface promotes quick understanding, enabling users to adapt smoothly. These results indicate that RM'S successfully reduces user effort, encouraging broader adoption among non-technical users.

Table 15. Facilitating Conditions

Facilitating Conditions	Mean	Rank	Verbal Interpretation
3.1 I have resources required to use the system.	3.32	3	Agree
3.2 I have the knowledge necessary.	3.28	7	Agree
3.3 Assistance is available when needed.	3.34	1	Agree
3.4 The system aligns with my work practices.	3.30	5	Agree
3.5 Technical support is responsive.	3.33	2	Agree
3.6 Hardware and internet support usage.	3.29	6	Agree
3.7 Organization encourages system use.	3.31	4	Agree
3.8 Training materials are helpful.	3.27	8	Agree
Overall Mean	**3.31**		**Agree**

The overall mean of 3.31, interpreted as "Agree," suggests that users are well-supported in utilizing RM'S. The highest-rated indicators — "Assistance is available when needed" (3.34) and "Technical support is responsive" (3.33) — indicate that users value the system's strong support framework and guidance availability. Respondents also acknowledged the system's alignment with their workflow, as well as the accessibility of resources and training materials. These findings demonstrate that RM'S provides adequate support mechanisms, enhancing its usability and reliability within the organization.

Table 16. Behavioral Intention

Behavioral Intention	**Mean**	**Rank**	**Verbal Interpretation**
4.1 Using the system is beneficial.	3.36	1	Agree
4.2 I enjoy using the system.	3.29	5	Agree
4.3 The system makes work more engaging.	3.26	7	Agree
4.4 I intend to continue using the system.	3.33	2	Agree
4.5 I would recommend the system to others.	3.30	4	Agree
4.6 I feel confident in its reliability.	3.31	3	Agree
4.7 I plan to explore more features.	3.27	6	Agree
4.8 The system motivates me to adopt similar technologies.	3.25	8	Agree
Overall Mean	**3.30**		**Agree**

The overall mean of 3.30 reflects that users "Agree" with positive behavioral intentions toward using RM'S. The indicators "Using the system is beneficial" (3.36) and "I intend to continue using the system" (3.33) scored highest, showing that users perceive RM'S as valuable and worth continued use. Respondents expressed enjoyment in using the system and a willingness to recommend it to others, demonstrating satisfaction and confidence in its reliability. These findings suggest a high potential for sustained adoption and advocacy of RM’S within REMHART’s operations.

Table 17. Summary Results of the Evaluation for UTAUT Model Metrics Tables

Items	Mean	Rank	Verbal Interpretation
Performance Expectancy	3.38	1	Agree
Effort Expectancy	3.29	3	Agree
Facilitating Conditions	3.31	2	Agree
Behavioral Intention	3.30	4	Agree
Overall Mean	**3.32**		**Agree**

The overall mean of 3.32 across the UTAUT metrics signifies that respondents "Agree" with the system's overall usefulness, accessibility, and ease of adoption. This result indicates that users have a positive perception of RM’S and recognize its value in supporting inventory management, sales monitoring, and demand forecasting activities. Among all dimensions, Performance Expectancy obtained the highest mean score of 3.38, demonstrating that users strongly believe the system enhances productivity, improves operational efficiency, and helps accomplish tasks more quickly and accurately.

Table 18. Implementation Results

Activities Targeted	Date	Progress	Notes	Outcome Remarks
Client Consultation and Requirement Discussion	Sept. 04, 2023	Completed	Gathered key system needs with REMHART.	Identified core functions.
Prototype Testing and Debugging	Apr. – Jun. 2024	Completed	Conducted user testing and bug fixes.	Improved performance and stability.
Deployment Letter Approval	Aug. 27, 2024	Approved	Deployment request reviewed by management.	Officially authorized.
System Deployment and Monitoring	Sept. 12, 2024	Completed	Installed and observed for stability.	Performed with minimal issues.
User Training and Orientation	Sept. 15–18, 2024	Completed	Trained staff and admins on system use.	Users operated system effectively.
Post-Implementation Review	Oct. 2024	Ongoing	Collected user feedback.	High satisfaction noted.

This portion illustrates the comprehensive results of the implementation activities and their impact on the system's overall performance and usability.

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

The study's significant findings are outlined in this last chapter, along with their importance. It presents the conclusions and valuable suggestions for additional study or improvements. As a thorough conclusion, it summarizes the study's findings and offers suggestions for possible future paths.

Summary

The capstone study titled “RM’S: An AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Enhanced Distribution at REMHART – A Unilever Products Distributor” focuses on developing and implementing a comprehensive digital platform designed to optimize REMHART’s business operations. The system addresses inefficiencies in inventory tracking, order processing, and sales forecasting by integrating artificial intelligence and automation into a centralized cloud-based solution.

The researchers conducted a systematic development process involving client consultations, prototype testing, deployment, and user evaluation. Using a descriptive-developmental research approach, the study gathered feedback from selected employees, administrators, and IT professionals through surveys aligned with ISO/IEC 25010 standards.

The findings revealed that RM'S system significantly improved inventory management accuracy, distribution efficiency, and e-commerce operations. AI-based forecasting enhanced decision-making by predicting product demand and restocking needs. The system's e-commerce feature also expanded REMHART's customer reach, while its real-time analytics dashboard enabled more strategic business planning.

Overall, the implementation of RM'S demonstrated a major advancement in automating the company's core processes, reducing manual workload, and supporting data-driven operations within the supply and distribution chain.

Conclusions

Based on the system evaluation and user feedback, the researchers arrived at the following conclusions:

1. The RM'S platform effectively streamlines REMHART's inventory and distribution processes, providing real-time tracking of Unilever products across multiple branches. This integration minimizes overstocking,

prevents shortages, and enhances the overall accuracy of stock monitoring.

2. The system’s AI-driven demand forecasting feature contributes to proactive inventory management. By analyzing sales trends and seasonal patterns, it allows REMHART management to anticipate customer demand, optimize restocking schedules, and reduce wastage.
3. The inclusion of an integrated e-commerce module expands REMHART’s business accessibility. Clients can now place orders online, improving convenience and enabling faster order processing through a seamless and user-friendly interface.
4. The system enhances managerial decision-making through real-time data visualization and analytics. The dashboard provides critical insights into sales performance, stock movement, and distribution patterns, supporting data-informed strategies.
5. User evaluations confirmed the system’s effectiveness and reliability in meeting operational needs. Employees and managers noted improvements in workflow efficiency, accuracy, and productivity, confirming that the RM’S platform aligns with REMHART’s operational goals and technological transition toward automation.

Recommendations

To further strengthen and sustain the performance of RM’S, the following recommendations are proposed:

1. Future improvements may include developing a mobile version of RM’S to allow administrators and employees to access inventory data, track deliveries, and manage orders conveniently via smartphones or tablets.
2. Regular user training sessions are recommended to ensure all staff remain proficient in using system features, including AI forecasting tools, report generation, and e-commerce management. This will help maximize system utilization and maintain operational consistency.
3. The development team should continuously collect user feedback to identify usability issues, feature gaps, and enhancement opportunities. Scheduled updates and iterative improvements will ensure that RM’S remains adaptable to the company’s evolving business needs.
4. As the system handles sensitive business and customer data, strong encryption, role-based access control, and automated backup mechanisms must be maintained to protect against data loss and breaches.

5. For greater scalability, future development could include integration with Unilever’s supplier databases and third-party logistics (3PL) systems to automate procurement, shipment tracking, and supply coordination.

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APPENDIX A

Project Team Assignment Form

Title	**RM'S: An Ai-Integrated Inventory And E-Commerce Management System For Real-Time Demand Forecasting And Enhanced Distribution At Remhart**

Name and signature	Project Role	Email Address and Number
Rhaj S. Madrona	o Programmer o System Analyst o Researchers o Team Leader	rhajmadrona@gmail.com 1234567890
Maria Hanah G. Mendoza	o Researcher o Quality Assurance o Technical Writer	hanahmendoza78@gmail.com 09276317620
Gwen M. Basconcillo	o Researcher o Quality Assurance o Technical Writer	gwenbasconcillo11@gmail.com 0966954800

APPENDIX B

Capstone Project Topic Proposal

<table>
<tr><th colspan="2">CAPSTONE/ RESEARCH PROJECT TOPIC PROPOSAL</th></tr>
<tr><td>Proposed Title</td><td>RM’S: AN AI-INTEGRATED INVENTORY AND E-COMMERCE MANAGEMENT SYSTEM FOR REAL-TIME DEMAND FORECASTING AND ENHANCED DISTRIBUTION AT REMHART</td></tr>
<tr><td>Name of Student/ Course.Year & Section</td><td>Rhaj S. Madrona
Maria Hanah G. Mendoza
Gwen M. Basconcillo
BSIT 4F3</td></tr>
<tr><td>Introduction</td><td>The development of an innovative system that transforms inventory and e-commerce management for REMHART uses the power of technology to address the unique challenges faced by businesses distributing fast-moving consumer goods. A centralized AI-integrated management system such as RM’S seeks to improve a realistic and organized inventory control system with real-time monitoring, automated replenishment, and demand forecasting, which can help identify potential shortages, optimize stock levels, and support data-driven business decisions.
RM’S envisions transforming the landscape of inventory and distribution management through a sophisticated and secure centralized platform. By integrating innovative technologies including AI-powered demand forecasting and e-commerce functionalities, this system aims to streamline the vast array of product and transaction data generated daily by REMHART’s operations. The ultimate vision is to empower administrators, sales representatives, and warehouse managers with comprehensive, real-time insights, thereby enhancing productivity, sustainability, and decision-making in REMHART’s distribution activities.</td></tr>
<tr><td>Statement of the Problem</td><td>The following statements of the research are the questions that will serve the purpose of the study that the researchers will focus on answering:</td></tr>
</table>

	1. *What are the key operational challenges faced by REMHART in inventory management and order fulfillment?* 2. *How can AI-driven demand forecasting improve stock replenishment accuracy for REMHART?* 3. *What features can be integrated into RM'S to enable seamless e-commerce and inventory management operations?* 4. *How effective is RM'S in terms of functional suitability, performance efficiency, usability, and reliability as measured by ISO 25010?* 5. *5. What is the level of user acceptance of RM'S as evaluated using the UTAUT model?*
Objectives of the Study	*The main objective of this study is to design and develop RM'S: An AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Enhanced Distribution at REMHART: A Unilever Products Distributor.* *Specifically, it aimed:* 1. *To incorporate an e-commerce feature where customers can browse, order, and purchase products online, with automatic inventory updates.* 2. *To create an inventory management module* 3. *that allows users to add, update, and organize product information with essential details.* 4. *To adopt AI-driven demand forecasting to predict future product requirements based on historical sales data.* 5. *To employ intelligent replenishment suggestions by analyzing sales trends; and* 6. *To apply location-based analysis to evaluate sales performance based on geographic locations.*
Review of Related Literatures and System	*According to Smith (2024), predictive analytics helps retailers forecast demand with greater accuracy by analyzing historical data and real-time market trends. The research demonstrates that AI-driven systems reduce stockouts and overstocking, minimizing costs and enhancing operational efficiency. By integrating machine learning algorithms, retailers can achieve dynamic inventory replenishment based on predicted demand patterns. The study highlights the limitations of*

	manual stock tracking and how AI-powered solutions significantly improve customer satisfaction by ensuring product availability at all times. The research suggests that predictive analytics offers a competitive edge in inventory management, particularly in industries with fluctuating demand such as retail and fast-moving consumer goods. Furthermore, Smith emphasizes that businesses adopting AI-powered forecasting tools experience a marked reduction in operational inefficiencies, as the technology continuously learns from new data and adjusts its predictions accordingly. However, the study also notes challenges including data quality issues and the need for proper training among staff to effectively use these technologies, suggesting that organizational readiness is as important as the technology itself. *Singh and Verma (2023) explored the integration of AI-powered inventory management systems in the pharmaceutical industry. Their research focused on how AI could address specific challenges in pharmaceutical distribution, such as shelf-life management, demand fluctuations, and stockouts of critical medications. The study found that predictive analytics could forecast the demand for drugs based on various factors including seasonality, market trends, historical consumption data, and patient admission rates in healthcare facilities. The researchers also highlighted the potential of AI in minimizing waste and reducing costs by ensuring that inventory levels are aligned with actual demand, which is particularly important given the high cost of many pharmaceutical products. One of the key findings was the efficiency gained from AI-based automated reordering systems which proactively restocked items based on predicted shortages, ensuring continuous availability of essential medicines. However, the study also raised concerns about data privacy, particularly regarding sensitive patient information which could be inadvertently exposed during the demand forecasting process. The authors recommended stricter cybersecurity measures and compliance with data protection regulations for companies adopting AI in inventory management within the healthcare sector.* *White and Brown (2023) explored the use of AI in optimizing warehouse management systems to improve inventory tracking and distribution processes across multiple industries. Their study*

	found that AI could streamline warehouse operations by automating tasks such as order picking, stock sorting, shelf management, and route optimization for delivery personnel. By integrating AI with robotic process automation, the system could process orders more quickly and accurately, reducing human errors and operational bottlenecks that often lead to delays in order fulfillment. The authors also highlighted how AI-powered warehouse management systems could monitor real-time stock levels and dynamically adjust picking and packing workflows based on current inventory conditions and incoming order volumes. Additionally, they pointed out that AI algorithms could optimize the use of warehouse space by predicting which items would be in high demand, ensuring that frequently ordered products were stored in the most accessible and strategically positioned locations within the warehouse. The study concluded that AI has the ability to significantly enhance warehouse productivity and transform operational efficiency in industries such as e-commerce, manufacturing, and retail distribution. *Lee, Park, Kim, and Choi (2023) examined smart demand forecasting systems in retail chains that combine time series analysis with AI algorithms to enhance forecasting accuracy and optimize stock levels. The system utilized both structured historical sales data and unstructured external data sources, including online consumer sentiment and regional demographic trends, to generate highly accurate demand predictions. The study found that retailers implementing this hybrid AI approach experienced a reduction in inventory holding costs by 18 percent while simultaneously improving product availability and reducing the frequency of stockout incidents. The research demonstrated that AI-powered demand forecasting could significantly improve supply chain agility, enabling retailers to respond more swiftly to changes in consumer preferences and market conditions. The authors emphasized that the key to successful AI implementation in retail lies in the quality and diversity of data inputs, as well as the ongoing calibration of forecasting models to reflect evolving market dynamics.* *Kumar and Gupta (2022) examined the integration of AI in warehouse management systems and its impact on operational performance. The AI-*

	driven platform helped manage inventory levels in real time by forecasting demand based on previous sales data and current market trends, enabling warehouse operators to maintain optimal stock levels with minimal human oversight. The study also highlighted how the system optimized storage space allocation and automated the reordering process, eliminating delays caused by manual purchase order generation. The research found a 35 percent increase in warehouse operational efficiency and a significant reduction in manual errors, showcasing the transformative potential of AI in modernizing traditional warehouse management practices. Kumar and Gupta further noted that the adoption of AI in warehouse management also contributed to improved employee satisfaction, as staff were relieved of repetitive tasks and could focus on more complex and rewarding work activities. *Rivera and Martin (2022) investigated real-time inventory tracking in the automotive sector using an AI-based system. The platform employed predictive analytics to anticipate parts demand and streamline restocking operations, which significantly reduced supply chain delays that could otherwise halt production lines. Implementation of the system enhanced production continuity and decreased equipment downtime by 20 percent, thereby improving overall operational efficiency and profitability. The study further demonstrated that AI-driven inventory management in the automotive sector not only reduced operational costs but also improved supplier relationships by providing more accurate and timely purchase orders. Rivera and Martin concluded that AI-powered real-time inventory tracking represents a critical competitive advantage for automotive manufacturers operating in highly dynamic and time-sensitive production environments.*
References	*Kumar, D., & Gupta, R. (2022). Integration of AI in warehouse management systems: A study on operational efficiency. Computers in Industry, 137, 103611. https://doi.org/10.1016/j.compind.2022.103611* *Lee, J., Park, S., Kim, B., & Choi, H. (2023). Smart demand forecasting in retail chains using time series analysis and AI. Expert Systems with Applications, 213, 119001. https://doi.org/10.1016/j.eswa.2023.119001*

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Prepared by:

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BSIT IV

Maria Hanah G. Mendoza
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Gwen M. Basconcillo
BSIT IV

APPENDIX C

Notice of Title Acceptance

C E R T I F I C A T I O N

The undersigned members comprising the panel for oral examination hereby approve the Research/Capstone Project entitled "**RM'S: An Ai-Integrated Inventory And E-Commerce Management System For Real-Time Demand Forecasting And Enhanced Distribution t Remhart**" including its team members composed of RHAJ S. MADRONA, MARIA HANAH G. MENDOZA, and GWEN M. BASCONCILLO.

ELMER S. FESTIJO, *MSIT*
Research Adviser

REGINE A. PONCE-MACHETE, *DIT*
Capstone Project Course Facilitator

DEZZA MARIE M. MAGSINO, *MSIT*
Program Research Coordinator

EPIE F. CUSTODIO, *DIT*
Program Chairperson

POLEMER M. CUARTO, *Ph. D.*
Director for Research and Publication Services

JOHN EDGAR S. ANTHONY, *DIT*
College Dean

APPENDIX D

Endorsement for Final Defense

CAPSTONE PROJECT FINAL DEFENSE ENDORSEMENT

This Research/Capstone Project entitled **"RM'S: An Ai-Integrated Inventory And E-Commerce Management System For Real-Time Demand Forecasting And Enhanced Distribution t Remhart"** prepared and submitted by **RHAJ S. MADRONA, MARIA HANAH G. MENDOZA,** and **GWEN M. BASCONCILLO** as a partial fulfillment of the requirement for the degree Bachelor of Science in Information Technology is hereby accepted and recommended for FINAL ORAL DEFENSE by me as their Capstone Project Adviser after thorough review of their manuscript and system output. Their oral presentation may be scheduled on the designated defense date.

____
ELMER S. FESTIJO, *MSIT*
Capstone Project Adviser

APPENDIX E

Grammarian's Certificate

Date: June 02, 2026

GRAMMARIAN'S CERTIFICATE

This is to certify that the undersigned has reviewed the grammatical construction, and the text organization of this Research/Capstone Project entitled **"RM'S: An Ai-Integrated Inventory And E-Commerce Management System For Real-Time Demand Forecasting And Enhanced Distribution At Remhart."**

Signed:

ALICE M. RAMOS, *Ph.D*
Grammarian

Conformed:

RHAJ S. MADRONA
Project Manager

APPENDIX F

Certificate of Originality

CERTIFICATE OF ORIGINALITY

This is to certify that the research work presented in this Research/Capstone Project, **RM'S: An Ai-Integrated Inventory And E-Commerce Management System For Real-Time Demand Forecasting And Enhanced Distribution At Remhart** for the degree Bachelor of Science in Information and Technology at the Mindoro State University embodies the result of original and scholarly work carried out by the undersigned. This Research/Capstone Project does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of a degree from any other higher education institution, except where proper referencing and acknowledgment were made.

BASCONCILLO, GWEN M.
Researcher

MENDOZA, MARIA HANAH G.
Researcher

MADRONA, RHAJ S.
Researcher

APPENDIX G

Program Listing

CONTROLLER

ADMIN CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Controllers\HomepageController;use App\Models\AdminModel;use App\Models\ClientModel;use App\Models\UserModel;use App\Models\ApplicantModel;use App\Models\Form1Model;use App\Models\Form2Model;use App\Models\Form3Model;use App\Models\Form4Model;use App\Models\Form5Model;use App\Models\AgentModel;use App\Models\ConfirmModel;use App\Models\ScheduleModel;use App\Models\CommiModel;use App\Models\SignatureModel;//use App\Models\NotifModel;use App\Controllers\NotifController;use App\Controllers\ReportsController;use App\Libraries\SmsLibrary;class AdminController extends BaseController{ private $commi; private $client; private $confirm; private $db; private $homecon; private $agent; private $user; private $applicant; private $admin; private $form1; private $form2; private $form3; private $form4; private $esign; private $form5; // private $notif; private $notifcont; private $reportscont; private $smsLibrary; protected $scheduleModel; // protected $cache; public function __construct() { $this->db = \Config\Database::connect(); $this->confirm = new ConfirmModel(); $this->user = new UserModel(); $this->applicant = new ApplicantModel(); $this->agent = new AgentModel(); $this->admin = new AdminModel(); $this->form1 = new Form1Model(); $this->form2 = new Form2Model(); $this->form3 = new Form3Model(); $this->form4 = new Form4Model(); $this->form5 = new Form5Model(); $this->homecon = new HomepageController(); $this->client = new ClientModel(); $this->scheduleModel = new ScheduleModel(); $this->esign = new SignatureModel(); $this->commi = new CommiModel(); // $this->notif = new NotifModel(); // $this->cache = \Config\Services::cache(); $this->notifcont = new NotifController(); $this->reportscont = new ReportsController(); $this->smsLibrary = new SmsLibrary(); } public function AdDash() { $totalAgents = count($this->agent->findAll()); $totalApplicants = count($this->applicant->findAll()); $pendingApplicants = $this->applicant->where('status', 'pending')->countAllResults(); $totalClients = $this->client->countAllResults(); $data = array_merge( $this->getData(), $this->getDataAd(), $this->topagent(), $this->getagent(), $this->topcommi(), $this->notifcont->notification(), $this->reportscont->usersreportdata() ); $data['totalAgents'] = $totalAgents; $data['totalApplicants'] = $totalApplicants; $data['pendingApplicants'] = $pendingApplicants; $data['totalClients'] = $totalClients; return view('Admin/AdDash', $data); } //Top 3 Agents private function topagent() { // Load the database service $builder = \Config\Database::connect()->table('agent a'); $builder->select('a.username, a.FA, a.agentprofile, a.agent_token, (SELECT COUNT(*) FROM agent b WHERE b.FA = a.agent_id) AS total_fA'); $builder->orderBy('total_fa', 'DESC'); $builder->limit(3); // Get the result as an array $result = $builder->get()->getResultArray(); // Pass the data to your view or perform any other actions $data['top'] = $result; return $data; } public function topcommi() { // Select the agent_id and sum of commissions for each agent $this->commi->select('agent_id, SUM(commi) AS total_commissions') ->groupBy('agent_id') ->orderBy('total_commissions', 'DESC') ->limit(3); // Get the query result $result = $this->commi->get()->getResultArray(); // Fetch additional agent data for the top agents $topAgents = []; foreach ($result as $row) { $agentId = $row['agent_id']; $agentData = $this->agent->where('agent_id', $agentId)->first(); if ($agentData) { $agentData['total_commissions'] = $row['total_commissions'];
```

```
$topAgents[] = $agentData;            }
}          // Prepare the data to be
returned          $data['top_commi'] =
$topAgents;          // Return the data
return $data;     }    private function
getagent()    {        $data['agent'] =
$this->agent->findAll();         return
$data;        }        public  function
ManageAgent()    {       // $agentModel
= new AgentModel();            $data =
array_merge($this->notifcont-
>notification(),   $this->usermerge());
$data['agent']      =      $this->agent-
>paginate(10, 'group1'); // Change 10 to
the   number   of   items   per   page
$data['pager'] = $this->agent->pager;
return        view('Admin/ManageAgent',
$data);     }    public function Forms()
{          $data = array_merge($this-
>getData(), $this->getDataAd(), $this-
>notifcont->notification());
return view('Admin/Forms', $data);    }
public function formsTable($form)     {
$data  =  array_merge($this->getData(),
$this->getDataAd(),   $this->notifcont-
>notification());         $data['user']
=   $this->user->where(['role   !='   =>
'admin'])->Where(['role      !='      =>
'client'])->findAll();
$data['link'] = $form;        $search =
$this->request-
>getPost('searchusers');             if
(!empty($search))                     {
$data['user']      =      $this->user-
>like('username',            $search)-
>where(['role    !='    =>    'admin'])-
>findAll();          }             return
view('Admin/formsTable',        $data);
// var_dump( $search);     }     public
function ManageApplicant()    {       //
$appmodel    =    new   ApplicantModel();
$data  =  array_merge($this->notifcont-
>notification(),   $this->usermerge());
// Add  a  where  condition  to  retrieve
only records with status = 'confirmed'
$applicants     =     $this->applicant-
>where('status',            'pending')-
>paginate(10,                'group1');
$data['applicant']    =    $applicants;
$data['pager']    =   $this->applicant-
>pager;                          return
view('Admin/ManageApplicant',   $data);
}    private function usermerge()     {
$session = session();         $userId =
$session->get('id');            $data =
$this->getDataAd();        $userModel =
new UserModel();      $data['user'] =
$userModel->find($userId);       return
$data;        }        public  function
userSearch()      {            $data =
array_merge($this->notifcont-
>notification(),   $this->usermerge());
$filterUser      =      $this->request-
>getPost('filterUser');
$applicants     =     $this->applicant-
>like('username',          $filterUser)-
>where('status',            'pending')-
>paginate(10,                'group1');
$data['applicant']    =    $applicants;
$data['pager']    =   $this->applicant-
>pager;                          return
view('Admin/ManageApplicant',   $data);
}    // Controller method for searching
agents by full name    public function
agentSearch()      {            $data =
array_merge($this->notifcont-
>notification(),   $this->usermerge());
$filterUser      =      $this->request-
>getPost('filterAgent');        $agents
=       $this->agent->like('username',
$filterUser)->paginate(10,   'group1');
$data['pager']  =  $this->agent->pager;
//        Use        $agentModel->pager
$data['agent'] = $agents;         return
view('Admin/ManageAgent', $data);      }
public  function  clientSearch()       {
$data  =  array_merge($this->notifcont-
>notification(),   $this->usermerge());
$filterUser      =      $this->request-
>getPost('filterClient');       $client
=       $this->client->like('username',
$filterUser)->paginate(10,   'group1');
$data['pager'] = $this->client->pager;
//        Use        $agentModel->pager
$data['clients'] = $client;       return
view('Admin/ManageClient', $data);     }
public  function  getDataAd()          {
$session = session();         $userId =
$session->get('id');
$data['admin']      =      $this->admin-
>where('admin_id', $userId)            -
>orderBy('id', 'desc')                 -
>first();          return $data;       }
public  function  AdProfile()          {
$data  =  array_merge($this->getData(),
$this->getDataAd(),   $this->notifcont-
>notification());                 return
view('Admin/AdProfile', $data);        }
public  function  AdSetting()          {
$data  =  array_merge($this->getData(),
$this->getDataAd(),   $this->notifcont-
>notification());                 return
view('Admin/AdSetting',  $data);       }
public function AdHelp()    {       $data
= array_merge($this->getData(), $this-
>getDataAd(),         $this->notifcont-
>notification());                 return
view('Admin/AdHelp',  $data);          }
public  function  promotion()          {
$data  =  array_merge($this->getData(),
$this->getDataAd(),   $this->notifcont-
>notification());      $search = $this-
>request->getPost('searchusers');
if          (!empty($search))          {
$data['applicant'] = $this->applicant-
>like('username',  $search)->findAll();
} else {          $data['applicant']
=     $this->applicant->where('status',
'pending')->paginate(10,     'group1');
$data['pager']    =   $this->applicant-
>pager;          }                return
view('Admin/promotion',  $data);       }
public   function   getData()          {
$session = session();         $userId =
$session->get('id');
$data['user']      =      $this->user-
```

```
>find($userId);       return $data;    }
public function viewAppForm($token)    {
$lifechangerForm    =    $this->form1-
>where('app_life_token',       $token)-
>first();        if (!$lifechangerForm)
{            return redirect()->back()-
>with('error', 'No Data Found');       }
$data['lifechangerform']    =    $this-
>form1->where('app_life_token',
$token)->first();        $data['sign']
=     $this->esign->where('user_token',
$token)->first();        $data['user']
= $this->user->where('token', $token)-
>first();                        return
view('Admin/Forms/details', $data);   }
public   function   viewAppForm2($token)
{              $aial  =  $this->form2-
>where('aial_token',  $token)->first();
if  (!$aial)  {                  return
redirect()->back()->with('error',   'No
Data Found');      }       $data['aial']
=     $this->form2->where('aial_token',
$token)->first();        $data['sign']
=     $this->esign->where('user_token',
$token)->first();                return
view('Admin/Forms/details2',    $data);
}                  public      function
viewAppForm3($token)    {        $gli =
$this->form3->where('app_gli_token',
$token)->first();        if (!$gli) {
return             redirect()->back()-
>with('error', 'No Data Found');       }
$data['gli']       =       $this->form3-
>where('app_gli_token',        $token)-
>first();        $data['sign'] = $this-
>esign->where('user_token',    $token)-
>first();                        return
view('Admin/Forms/details3',    $data);
}                  public      function
viewAppForm4($token)    {        $aonff
=                        $this->form4-
>where('app_aonff_token',      $token)-
>first();             if  (!$aonff)  {
return             redirect()->back()-
>with('error', 'No Data Found');       }
$data['aonff']      =      $this->form4-
>where('app_aonff_token',      $token)-
>first();        $data['sign'] = $this-
>esign->where('user_token',    $token)-
>first();                        return
view('Admin/Forms/details4',    $data);
}                  public      function
viewAppForm5($token)    {        $sou =
$this->form5->where('app_sou_token',
$token)->first();        if (!$sou) {
return             redirect()->back()-
>with('error', 'No Data Found');       }
$data['sou']       =       $this->form5-
>where('app_sou_token',        $token)-
>first();        $data['sign'] = $this-
>esign->where('user_token',    $token)-
>first();                        return
view('Admin/Forms/details5',    $data);
}     public function random($length =
200)       {            $characters =
'0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ'
;        $code = '';        for ($i =
0; $i < $length; $i++) {         $code
.=                $characters[rand(0,
strlen($characters)  - 1)];          }
return  $code;     }     public function
newAgent($app_token)                  {
$data['applicant'] = $this->applicant-
>where('app_token',        $app_token)-
>first();                $username   =
$data['applicant']['username'];
$agentCode = $this->random();      $ref
=         $data['applicant']['refcode'];
$agent            =        $this->agent-
>where('AgentCode',     $ref)->first();
$FA = $agent['agent_id'];       $appdata
=  [                   'agent_id'  =>
$data['applicant']['applicant_id'],
'username'                          =>
$data['applicant']['username'],
'email' => $data['applicant']['email'],
'firstname'                         =>
$data['applicant']['firstname'],
'lastname'                          =>
$data['applicant']['lastname'],
'middlename'                        =>
$data['applicant']['middlename'],
'province'                          =>
$data['applicant']['province'],
'region'                            =>
$data['applicant']['region'],
'city'  =>  $data['applicant']['city'],
'birthday'                          =>
$data['applicant']['birthday'],
'barangay'                          =>
$data['applicant']['barangay'],
'street'                            =>
$data['applicant']['street'],
'number'                            =>
$data['applicant']['number'],
'agentprofile'                      =>
$data['applicant']['profile'],
'zipcode'                           =>
$data['applicant']['zipcode'],
'agent_token'                       =>
$data['applicant']['app_token'],
'AgentCode' => $agentCode,          'FA'
=> $FA,          ];          // return
redirect()->back();                   //
var_dump($appdata);       $this->agent-
>save($appdata);                  $this-
>applicant->set('status', 'confirmed')-
>where('app_token',        $app_token)-
>update();                $this->user-
>set('role',   'agent')->where('token',
$app_token)->update();
$verificationLink = site_url("login");
$emailSubject      =       'Promotion';
$emailMessage  =  "Congratulations  on
your promotion! We're thrilled to see
your hard work and dedication pay off.
Please click the link below to login and
access   your   new   responsibilities:
$verificationLink";               $this-
>homecon-
>sendVerificationEmail($data['applican
t']['email'],           $emailSubject,
$emailMessage);                $to   =
$data['applicant']['number'];
$from = '447491163443';         $text =
"Congratulations  on  your  promotion!
```

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We're thrilled to see your hard work and dedication pay off.        Please click the link below to login and access your new responsibilities: $verificationLink";        $response = $this->smsLibrary->sendSms($to,  $from, $text);              return redirect()->to('promotion')->with('success', "$username was Promoted To Agent");    }   public   function   svad()         { $session = session();         $userId = $session->get('id');      // Initialize $data array        $data = [];        // Get the old image file name from the database      $oldAdmin = $this->admin->select('adminProfile')->where('admin_id',   $userId)->first(); // Check if a file is uploaded       if ($imageFile      =      $this->request->getFile('profile')) {              // Check if the file is valid          if ($imageFile->isValid()) {           // Generate a unique name for the uploaded image                      $imageName = $imageFile->getRandomName(); // Set the path to the upload directory $uploadPath  =  FCPATH  .  'uploads/'; // Move the uploaded image to the upload directory             if ($imageFile->move($uploadPath,    $imageName))    { // Image upload successful, store the image   filename   in   the   database $img['adminProfile']   =   $imageName; // Delete the old image file if it exists                                  if (!empty($oldAdmin['adminProfile']))   { $oldFilePath    =    $uploadPath    . $oldAdmin['adminProfile']; if    (file_exists($oldFilePath))    { unlink($oldFilePath); }                  }                      } else {                       $error = $imageFile->getError();                 } }       }        // Add other form data to the data array          $data += [ 'lastname'      =>      $this->request->getVar('lastname'), 'firstname'     =>     $this->request->getVar('firstname'), 'middlename'     =>     $this->request->getVar('middlename'), 'username'      =>      $this->request->getVar('username'),            'number' =>   $this->request->getVar('number'), 'email'       =>       $this->request->getVar('email'),            'birthday' =>  $this->request->getVar('birthday'), 'region'      =>      $this->request->getVar('region_text'), 'province'      =>      $this->request->getVar('province_text'), 'city'       =>       $this->request->getVar('city_text'), 'barangay'      =>      $this->request->getVar('barangay_text'), 'street'      =>      $this->request->getVar('street'),            'zipcode' =>   $this->request->getVar('zipcode'),
];       // Check if $data array is not empty  before  updating  the  database if (!empty($data)) {          // Update the admin data          $this->admin->set($data)->where('admin_id', $userId)->update();      }      return redirect()->to('/AdSetting');        } public  function  confirm($token)     { $data['applicant']  =  $this->confirm->where('token',      $token)->first(); //   $form['applicant']   =   $this->applicant->where('app_token', $token)->first();        $verificationToken = bin2hex(random_bytes(25));          if ($data['applicant']['role']          != 'client') {             $appdata = [ 'applicant_id'                        => $data['applicant']['applicant_id'], 'username'                            => $data['applicant']['username'], 'number'                              => $data['applicant']['number'], 'firstname'                           => $data['applicant']['firstname'], 'lastname'                            => $data['applicant']['lastname'], 'middlename'                          => $data['applicant']['middlename'], 'email' => $data['applicant']['email'], 'refcode'                             => $data['applicant']['refcode'], 'app_token'                           => $data['applicant']['token'], ];                 $this->applicant->save($appdata);             $this->confirm->delete($data['applicant']['id']); $con    =    ['confirm'    =>    'true', 'verification_token'                  => $verificationToken];              $this->user->set($con)->where('token', $token)->update();          } else { $lastApplicationNo   =   $this->client->selectMax('applicationNo')->get()->getRowArray()['applicationNo']; $newApplicationNo = $lastApplicationNo + 1;               $clientData = [ 'client_id'                           => $data['applicant']['applicant_id'], 'username'                            => $data['applicant']['username'], 'number'                              => $data['applicant']['number'], 'firstName'                           => $data['applicant']['firstname'], 'lastName'                            => $data['applicant']['lastname'], 'middleName'                          => $data['applicant']['middlename'], 'email' => $data['applicant']['email'], 'refcode'                             => $data['applicant']['refcode'], 'client_token'                        => $data['applicant']['token'], //              'plan'              => $data['applicant']['plan'], 'applicationNo'  =>  $newApplicationNo, ];                      $this->client-
```

```
>save($clientData);              $this-
>confirm-
>delete($data['applicant']['id']);
$con    =    ['confirm'    =>    'true',
'verification_token'                  =>
$verificationToken];             $this-
>user->set($con)->where('token',
$token)->update();                     }
$verificationLink  =  site_url("verify-
email/{$verificationToken}");
$emailSubject      =      'Registration
Confirmation';          $emailMessage =
"Your account has been confirmed. Please
click  the  link  verify  your  account:
{$verificationLink}";            $this-
>homecon-
>sendVerificationEmail($data['applican
t']['email'],           $emailSubject,
$emailMessage);      return redirect()-
>to('/confirmation')->with('success',
'Acount Confirmed!');      }     public
function    deny($token)               {
$data['applicant']  =  $this->confirm-
>where('token',       $token)->first();
$id     =     $data['applicant']['id'];
$this->confirm->delete($id);          //
$verificationLink  =  site_url("verify-
email/{$verificationToken}");
$emailSubject    =    'Register   Deny';
$emailMessage = "Your Account was Deny
Where sorry to inform you";      $this-
>homecon-
>sendVerificationEmail($data['applican
t']['email'],           $emailSubject,
$emailMessage);       // var_dump($id);
return redirect()->to('/confirmation')-
>with('warning', 'Acount Denied');    }
public  function  confirmation()       {
$data  =  array_merge($this->getData(),
$this->getDataAd(),   $this->notifcont-
>notification());        $con = $this-
>confirm->paginate(10,       'group1');
$data['applicant']       =        $con;
$data['pager'] = $this->confirm->pager;
$search       =        $this->request-
>getPost('searchusers');             if
(!empty($search))                      {
$data['applicant']  =  $this->confirm-
>like('username',           $search)-
>paginate(10, 'group1');      }      //
var_dump($data);                  return
view('Admin/confirmation', $data);    }
public function sched()    {       $data
= array_merge($this->getData(), $this-
>getDataAd(),          $this->notifcont-
>notification());
$data['schedules']       =       $this-
>scheduleModel->findAll();        return
view('Admin/Schedule',  $data);        }
public  function  schedsave()          {
$input   =   $this->request->getPost();
$validationRules = [            'title'
=> 'required',            'description'
=> 'required',         'start_datetime'
=>                'required|valid_date',
'end_datetime' => 'required|valid_date'
];                         if   (!$this-
>validate($validationRules))           {
return redirect()->back()->withInput()-
>with('errors',      $this->validator-
>getErrors());        }         $data =
[           'title' => $input['title'],
'description' => $input['description'],
'start_datetime'                     =>
$input['start_datetime'],
'end_datetime'                       =>
$input['end_datetime'],              ];
$this->scheduleModel->insert($data);
return              redirect()->back()-
>with('success',  'Schedule  Added!.');
}   public function schededit($id)    {
$data  =  array_merge($this->getData(),
$this->getDataAd(),   $this->notifcont-
>notification());
$data['schedules']       =       $this-
>scheduleModel->findAll();
$schedule    =    $this->scheduleModel-
>find($id);           if  (!$schedule) {
return              redirect()->back()-
>with('error', 'Schedule not found.');
}        $data['schedule'] = $schedule;
return  view('Admin/Schedule',  $data);
}     public  function  schedupdate($id)
{            $input  = $this->request-
>getPost();        $validationRules = [
'title'         =>          'required',
'description'      =>       'required',
'start_datetime'                     =>
'required|valid_date',
'end_datetime' => 'required|valid_date'
];                          if   (!$this-
>validate($validationRules))           {
return  redirect('sched')->withInput()-
>with('errors',       $this->validator-
>getErrors());        }         $data =
[           'title' => $input['title'],
'description' => $input['description'],
'start_datetime'                     =>
$input['start_datetime'],
'end_datetime'                       =>
$input['end_datetime'],              ];
$this->scheduleModel->update($id,
$data);       return redirect('sched')-
>with('success',   'Schedule   updated
successfully.');    }   public function
scheddelete($id)     {        $schedule
=      $this->scheduleModel->find($id);
if (!$schedule) {                return
redirect()->back()->with('error',
'Schedule  not  found.');             }
$this->scheduleModel->delete($id);
return                redirect('sched')-
>with('success',   'Schedule   deleted
successfully.');    }   public function
ManageClients()      {          $data =
array_merge($this->notifcont-
>notification(),   $this->usermerge());
$data['clients']     =    $this->client-
>paginate(10, 'group1'); // Change 10 to
the    number    of    items    per    page
$data['pager'] = $this->client->pager;
return       view('Admin/ManageClient',
$data);        }       private  function
getclients()    {      $data['clients']
= $this->client->findAll();       return
$data;    }}
```

AGENT CONTROLLER

```
<?phpnamespace App\Controllers;use
App\Controllers\BaseController;use
\App\Models\AgentModel;use
\App\Models\UserModel;use
App\Models\ApplicantModel;use
App\Controllers\AppController;use
App\Models\ScheduleModel;use
App\Models\Scheduler;use
App\Models\ClientModel;use
App\Models\PlanModel;use
App\Models\ClientPlanModel;use
App\Models\CommiModel;use
App\Controllers\FilesController;class
AgentController extends BaseController{
private $commission; private
$client_plan; private $plan;
private $client; private $sched;
private $appcon; private $user;
private $applicant; private $agent;
protected $scheduleModel; private
$filescont; // protected $cache;
public function __construct() {
$this->appcon = new AppController();
$this->agent = new AgentModel();
$this->user = new UserModel();
$this->applicant = new
ApplicantModel(); $this-
>scheduleModel = new ScheduleModel();
$this->sched = new Scheduler();
$this->client = new ClientModel();
$this->plan = new PlanModel();
$this->client_plan = new
ClientPlanModel(); $this-
>commission = new CommiModel();
$this->filescont = new
FilesController(); // $this-
>cache = \Config\Services::cache(); }
public function AgDash() { $data
= array_merge( $this-
>getData(), // Assuming this method
fetches general user data
$this->getDataAge() // Assuming this
method fetches agent-specific data
); $agentid =
$data['agent']['agent_id'];
$agentcode =
$data['agent']['AgentCode'];
$data['FA'] = $this->agent->where('FA',
$agentid)->findAll();
$data['applicants'] = $this->applicant-
>where('refcode', $agentcode)-
>countAllResults();
$data['ranking'] = $this->agent-
>where('FA', $ } public function
getData() { $session =
session(); // $data['class'] =
'Awaiting'; return
view('Agent/cliSched', $data); }
public function inprog() {
$session = session(); $agent =
$session->get('id'); $data =
array_merge($this->getData(), $this-
>appcon->getDataApp(), $this-
>getDataAge()); $data['schedule']
= $this->sched->where('agent', $agent)-
>where('status', 'inprogress')-
>findAll(); $data['plans'] =
$this->plan->findAll();
$data['client'] = $this->client-
>findAll(); $data['status'] = 'In
Progress'; $data['class'] = 'In
Progress'; return
view('Agent/cliSched', $data); //
var_dump($agent); } public function
comp() { $session = session();
$agent = $session->get('id');
$data = array_merge($this->getData(),
$this->appcon->getDataApp(), $this-
>getDataAge()); $data['schedule']
= $this->sched->where('agent', $agent)-
>where('status', 'completed')-
>findAll(); $data['plans'] =
$this->plan->findAll();
$data['client'] = $this->client-
>findAll(); $data['status'] =
'Completed'; $data['class'] =
'Completed'; return
view('Agent/cliSched', $data); //
var_dump($agent); } // public
function compost() // { // $data
= []; // $uploadPath = FCPATH .
'uploads/clients/receipts/'; // //
Ensure the upload directory exists //
if (!is_dir($uploadPath)) { //
mkdir($uploadPath, 0777, true); //
} // if ($imageFile = $this-
>request->getFile('receipt')) { //
if ($imageFile->isValid()) { //
$imageName = $imageFile-
>getRandomName(); // if
($imageFile->move($uploadPath,
$imageName)) { //
$data['receipt'] = $imageName; //
} else { // $error =
$imageFile->getError(); //
// Handle the error accordingly //
} // } // } //
// Calculate commission based on the
mode of payment // $cal['plan']
= $this->plan->where('token', $this-
>request->getVar('plan'))->first();
// if ($cal['plan']) { //
$price = $cal['plan']['price']; //
$percentage =
$cal['plan']['com_percentage'] / 100;
// switch ($this->request-
>getVar('typeofpayment')) { //
case 'Annual': //
$commissionAmount = $price *
$percentage; //
$amountPaid = $price; // Amount Paid =
Annual Price - Commission //
break; // case 'Semi-
Annual': //
$commissionAmount = ($price / 2) *
$percentage; //
$amountPaid = ($price / 2); // Amount
Paid = Half of Annual Price //
break; // case 'Quarterly':
// $commissionAmount =
($price / 4) * $percentage; //
$amountPaid = ($price / 4); // Amount
Paid = Quarter of Annual Price //
break; // case 'Monthly':
```

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//                $commissionAmount =
($price / 12) * $percentage;       //
$amountPaid = ($price / 12); // Amount
Paid = Twelfth of Annual Price      //
break;    //           default:    //
$commissionAmount   =   0;          //
$amountPaid = 0;     //          }    //
}           //               $token   =
bin2hex(random_bytes(25));           //
$tokens   =   bin2hex(random_bytes(50));
//     $data += [    //         'plan'
=> $this->request->getVar('plan'),   //
'agent'        =>        $this->request-
>getVar('agent'),                     //
'client_id'      =>      $this->request-
>getVar('client_id'),                 //
'mode_payment'    =>    $this->request-
>getVar('typeofpayment'),             //
'term'        =>         $this->request-
>getVar('term'),                      //
'applicationNo'    =>    $this->request-
>getVar('applicationNo'),             //
'status' => 'paid',    //        'token'
=> $token,     //          'commission'
=> $commissionAmount,    //     ];    //
$this->client_plan->save($data);      //
$this->sched->set(['status'          =>
'completed'])->where('id',       $this-
>request->getVar('schedId'))-
>update();    //     $this->commission-
>save([   //       'token' => $tokens,
//         'commi' => $commissionAmount,
//         'agent_id' => $this->request-
>getVar('agent'),                     //
'client_id'      =>      $this->request-
>getVar('client_id'),                 //
'amount_paid'  =>  $amountPaid,       //
'receipts' => $imageName,    //     ]);
//              return    redirect()-
>to('cliSched')->with('success',
'Transaction  Completed');         //  }
public  function  compost()           {
$data = [];         $uploadPath = FCPATH
. 'uploads/clients/receipts/';       //
Ensure  the  upload  directory  exists
if        (!is_dir($uploadPath))       {
mkdir($uploadPath, 0777, true);       }
if    ($imageFile    =    $this->request-
>getFile('receipt')) {                if
($imageFile->isValid())                {
$imageName         =         $imageFile-
>getRandomName();                     if
($imageFile->move($uploadPath,
$imageName))                           {
$data['receipt']      =      $imageName;
} else {                       $error =
$imageFile->getError();
//   Handle   the   error   accordingly
}             }           }            //
Calculate commission based on the mode
of payment        $cal['plan'] = $this-
>plan->where('token',   $this->request-
>getVar('plan'))->first();           if
($cal['plan']) {             $price =
$cal['plan']['price'];
$percentage                            =
$cal['plan']['com_percentage']  /  100;
switch                ($this->request-
>getVar('typeofpayment'))              {
case                           'Annual':
$commissionAmount     =     $price     *
$percentage;                 $amountPaid
= $price;                   $addDuration
= '+1 year';                      break;
case                      'Semi-Annual':
$commissionAmount  =  ($price  /  2)  *
$percentage;                 $amountPaid
=          ($price         /          2);
$addDuration     =     '+6     months';
break;              case 'Quarterly':
$commissionAmount  =  ($price  /  4)  *
$percentage;                 $amountPaid
=          ($price         /          4);
$addDuration     =     '+3     months';
break;                 case 'Monthly':
$commissionAmount  =  ($price  /  12)  *
$percentage;                 $amountPaid
=         ($price         /         12);
$addDuration     =     '+1      month';
break;                          default:
$commissionAmount            =         0;
$amountPaid               =            0;
$addDuration = null;           }       }
//     Calculate     the     due     date
$currentDate = new \DateTime();       if
($addDuration) {          $currentDate-
>modify($addDuration);                 }
$duedate  =  $currentDate->format('Y-m-
d');                          $token   =
bin2hex(random_bytes(25));
$tokens   =   bin2hex(random_bytes(50));
$data += [            'plan' => $this-
>request->getVar('plan'),
'agent'        =>        $this->request-
>getVar('agent'),           'client_id'
=> $this->request->getVar('client_id'),
'mode_payment'    =>    $this->request-
>getVar('typeofpayment'),
'term'        =>         $this->request-
>getVar('term'),
'applicationNo'    =>    $this->request-
>getVar('applicationNo'),
'status' => 'paid',              'token'
=> $token,            'commission' =>
$commissionAmount,              'duedate'
=> $duedate,          ];           $this-
>client_plan->save($data);        $this-
>sched->set(['status' => 'completed'])-
>where('id',           $this->request-
>getVar('schedId'))->update();
$this->commission->save([
'token' => $tokens,              'commi'
=>                    $commissionAmount,
'agent_id'      =>      $this->request-
>getVar('agent'),           'client_id'
=> $this->request->getVar('client_id'),
'amount_paid'      =>      $amountPaid,
'receipts'        =>      $imageName,
'duedate'  =>  $duedate,            ]);
return      redirect()->to('cliSched')-
>with('success',            'Transaction
Completed');     }    // public function
compost()    // {    //     $data = [];
//     if ($imageFile = $this->request-
>getFile('receipt')) {    //         if
($imageFile->isValid())   {           //
```

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$imageName = $imageFile->getRandomName(); // $uploadPath = FCPATH . 'uploads/clients/receipts/'; // if ($imageFile->move($uploadPath, $imageName)) { // $data['receipt'] = $imageName; // } else { // $error = $imageFile->getError(); // } // } // } // $token = bin2hex(random_bytes(25)); // $tokens = bin2hex(random_bytes(50)); // $schedId = $this->request->getVar('schedId'); // $plan = $this->request->getVar('plan'); // $email = $this->request->getVar('email'); // $plantoken = $this->request->getVar('plan'); // $modepayment = $this->request->getVar('typeofpayment'); // $cal['plan'] = $this->plan->where('token', $plantoken)->first(); // if ($cal['plan']) { // switch ($modepayment) { // case 'Annual': // $commissionAmount = $cal['plan']['price'] / 1 * ($cal['plan']['com_percentage'] / 100); // break; // case 'Semi-Annual': // $commissionAmount = $cal['plan']['price'] / 2 * ($cal['plan']['com_percentage'] / 100); // break; // case 'Quarterly': // $commissionAmount = $cal['plan']['price'] / 4 * ($cal['plan']['com_percentage'] / 100); // break; // case 'Monthly': // $commissionAmount = $cal['plan']['price'] / 12 * ($cal['plan']['com_percentage'] / 100); // break; // default: // $commissionAmount = 0; // } // } // $data += [ // 'plan' => $this->request->getVar('plan'), // 'agent' => $this->request->getVar('agent'), // 'client_id' => $this->request->getVar('client_id'), // 'mode_payment' => $this->request->getVar('typeofpayment'), // 'term' => $this->request->getVar('term'), // 'applicationNo' => $this->request->getVar('applicationNo'), // 'status' => 'paid', // 'token' => $token, // 'commission' => $commissionAmount, // ]; // $this->client_plan->save($data); // $s = ['status' => 'completed']; // $this->sched->set($s)->where('id', $schedId)->update(); // $commi = [ // 'token' => $tokens, // 'commi' => $commissionAmount, // 'agent_id' => $this->request->getVar('agent'), // 'client_id' => $this->request->getVar('client_id'), // ]; // $this->commission->save($commi); // return redirect()->to('cliSched')->with('success', 'Transaction Completed'); // } public function upstatusplan($token) { $stats = []; if ($imageFile = $this->request->getFile('receipt')) { if ($imageFile->isValid()) { $imageName = $imageFile->getRandomName(); $uploadPath = FCPATH . 'uploads/clients/receipts/'; if ($imageFile->move($uploadPath, $imageName)) { $data['receipt'] = $imageName; } else { $error = $imageFile->getError(); // Handle the error as needed return redirect()->back()->with('error', 'Image upload error: ' . $error); } } else { return redirect()->back()->with('error', 'Invalid file upload'); } } $tokens = bin2hex(random_bytes(50)); $data['commi'] = $this->client_plan->where('token', $token)->first(); $data['percentage'] = $this->plan->where('token', $data['commi']['plan'])->first(); $annualpay = $data['percentage']['price']; $per = $data['percentage']['com_percentage']; $paymentmode = $data['commi']['mode_payment']; $oldcommi = $data['commi']['commission']; $agentid = $data['commi']['agent']; $clientid = $data['commi']['client_id']; // Calculate new commission based on payment mode $newcommi = $oldcommi; $amountPaid = 0; // Initialize amount_paid variable switch ($paymentmode) { case 'Annual': $newcommi += $annualpay * ($per / 100); $amountPaid = $annualpay; break; case 'Semi-Annual': $newcommi += $annualpay * ($per / 100) / 2; $amountPaid = $annualpay / 2; break; case 'Quarterly': $newcommi += $annualpay * ($per / 100) / 4; $amountPaid = $annualpay / 4; break; case 'Monthly': $newcommi += $annualpay * ($per / 100) / 12; $amountPaid = $annualpay / 12; break; } if ($paymentmode == 'Annual') { $commi = $annualpay * ($per / 100); } elseif ($paymentmode == 'Semi-Annual') { $commi = $annualpay * ($per / 100) / 2; }
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elseif ($paymentmode == 'Quarterly') { $commi = $annualpay * ($per / 100) / 4; } elseif ($paymentmode == 'Monthly') { $commi = $annualpay * ($per / 100) / 12; }        // Update the commission and status in the database        $stats += [                'status' => 'paid', 'commission' => $newcommi,          ]; // Merge image data into the stats array if an image was uploaded          if (isset($data['receipt']))          { $stats['receipt'] = $data['receipt']; }                $this->client_plan->set($stats)->where('token',  $token)->update();          // pwede ko ditong lagyan ng add para ma filters yung monthly nya            $commiS = [ 'token' => $tokens,           'commi' => $commi,           'agent_id' => $agentid,            'client_id' => $clientid,            'receipts' => $imageName,           'amount_paid' => $amountPaid,         ];          $this->commission->save($commiS); return             redirect()->back()->with('success',    'Plan    updated successfully');    }    public function con($dec)       {            $id = base64_decode($dec);         $status = $this->sched->where('id',  $id)->get()->getRow()->status;         $clientEmail =    $this->sched->where('id',    $id)->get()->getRow()->email;            if ($status === 'pending') {         $con =    ['status'    =>    'inprogress']; $this->sched->set($con)->where('id', $id)->update();                          } $emailSubject    =    "Appointment    In Progress";       $emailMessage = "Your Appointment    has    been    confirm"; $this->sendVerificationEmail($clientEmail, $emailSubject, $emailMessage);       // var_dump($clientEmail);          return redirect()->to('cliSched')->with('success',    'Transaction    In Progress');     }     public function client()   {     $session = session(); $agId     =    $session->get('id'); $data['client'] = $this->client_plan->where('agent',     $agId)->findAll(); // var_dump($data);       $clientIds = array_column($data['client'], 'client_id');          $clientId = array_unique($clientIds);      $filter =                 $this->request->getVar('filterclient');         if (!empty($clientId))  {  //  Check  if $clientId array is not empty        if (!empty($filter)) {          $clients =      $this->client->like('username', $filter)->whereIn('client_id', $clientId)->paginate(10,     'group1'); } else {             $clients = $this->client->whereIn('client_id', $clientId)->paginate(10,     'group1'); }       } else {          $clients = [];            }            $data = array_merge($this->getData(),    $this->appcon->getDataApp(),         $this->getDataAge());        $data['pager'] = $this->client->pager; $data['client'] = $clients;         // var_dump($client);               return view('Agent/clients',  $data);       } public  function  mycommi()       { $data = array_merge(          $this->getData(),  //  Assuming  this  method fetches    general    user    data $this->appcon->getDataApp(),        // Assuming    this    method    fetches application-related data       $this->getDataAge() // Assuming this method fetches agent-specific data         ); $data['mycommission']      =      $this->client_plan->select('client.username as  client_name,  client.client_token, client_plan.token     as     tokin, client_plan.*, plan.*')              ->join('client',  'client.client_id  = client_plan.client_id')              ->join('plan',      'plan.token     = client_plan.plan')              ->where('client_plan.agent', $data['agent']['agent_id'])            ->findAll();               return view('Agent/mycommi',  $data);     } private               function sendVerificationEmail($to,   $subject, $message)    {          // Load Email Library                 $email   = \Config\Services::email();           // SMTP Configuration (replace with your actual SMTP settings)        $config = [             'protocol' => 'smtp', 'SMTPHost'    =>    'smtp.gmail.com', 'SMTPUser'                       => 'alejandrogino950@gmail.com', // Your Gmail address          'SMTPPass' => 'kiewkcfnftnigkvh',         'SMTPPort' => 587,        'SMTPCrypto' => 'tls', 'mailType' => 'html',         'charset' => 'utf-8'         ];          $email->initialize($config);      // Set Email Parameters          $email->setTo($to); $email->setFrom('alejandrogino950@gmail.com', 'ALLIANZ  PNB  CALAPAN');  //  Set  the "From" address and name        $email->setSubject($subject);          $email->setMessage($message);         // Try sending the email          if ($email->send()) {          echo 'Email sent successfully.';             } else { echo    'Error:    '    .    $email->printDebugger(['headers']);          } }}
```

APP CONTROLLER

```
<?phpnamespace     App\Controllers;use App\Controllers\BaseController;use App\Models\ApplicantModel;use App\Models\Form1Model;use App\Models\Form2Model;use App\Models\Form3Model;use
```

```
App\Models\Form4Model;use
App\Models\Form5Model;use
App\Models\SignatureModel;use
\App\Models\UserModel;use
\App\Models\AgentModel;use
\App\Models\NotifModel;use
\App\Controllers\NotifController;class
AppController extends BaseController{
private $agent;       private $form1;
private $form2;       private $form3;
private $form4;       private $form5;
private $esign;        private $user;
private $applicant;    private $notif;
private $notificationcontroller;    //
protected $cache;      public function
__construct()    {         $this->agent
= new AgentModel();        $this->form1
= new Form1Model();        $this->form2
= new Form2Model();        $this->form3
= new Form3Model();        $this->form4
= new Form4Model();        $this->form5
= new Form5Model();        $this->esign
= new SignatureModel();          $this-
>user = new UserModel();         $this-
>applicant   =   new   ApplicantModel();
$this->notif    =    new    NotifModel();
$this->notificationcontroller   =   new
NotifController();           // $this-
>cache = \Config\Services::cache();   }
public   function   AppDash()          {
$data  =  array_merge($this->getData(),
$this->getDataApp());             return
view('Applicant/AppDash',       $data);
// var_dump($data);      }        public
function AppProfile()    {         $data
= array_merge($this->getData(), $this-
>getDataApp());                   return
view('Applicant/AppProfile',    $data);
}     public function AppForms()       {
$data  =  array_merge($this->getData(),
$this->getDataApp());             return
view('Applicant/forms', $data);       }
private   function   getData()         {
$session = session();         // Check
if the user is logged in            if
(!$session->get('id')) {             //
Redirect or handle the case where the
user is not logged in           return
redirect()->to('login');              }
// Get the user ID from the session
$userId = $session->get('id');       //
Find the user by ID       $data['user']
=           $this->user->find($userId);
return $data;     }     public function
getDataApp()      {          $session =
session();         $userId = $session-
>get('id');        $data['applicant'] =
$this->applicant-
>where('applicant_id',         $userId)
->orderBy('id', 'desc')               -
>first();          return $data;      }
public  function  getform1Data()       {
$session = session();        $userId =
$session->get('id');
$data['lifechangerform']    =    $this-
>form1->where('user_id',       $userId)
->first();         return $data;      }
public  function  getform2Data()       {
$session = session();         $userId =
$session->get('id');
$data['aial']       =      $this->form2-
>where('user_id', $userId)             -
>first();          return $data;      }
public  function  getform3Data()       {
$session = session();         $userId =
$session->get('id');       $data['gli']
=   $this->form3->where('applicant_id',
$userId)                    ->first();
return $data;     }     public function
getform4Data()     {          $session =
session();         $userId = $session-
>get('id');          $data['aonff'] =
$this->form4->where('applicant_id',
$userId)                    ->first();
return $data;     }     public function
getform5Data()     {          $session =
session();         $userId = $session-
>get('id');      $data['sou'] = $this-
>form5->where('applicant_id',  $userId)
->first();          return $data;     }
public   function   esign()            {
$session = session();         $userId =
$session->get('id');
$data['sign']       =      $this->esign-
>where('user_id', $userId)             -
>first();          return $data;      }
public  function  AppSetting()         {
$data  =  array_merge($this->getData(),
$this->getDataApp());             return
view('Applicant/AppSetting',    $data);
}     public function signature()      {
$data  =  array_merge($this->getData(),
$this->getDataApp(),   $this->esign());
return      view('Applicant/signature',
$data);        }        public  function
signsave()      {           // Retrieve
user_id from the session       $session
= session();       $userId = $session-
>get('id');         $token = $session-
>get('token');          // Decode the
base64 signature data      $base64Image
=       $this->request->getVar('sign');
if (!$base64Image) {            return
redirect()->back()->with('error',   'No
signature data received.');          }
// Decode the base64 data to get the
image content             $imageData =
base64_decode(preg_replace('#^data:ima
ge/\w+;base64,#i', '', $base64Image));
if    ($imageData    ===    false)    {
return             redirect()->back()-
>with('error',    'Invalid    signature
data.');         }        // Define the
upload      path       and       filename
$uploadPath       =       FCPATH        .
'uploads/signatures/';        $filename
=  'signature_'  .  time()  .  '.png';
// Ensure the directory exists        if
(!is_dir($uploadPath)                &&
!mkdir($uploadPath,   0755,   true))  {
return             redirect()->back()-
>with('error',   'Failed   to   create
directory  for  signature  uploads.');
}          // Create an image resource
from       the       decoded       data
$signatureImage                       =
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imagecreatefromstring($imageData);
if (!$signatureImage) {            return
redirect()->back()->with('error',
'Failed to create image from signature
data.');          }          // Get the
dimensions  of  the  signature  image
$width    =   imagesx($signatureImage);
$height   =   imagesy($signatureImage);
// Create a white background image with
the           same              dimensions
$backgroundImage                          =
imagecreatetruecolor($width,  $height);
$white                                    =
imagecolorallocate($backgroundImage,
255,   255,   255);  //   White   color
imagefill($backgroundImage,    0,    0,
$white); // Fill  the  background  with
white       // Merge the signature image
onto       the       white      background
imagecopy($backgroundImage,
$signatureImage,  0,  0,  0,  0,  $width,
$height);        // Save the final image
with the white background        $saved
=            imagepng($backgroundImage,
$uploadPath . $filename);       // Clean
up                               resources
imagedestroy($signatureImage);
imagedestroy($backgroundImage);
if ($saved) {            // Prepare data
for                   insertion/updation
$signatureData = [             'user_id'
=> $userId,                'user_token'
=> $token,                  'signature'
=> $filename,           ];            //
Check if the user_id exists in the e-
signature                          table
$existingSignature    =    $this->esign-
>where('user_id',    $userId)->first();
if ($existingSignature) {             //
Update      the      existing      record
$this->esign->where('user_id',
$userId)->set($signatureData)-
>update();                // Remove the
old  signature  file  if  it  exists
$oldSignature                             =
$existingSignature['signature'];
if       (!empty($oldSignature))        {
$oldFilePath      =     $uploadPath     .
$oldSignature;                           if
(file_exists($oldFilePath))             {
unlink($oldFilePath);                   }
}           } else {                $this-
>esign->save($signatureData);
}              // Redirect or display a
success  message                    return
redirect()->back()->with('success',
'Signature saved successfully');        }
else {              // Handle error in
saving the image                    return
redirect()->back()->with('error',
'Failed  to  save  signature  image');
}     }     public function svap()     {
$session = session();          $userId =
$session->get('id');       // Initialize
$data array        $data = [];        //
Get the old image file name from the
database              $oldApp = $this-
>applicant->select('profile')-
>where('applicant_id',        $userId)-
>first();          // Check if a file is
uploaded        if ($imageFile = $this-
>request->getFile('profile'))          {
// Check if the file is valid          if
($imageFile->isValid()  &&  !$imageFile-
>hasMoved()) {              // Generate
a unique  name  for  the uploaded image
$imageName          =        $imageFile-
>getRandomName();                 // Set
the  path  to  the  upload  directory
$uploadPath  =  FCPATH   .   'uploads/';
// Move the uploaded image to the upload
directory                if ($imageFile-
>move($uploadPath,    $imageName))    {
// Image upload successful, store the
image   filename   in   the   database
$data['profile']      =      $imageName;
// Delete  the  old  image  file  if  it
exists                                if
(!empty($oldApp['profile']))           {
$oldFilePath     =     $uploadPath     .
$oldApp['profile'];
if     (file_exists($oldFilePath))     {
unlink($oldFilePath);
}                   }                   }
else {                        $error =
$imageFile->getError();
echo $error;              }            }
}         // Add other form data to the
data array       $data += [          //
'applicantfullname' => $this->request-
>getVar('applicantfullname'),
'lastname'       =>       $this->request-
>getVar('lastname'),
'firstname'      =>       $this->request-
>getVar('firstname'),
'middlename'     =>       $this->request-
>getVar('middlename'),
'username'       =>       $this->request-
>getVar('username'),             'number'
=>    $this->request->getVar('number'),
'email'         =>        $this->request-
>getVar('email'),              'birthday'
=>  $this->request->getVar('birthday'),
'region'         =>       $this->request-
>getVar('region_text'),
'province'       =>       $this->request-
>getVar('province_text'),
'city'          =>        $this->request-
>getVar('city_text'),
'barangay'       =>       $this->request-
>getVar('barangay_text'),
'street'         =>       $this->request-
>getVar('street'),              'zipcode'
=>   $this->request->getVar('zipcode'),
];        // Check if $data array is not
empty  before  updating  the  database
if (!empty($data)) {           // Update
the applicant data                 $this-
>applicant->set($data)-
>where('applicant_id',        $userId)-
>update();            }           return
redirect()->to('/AppSetting');         }
public   function   AppHelp()          {
$data  =  array_merge($this->getData(),
$this->getDataApp());              return
view('Applicant/AppHelp', $data);      }
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public function AppForm1() { $data['agents'] = $this->agent->findAll(); // Merge arrays while retaining the 'agents' key $data = array_merge($this->getData(), $this->getDataApp(), $this->getform1Data(), $this->esign(), $this->refbywhom(), $data); return view('Applicant/AppForm1', $data); } public function refbywhom() { $session = session(); $userId = $session->get('id'); $refcode = $this->applicant->select('refcode')->where('applicant_id', $userId)->first()['refcode'] ?? null; // Proceed only if refcode is found $referalby = $refcode ? $this->agent->select("CONCAT(firstname, ' ', LEFT(middlename, 1), '. ', lastname) as fullname") ->where('AgentCode', $refcode)->first()['fullname'] ?? null : null; // Return an array with the referral name return ['referalby' => $referalby]; } // public function form1sv() // { // $session = session(); // // Retrieve user_id from the session // $userId = $session->get('id'); // $token = $session->get('token'); // // Decode the base64 signature data // $base64Image = $this->request->getVar('sign'); // $imageData = base64_decode(preg_replace('#^data:image/\w+;base64,#i', '', $base64Image)); // // Define the upload path and filename // $uploadPath = FCPATH . 'uploads/signatures/'; // Adjust the path as needed // $filename = 'signature_' . time() . '.png'; // Generate a unique filename // // Check if user_id exists in the database // $existingUser = $this->form1->select('signature')->where('user_id', $userId)->first(); // if (file_put_contents($uploadPath . $filename, $imageData)) { // // Image saved successfully, prepare data for insertion/updation // $data = [ // 'user_id' => $userId, // 'app_life_token' => $token, // 'position' => $this->request->getVar('positionApplying'), // 'preferredArea' => $this->request->getVar('preferredArea'), // 'referral' => $this->request->getVar('referral') ?? 'No', // 'referralBy' => $this->request->getVar('referralBy'), // 'onlineAd' => $this->request->getVar('onlineAd') ?? 'No', // 'walkIn' => $this->request->getVar('walkIn') ?? 'No', // 'othersRef' => $this->request->getVar('othersRef') ?? 'No', // 'fname' => $this->request->getVar('fullname'), // 'nickname' => $this->request->getVar('nickname'), // 'birthdate' => $this->request->getVar('birthdate'), // 'placeOfBirth' => $this->request->getVar('placeOfBirth'), // 'gender' => $this->request->getVar('gender'), // 'bloodType' => $this->request->getVar('bloodType'), // 'homeAddress' => $this->request->getVar('homeAddress'), // 'mobileNo' => $this->request->getVar('mobileNo'), // 'landline' => $this->request->getVar('landline') ?? 'N/A', // 'email' => $this->request->getVar('email'), // 'citizenship' => $this->request->getVar('citizenship'), // 'othersCitizenship' => $this->request->getVar('othersCitizenship') ?? 'N/A', // 'naturalizationInfo' => $this->request->getVar('naturalizationInfo') ?? 'N/A', // 'maritalStatus' => $this->request->getVar('maritalStatus'), // 'maidenName' => $this->request->getVar('maidenName') ?? 'N/A', // 'spouseName' => $this->request->getVar('spouseName') ?? 'N/A', // 'sssNo' => $this->request->getVar('sssNo'), // 'tin' => $this->request->getVar('tin'), // 'lifeInsuranceExperience' => $this->request->getVar('lifeInsuranceExperience') ?? 'No', // 'traditional' => $this->request->getVar('traditional') ?? 'No', // 'variable' => $this->request->getVar('variable') ?? 'No', // 'recentInsuranceCompany' => $this->request->getVar('recentInsuranceCompany') ?? 'N/A', // 'highSchool' => $this->request->getVar('highSchool') ?? 'N/A', // 'highSchoolCourse' => $this->request->getVar('highSchoolCourse') ?? 'N/A', // 'highSchoolYear' => $this->request->getVar('highSchoolYear') ?? 'N/A', // 'college' => $this->request->getVar('college') ?? 'N/A', // 'collegeCourse' => $this->request->getVar('collegeCourse') ?? 'N/A', // 'collegeYear' => $this->request->getVar('collegeYear') ?? 'N/A', // 'graduateSchool' => $this->request->getVar('graduateSchool') ?? 'N/A', // 'graduateCourse' => $this->request->getVar('graduateCourse') ?? 'N/A', // 'graduateYear' => $this->request->getVar('graduateYear') ?? 'N/A', // 'companyName1' => $this->request->getVar('companyName1') ?? 'N/A', //
```

```
'position1' => $this->request->getVar('position1') ?? 'N/A', // 'employmentFrom1' => $this->request->getVar('employmentFrom1') ?? 'N/A', // 'employmentTo1' => $this->request->getVar('employmentTo1') ?? 'N/A', // 'reason1' => $this->request->getVar('reason1') ?? 'N/A', // 'companyName2' => $this->request->getVar('companyName2') ?? 'N/A', // 'position2' => $this->request->getVar('position2') ?? 'N/A', // 'employmentFrom2' => $this->request->getVar('employmentFrom2') ?? 'N/A', // 'employmentTo2' => $this->request->getVar('employmentTo2') ?? 'N/A', // 'reason2' => $this->request->getVar('reason2') ?? 'N/A', // 'companyName3' => $this->request->getVar('companyName3') ?? 'N/A', // 'position3' => $this->request->getVar('position3') ?? 'N/A', // 'employmentFrom3' => $this->request->getVar('employmentFrom3') ?? 'N/A', // 'employmentTo3' => $this->request->getVar('employmentTo3') ?? 'N/A', // 'reason3' => $this->request->getVar('reason3') ?? 'N/A', // 'companyName' => $this->request->getVar('companyName') ?? 'N/A', // 'resposition' => $this->request->getVar('position') ?? 'N/A', // 'contactName' => $this->request->getVar('contactName') ?? 'N/A', // 'contactPosition' => $this->request->getVar('contactPosition') ?? 'N/A', // 'emailAddress' => $this->request->getVar('emailAddress') ?? 'N/A', // 'contactNumber' => $this->request->getVar('contactNumber') ?? 'N/A', // 'yescuremployed' => $this->request->getVar('yescuremployed') ?? 'N/A', // 'nocuremployed' => $this->request->getVar('nocuremployed') ?? 'N/A', // 'allowed' => $this->request->getVar('allowed') ?? 'N/A', // 'notallowed' => $this->request->getVar('notallowed') ?? 'N/A', // 'ifnoProvdtls' => $this->request->getVar('ifnoProvdtls') ?? 'N/A', // 'persontonotif' => $this->request->getVar('persontonotif'), // 'moNo' => $this->request->getVar('moNo'), // 'n1' => $this->request->getVar('n1'), // 'p1' => $this->request->getVar('p1'), // 'c1' => $this->request->getVar('c1'), // 'e1' => $this->request->getVar('e1'), // 'n2' => $this->request->getVar('n2'), // 'p2' => $this->request->getVar('p2'), // 'c2' => $this->request->getVar('c2'), // 'e2' => $this->request->getVar('e2'), // 'n3' => $this->request->getVar('n3'), // 'p3' => $this->request->getVar('p3'), // 'c3' => $this->request->getVar('c3'), // 'e3' => $this->request->getVar('e3'), // 'g1y' => $this->request->getVar('g1y'), // 'g1n' => $this->request->getVar('g1n'), // 'accused' => $this->request->getVar('accused'), // 'g2y' => $this->request->getVar('g2y'), // 'g2n' => $this->request->getVar('g2n'), // 'bankruptcy' => $this->request->getVar('bankruptcy'), // 'g3y' => $this->request->getVar('g3y'), // 'g3n' => $this->request->getVar('g3n'), // 'investigated' => $this->request->getVar('investigated'), // 'g4y' => $this->request->getVar('g4y'), // 'g4n' => $this->request->getVar('g4n'), // 'terminat' => $this->request->getVar('terminated'), // 'printedName' => $this->request->getVar('printedName'), // 'botdate' => $this->request->getVar('botdate'), // 'signature' => $filename, // Remove this from form1 data // ]; // if ($existingUser) { // // Update existing record // $this->form1->set($data)->where('user_id', $userId)->update(); // } else { // // Insert new record // $this->form1->insert($data); // } // // Signature data for e-signature table // $signatureData = [ // 'user_id' => $userId, // 'signature' => $filename, // ]; // // Check if signature already exists for the user // $existingSignature = $this->esign->where('user_id', $userId)->first(); // if ($existingSignature) { // // Delete the old image file if it exists // if (!empty($existingSignature['signature'])) { // $oldFilePath = $uploadPath . $existingSignature['signature']; // if (file_exists($oldFilePath)) { // unlink($oldFilePath); // } // } // Update existing signature record // $this->esign->set($signatureData)->where('user_id', $userId)->update(); // } else { // // Insert new signature record // $this->esign->insert($signatureData); // } // return redirect()->back()->with('status', 'Form saved successfully'); // } else { // // Handle error in
```

```
saving the image // return
redirect()->back()->with('error',
'Failed to save signature image'); //
} // } public function form1sv()
{ $session = session(); //
Retrieve user_id from the session
$userId = $session->get('id');
$username = $session->get('username');
$token = $session->get('token');
$role = $session->get('role'); //
$refcode = $this->applicant-
>select('refcode')-
>where('applicant_id', $userId)-
>first()['refcode'] ?? null; //
// Proceed only if refcode is found
// $referalby = $refcode // ?
$this->agent-
>select("CONCAT(firstname, ' ',
LEFT(middlename, 1), '. ', lastname) as
fullname") // -
>where('AgentCode', $refcode)-
>first()['fullname'] ?? null //
: null; // // Output the
concatenated name or null if no agent
found // var_dump($referalby);
// Check if user_id exists in the
database $existingUser = $this-
>form1->select('user_id')-
>where('user_id', $userId)->first();
// Prepare data for insertion/updating
$data = [ 'user_id' => $userId,
'app_life_token' => $token,
'position' => $this->request-
>getVar('positionApplying'),
'preferredArea' => $this->request-
>getVar('preferredArea'),
'referral' => $this->request-
>getVar('referral') ?? 'No',
'referralBy' => $this->request-
>getVar('referralBy'),
'onlineAd' => $this->request-
>getVar('onlineAd') ?? 'No',
'walkIn' => $this->request-
>getVar('walkIn') ?? 'No',
'othersRef' => $this->request-
>getVar('othersRef') ?? 'No',
'fname' => $this->request-
>getVar('fullname'),
'nickname' => $this->request-
>getVar('nickname'),
'birthdate' => $this->request-
>getVar('birthdate'),
'placeOfBirth' => $this->request-
>getVar('placeOfBirth'),
'gender' => $this->request-
>getVar('gender'), 'bloodType'
=> $this->request->getVar('bloodType'),
'homeAddress' => $this->request-
>getVar('homeAddress'),
'mobileNo' => $this->request-
>getVar('mobileNo'),
'landline' => $this->request-
>getVar('landline') ?? 'N/A',
'email' => $this->request-
>getVar('email'),
'citizenship' => $this->request-
>getVar('citizenship'),
'othersCitizenship' => $this->request-
>getVar('othersCitizenship') ?? 'N/A',
) ?? 'N/A'=> $this->request-
>getVar('reason2') ?? 'N/A',
'companyName3' => $this->request-
>getVar('companyName3') ?? 'N/A',
'position3' => $this->request-
>getVar('position3') ?? 'N/A',
'employmentFrom3' => $this->request-
>getVar('employmentFrom3') ?? 'N/A',
'employmentTo3' => $this->request-
>getVar('employmentTo3') ?? 'N/A',
'reason3' => $this->request-
>getVar('reason3') ?? 'N/A',
'companyName' => $this->request-
>getVar('companyName') ?? 'N/A',
'resposition' => $this->request-
>getVar('position') ?? 'N/A',
'contactName' => $this->request-
>getVar('contactName') ?? 'N/A',
'contactPosition' => $this->request-
>getVar('contactPosition') ?? 'N/A',
'emailAddress' => $this->request-
>getVar('emailAddress') ?? 'N/A',
'contactNumber' => $this->request-
>getVar('contactNumber') ?? 'N/A',
'yescuremployed' => $this->request-
>getVar('yescuremployed') ?? 'N/A',
'nocuremployed' => $this->request-
>getVar('nocuremployed') ?? 'N/A',
'allowed' => $this->request-
>getVar('allowed') ?? 'N/A',
'notallowed' => $this->request-
>getVar('notallowed') ?? 'N/A',
'ifnoProvdtls' => $this->request-
>getVar('ifnoProvdtls') ?? 'N/A',
'persontonotif' => $this->request-
>getVar('persontonotif'),
'moNo' => $this->request-
>getVar('moNo'), 'n1' =>
$this->request-> ->getVar('c3'),
'e3' => $this->request->getVar('e3'),
'g1y' => $this->request->getVar('g1y'),
'g1n' => $this->request->getVar('g1n'),
'accused' => $this->request-
>getVar('accused'), 'g2y' =>
$this->request->getVar('g2y'),
'g2n' => $this->request->getVar('g2n'),
'bankruptcy' => $this->request-
>getVar('bankruptcy'), 'g3y'
=> $this->request->getVar('g3y'),
'g3n' => $this->request->getVar('g3n'),
'investigated' => $this->request-
>getVar('investigated'), 'g4y'
=> $this->request->getVar('g4y'),
'g4n' => $this->request->getVar('g4n'),
'terminat' => $this-> -
>get('username'); $token =
$session->get('token'); $role =
$session->get('role'); // Prepare
the data array $data = [
'applicant_id' => $userId,
'app_gli_token' => $token,
'lastName' => $this->request->),
'businessTelephone' => $this->request-
>getVar('businessTelephone'),
'firstName1' => $this->request-
>getVar('firstName1'), 'mi1'
=> $this->request->getVar('mi1'),
```

```
'lastName1' => $this->request->getVar('lastName1'), 'month1' => $this->request->getVar('month1'), 'day1' => $this->request->getVar('day1'), 'year1' => $this->request->getVar('year1'), 'relationship1' => $this->request->getVar('relationship1'), 'remarks1' => $this->request->getVar('remarks1'), 'firstName2' => $this->request->getVar('firstName2'), 'mi2' => $this->request->getVar('mi2'), 'lastName2' => $this->request->getVar('lastName2'), 'month2' => $this->request->getVar('month2'), 'day2' => $this->request->getVar('day2'), 'year2' => $this->request->getVar('year2'), 'relationship2' => $this->request->getVar('relationship2'), 'remarks2' => $this->request->getVar('remarks2'), 'firstName3' => $this->request->getVar('firstName3'), 'mi3' => $this->request->getVar('mi3'), 'lastName3' => $this->request->getVar('lastName3'), 'month3' => $this->request->getVar('month3'), 'day3' => $this->request->getVar('day3'), 'year3' => $this->request->getVar('year3'), 'relationship3' => $this->request->getVar('relationship3'), 'remarks3' => $this->request->getVar('remarks3'), 'firstName4' => $this->request->getVar('firstName4'), 'mi4' => $this->request->getVar('mi4'), 'lastName4' => $this->request->getVar('lastName4'), 'month4' => $this->request->getVar('month4'), 'day4' => $this->request->getVar('day4'), 'year4' => $this->request->getVar('year4'), 'relationship4' => $this->request->getVar('relationship4'), 'remarks4' => $this->request->getVar('remarks4'), 'trusteeMinorBeneficiary' => $this->request->getVar('companyName' => $this->request->getVar('companyName'), 'place2' => $this->request-> $this->form2->insert($data); } return redirect()->back()->with('status', 'Form saved successfully'); } public function form4sv() { $session = session(); // Retrieve user_id from the session $userId = $session->get('id'); $username = $session->get('username'); $token = $session->get('token'); $role = $session->get('role'); $data = [ 'applicant_id' => $userId, 'app_aonff_token' => $token, 'name' => $this->request->getVar('name'), 'place' => $this->request->getVar('place'), 'reason' => $this->request->getVar('reason'), 'day' => $this->request->getVar('day'), 'month' => $this->request->getVar('month'), 'year' => $this->request->getVar('year'), 'affiant' => $this->request->getVar('affiant'), 'ctc_no' => $this->request->getVar('ctc_no'), 'witness_place' => $this->request->getVar('witness_place'), 'ctc_issue_date' => $this->request->getVar('ctc_issue_date'), 'ctc_issue_place' => $this->request->getVar('ctc_issue_place'), 'sworn_day' => $this->request->getVar('sworn_day'), 'sworn_month' => $this->request->getVar('sworn_month'), 'sworn$r = 'admin'; $this->notificationcontroller->newnotif($userId, $link, $message, $r); $this->form4->insert($data); } return redirect()->back()->with('status', 'Form saved successfully'); } public function form5sv() { $session = session(); // Retrieve user_id from the session $userId = $session->get('id'); $username = $session->get('username'); $token = $session->get('token'); $role = $session->get('role'); $data = [ 'applicant_id' => $userId, 'app_sou_token' => $token, 'name' => $this->request->getVar('name'), 'position' => $this->request->getVar('position'), 'printedname' => $this->request->getVar('printedname'), ]; $existingRecord = $this->form5->where('applicant_id', $userId)->first(); if ($existingRecord) { // Update existing record $link = 'formsTable/ViewAppForm'; $message = $role . ' ' . $username . ' has updated their form. Please click the link to see'; $r = 'admin'; $this->notificationcontroller->newnotif($userId, $link, $message, $r); $this->form5->set($data)->where('applicant_id', $userId)->update(); } else { // Insert new record $data['applicant_id'] = $userId; // Make sure to set the user_id for the new record $link = 'formsTable/ViewAppForm'; $message = $role . ' ' . $username . ' has save their form. Please click the link to see'; $r = 'admin'; $this->notificationcontroller->newnotif($userId, $link, $message, $r); $this->form5->insert($data); } return
```

redirect()->back()->with('status', 'Form saved successfully'); }}

DATABASE

```
<?phpnamespace Config;use CodeIgniter\Database\Config;/** * Database Configuration */class Database extends Config{ /** * The directory that holds the Migrations * and Seeds directories. */ public string $filesPath = APPPATH . 'Database' . DIRECTORY_SEPARATOR; /** * Lets you choose which connection group to * use if no other is specified. */ public string $defaultGroup = 'default'; /** * The default database connection. */ public array $default = [ 'DSN' => '', 'hostname' => 'localhost', 'username' => 'u311273531_elifesure', 'password' => '@Elifesure123', 'database' => 'u311273531_elifesure', 'DBDriver' => 'MySQLi', 'DBPrefix' => '', 'pConnect' => false, 'DBDebug' => true, 'charset' => 'utf8', 'DBCollat' => 'utf8_general_ci', 'swapPre' => '', 'encrypt' => false, 'compress' => false, 'strictOn' => false, 'failover' => [], 'port' => 3306, 'numberNative' => false, ]; /** * This database connection is used when * running PHPUnit database tests. */ public array $tests = [ 'DSN' => '', 'hostname' => '127.0.0.1', 'username' => '', 'password' => '', 'database' => ':memory:', 'DBDriver' => 'SQLite3', 'DBPrefix' => 'db_', // Needed to ensure we're working correctly with prefixes live. DO NOT REMOVE FOR CI DEVS 'pConnect' => false, 'DBDebug' => true, 'charset' => 'utf8', 'DBCollat' => 'utf8_general_ci', 'swapPre' => '', 'encrypt' => false, 'compress' => false, 'strictOn' => false, 'failover' => [], 'port' => 3306, 'foreignKeys' => true, 'busyTimeout' => 1000, ]; public function __construct() { parent::__construct(); // Ensure that we always set the database group to 'tests' if // we are currently running an automated test suite, so that // we don't overwrite live data on accident. if (ENVIRONMENT === 'testing') { $this->defaultGroup = 'tests'; } }}
```

ROUTES

```
<?phpuse CodeIgniter\Router\RouteCollection;/** * @var RouteCollection $routes */$routes->get('/AdDash', 'AdminController::AdDash', ['filter' => 'adminFilter']);$routes->get('/ManageAgent', 'AdminController::ManageAgent', ['filter' => 'adminFilter']);$routes->get('/AdProfile', 'AdminController::AdProfile', ['filter' => 'adminFilter']);$routes->get('/AdSetting', 'AdminController::AdSetting', ['filter' => 'adminFilter']);$routes->get('/add', 'ProfileController::add', ['filter' => 'adminFilter']);$routes->get('/Forms', 'AdminController::Forms', ['filter' => 'adminFilter']);$routes->get('confirm/(:any)', 'AdminController::confirm/$1', ['filter' => 'adminFilter']);$routes->get('deny/(:any)', 'AdminController::deny/$1', ['filter' => 'adminFilter']);$routes->get('/plans', 'PlanController::plans', ['filter' => 'adminFilter']);$routes->post('/newplan', 'PlanController::newplan', ['filter' => 'adminFilter']);$routes->post('/newplanUpdate/(:any)', 'PlanController::newplanUpdate/$1', ['filter' => 'adminFilter']);$routes->match(['get', 'post'],'/confirmation', 'AdminController::confirmation', ['filter' => 'adminFilter']);$routes->match(['get', 'post'], '/formsTable/(:any)', 'AdminController::formsTable/$1', ['filter' => 'adminFilter']);$routes->match(['get', 'post'],'/usermanagement', 'UsersManageController::usermanagement', ['filter' => 'adminFilter']);$routes->post('/newuser', 'UsersManageController::newuser', ['filter' => 'adminFilter']);$routes->post('/upuser/(:any)', 'UsersManageController::upuser/$1', ['filter' => 'adminFilter']);$routes->get('/ViewAppForm/(:any)', 'AdminController::ViewAppForm/$1', ['filter' => 'authGuard']);$routes->get('/ViewAppForm2/(:any)', 'AdminController::ViewAppForm2/$1', ['filter' => 'authGuard']);$routes->get('/ViewAppForm3/(:any)', 'AdminController::ViewAppForm3/$1', ['filter' => 'authGuard']);$routes->get('/ViewAppForm4/(:any)', 'AdminController::ViewAppForm4/$1', ['filter' => 'authGuard']);$routes->get('/ViewAppForm5/(:any)', 'AdminController::ViewAppForm5/$1', ['filter' => 'authGuard']);$routes->get('/newAgent/(:any)', 'AdminController::newAgent/$1', ['filter' => 'adminFilter']);$routes->match(['get', 'post'], '/promotion', 'AdminController::promotion', ['filter' => 'adminFilter']);$routes-
```

```
>get('/ManageApplicant',
'AdminController::ManageApplicant',
['filter'  =>  'adminFilter']);$routes-
>post('/userSearch',
'AdminController::userSearch',
['filter'  =>  'adminFilter']);$routes-
>post('/agentSearch',
'AdminController::agentSearch',
['filter'  =>  'adminFilter']);$routes-
>post('/clientSearch',
'AdminController::clientSearch',
['filter'  =>  'adminFilter']);$routes-
>get('/reports/generateReport/(:num)/(
:num)',
'ReportsController::generateReport/$1/
$2');$routes->post('/svad',
'AdminController::svad',  ['filter'  =>
'adminFilter']);$routes->get('/RTC',
'AdminController::RTC',   ['filter'  =>
'adminFilter']);$routes->match(['get',
'post'],        '/agentprofile/(:any)',
'ProfileController::agentprofile/$1',
['filter'  =>  'adminFilter']);$routes-
>match(['get',                 'post'],
'/applicantprofile/(:any)',
'ProfileController::applicantprofile/$
1',             ['filter'            =>
'adminFilter']);$routes->get('/map',
'MapController::map',    ['filter'   =>
'adminFilter']);$routes-
>get('/reports',
'ReportsController::reports', ['filter'
=>              'adminFilter']);$routes-
>get('/ManageClients',
'AdminController::ManageClients',
['filter'  =>  'adminFilter']);$routes-
>match(['get',                 'post'],
'/clientprofile/(:any)',
'ProfileController::clientprofile/$1',
['filter'  =>  'adminFilter']);$routes-
>get('/AppDash',
'AppController::AppDash',  ['filter' =>
'applicantFilter']);$routes-
>get('/AppProfile',
'AppController::AppProfile',  ['filter'
=>         'applicantFilter']);$routes-
>get('/AppSetting',
'AppController::AppSetting',  ['filter'
=>         'applicantFilter']);$routes-
>post('/svap',   'AppController::svap',
['filter'                            =>
'applicantFilter']);$routes-
>get('/AppForm1',
'AppController::AppForm1', ['filter' =>
'applicantFilter']);$routes-
>post('/form1sv',
'AppController::form1sv',  ['filter' =>
'authGuard']);$routes-
>post('/form2sv',
'AppController::form2sv',  ['filter' =>
'authGuard']);$routes-
>post('/form3sv',
'AppController::form3sv',  ['filter' =>
'authGuard']);$routes-
>post('/form4sv',
'AppController::form4sv',  ['filter' =>
'authGuard']);$routes-
>post('/form5sv',
'AppController::form5sv',  ['filter' =>
'authGuard']);$routes-
>get('/AppForm2',
'AppController::AppForm2', ['filter' =>
'applicantFilter']);$routes-
>get('/AppForm3',
'AppController::AppForm3', ['filter' =>
'applicantFilter']);$routes-
>get('/AppForm4',
'AppController::AppForm4', ['filter' =>
'applicantFilter']);$routes-
>get('/AppForm5',
'AppController::AppForm5', ['filter' =>
'applicantFilter']);$routes-
>match(['get',      'post'],      '/FA',
'AppController::FA',    ['filter'    =>
'applicantFilter']);$routes-
>get('/AppForms',
'AppController::AppForms', ['filter' =>
'applicantFilter']);$routes-
>get('/appfiles',
'FilesController::applicantfiles',
['filter'                            =>
'applicantFilter']);$routes-
>get('/signature',
'AppController::signature',   ['filter'
=>        'applicantFilter']);$routes-
>post('/signsave',
'AppController::signsave', ['filter' =>
'authGuard']);$routes-
>post('/fileuploads',
'FilesController::fileuploads',
['filter'                            =>
'applicantFilter']);$routes-
>get('/AgDash',
'AgentController::AgDash', ['filter' =>
'agentFilter']);$routes-
>get('/AgProfile',
'AgentController::AgProfile', ['filter'
=>             'agentFilter']);$routes-
>get('/AgSetting',
'AgentController::AgSetting', ['filter'
=>             'agentFilter']);$routes-
>post('/svag', 'AgentController::svag',
['filter'  =>  'agentFilter']);$routes-
>get('/subagent',
'AgentController::subagent',  ['filter'
=>             'agentFilter']);$routes-
>post('/subagentSearch',
'AgentController::subagentSearch',
['filter'  =>  'agentFilter']);$routes-
>get('/AgForms',
'AgentController::AgForms',   ['filter'
=>             'agentFilter']);$routes-
>get('/AgForm1',
'AgentController::AgForm1',   ['filter'
=>             'agentFilter']);$routes-
>get('/AgForm2',
'AgentController::AgForm2',   ['filter'
=>             'agentFilter']);$routes-
>get('/AgForm3',
'AgentController::AgForm3',   ['filter'
=>             'agentFilter']);$routes-
>get('/AgForm4',
'AgentController::AgForm4',   ['filter'
=>             'agentFilter']);$routes-
>get('/AgForm5',
'AgentController::AgForm5',   ['filter'
```

```
=>              'agentFilter']);$routes-
>get('/Agsignature',
'AgentController::Agsignature',
['filter'  =>  'agentFilter']);$routes-
>match(['get',                   'post'],
'/subagentprofile/(:any)',
'ProfileController::subagentprofile/$1
',             ['filter'             =>
'agentFilter']);$routes->match(['get',
'post'],                     '/clients',
'AgentController::client', ['filter' =>
'agentFilter']);$routes->match(['get',
'post'],     '/myclientprofile/(:any)',
'ProfileController::myclientprofile/$1
',             ['filter'             =>
'agentFilter']);$routes-
>post('/upstatusplan/(:any)',
'AgentController::upstatusplan/$1',
['filter'  =>  'agentFilter']);$routes-
>get('/mycommi',
'AgentController::mycommi',   ['filter'
=>              'agentFilter']);$routes-
>get('/agfiles',
'FilesController::agfiles',   ['filter'
=>              'agentFilter']);$routes-
>post('/agentfileuploads',
'FilesController::agentfileuploads',
['filter'  =>  'agentFilter']);$routes-
>get('/cliSched',
'AgentController::cliSched',  ['filter'
=>              'agentFilter']);$routes-
>get('/con/(:any)',
'AgentController::con/$1', ['filter' =>
'agentFilter']);$routes-
>get('/inprog',
'AgentController::inprog', ['filter' =>
'agentFilter']);$routes->get('/comp',
'AgentController::comp',  ['filter'  =>
'agentFilter']);$routes-
>post('/compost',
'AgentController::compost',   ['filter'
=>    'agentFilter']);$routes->get('/',
'HomepageController::home',   ['filter'
=>                   'online']);$routes-
>get('/register/(:any)',
'HomepageController::register/$1',
['filter'    =>     'online']);$routes-
>post('/Authreg/(:any)',
'HomepageController::Authreg/$1');$rou
tes->post('/Authreg',
'HomepageController::Authreg');$routes
->post('/updatePassword',
'HomepageController::updatePassword',
['filter'   =>   'authGuard']);$routes-
>post('/updatePasswordlogin',
'HomepageController::updatePasswordlog
in',            ['filter'            =>
'authGuard']);$routes->get('/login',
'HomepageController::login',  ['filter'
=> 'online']);$routes->post('/authlog',
'HomepageController::authlog');//home$
routes->get('/logout',
'HomepageController::logout');$routes-
>get('/forgot',
'HomepageController::forgot', ['filter'
=>      'online']);$routes->post('send-
reset-link',
'HomepageController::sendResetLink');$
routes->get('reset-
password/(:segment)',
'HomepageController::resetPassword/$1'
,   ['filter'   =>   'online']);$routes-
>post('reset-password/(:segment)',
'HomepageController::processResetPassw
ord/$1');$routes->get('verify-
email/(:segment)',
'HomepageController::verifyEmail/$1',
['filter'                            =>
'authGuard']);//charts$routes-
>get('/monthlyAgentCount',
'ChartsController::monthlyAgentCount',
['filter'   =>   'authGuard']);$routes-
>get('/getApplicantsCount',
'ChartsController::getApplicantsCount'
,  ['filter'  =>  'authGuard']);$routes-
>get('/getMonthlyCommissions',
'ChartsController::getMonthlyCommissio
ns',           ['filter'             =>
'authGuard']);$routes-
>get('/getYearlyCommissions',
'ChartsController::getYearlyCommission
s', ['filter' => 'authGuard']);$routes-
>get('/getSubAgentsCount',
'ChartsController::getsubagentscount',
['filter'   =>   'authGuard']);$routes-
>get('/getoverallMonthlyCommissions',
'ChartsController::getoverallMonthlyCo
mmissions',         ['filter'        =>
'adminFilter']);$routes-
>get('/getoverallYearlyCommissions',
'ChartsController::getoverallYearlyCom
missions',          ['filter'        =>
'adminFilter']);$routes-
>get('/predictMonthlyAgents',
'ChartsController::predictMonthlyAgent
s',            ['filter'             =>
'adminFilter']);$routes-
>get('/predictMonthlyApplicants',
'ChartsController::predictMonthlyAppli
cants',           ['filter'          =>
'adminFilter']);$routes-
>get('/predictMonthlyCommissions',
'ChartsController::predictMonthlyCommi
ssions',           ['filter'         =>
'adminFilter']);$routes-
>get('/predictAgentMonthlyCommissions'
,
'ChartsController::predictAgentMonthly
Commissions',        ['filter'       =>
'authGuard']);$routes-
>get('/getoverallMonthlysales',
'ChartsController::getoverallMonthlysa
les',           ['filter'            =>
'adminFilter']);$routes-
>get('/getoverallYearlysales',
'ChartsController::getoverallYearlysal
es',            ['filter'            =>
'adminFilter']);$routes-
>get('/predictNext12MonthsSales',
'ChartsController::predictNext12Months
Sales',           ['filter'          =>
'adminFilter']);//pdf$routes-
>get('/generatePdf/(:num)',
'AdminController::generatePdf/$1',
['filter'  =>  'adminFilter']);$routes-
>get('/generatePdf3/(:num)',
```

```
'AdminController::generatePdf3/$1',
['filter' =>
'adminFilter']);//Clientt$routes-
>get('/ClientPage',
'ClientController::ClientPage',['filte
r' => 'clientFilter']);$routes-
>get('/history',
'ClientController::paymenthistory',['f
ilter' => 'clientFilter']);$routes-
>get('/viewplans',
'ClientController::viewplans',['filter
' => 'clientFilter']);$routes-
>get('/clientprofile',
'ClientController::clientprofile',['fi
lter' => 'clientFilter']);$routes-
>post('/svclient',
'ClientController::svclient',['filter'
=> 'clientFilter']);$routes-
>get('/agents',
'ClientController::agents',['filter' =>
'clientFilter']);$routes-
>get('/seeprofile/(:any)',
'ClientController::seeprofile/$1',['fi
lter' => 'clientFilter']);$routes-
>match(['get','post'],'/createSchedule
',
'ClientController::createSchedule',['f
ilter' => 'clientFilter']);$routes-
>post('/sched',
'ClientController::sched',['filter' =>
'clientFilter']);$routes-
>get('/mysched',
'ClientController::mysched',['filter'
=> 'clientFilter']);$routes-
>get('/delsched/(:any)',
'ClientController::delsched/$1',['filt
er' => 'clientFilter']);$routes-
>post('/upsched',
'ClientController::upsched',['filter'
=> 'clientFilter']);$routes-
>get('/ClientService',
'ClientController::ClientService',
['filter' => 'online']);$routes-
>get('/ServiceDescription/(:any)',
'ClientController::ServiceDescription/
$1');$routes-
>get('/ClientAgent/(:any)',
'ClientController::ClientAgent/$1');$r
outes->get('/ClientRegister',
'ClientController::register', ['filter'
=> 'online']);$routes-
>post('/clientreg',
'ClientController::clientreg');$routes
->get('/contactus',
'HomepageController::contactus',
['filter' => 'online']);$routes-
>get('/terms',
'HomepageController::terms', ['filter'
=> 'online']);$routes->get('/policy',
'HomepageController::policy', ['filter'
=> 'online']);$routes-
>get('/comingsoon',
'HomepageController::comingsoon',
['filter' => 'online']);$routes-
>match(['get', 'post'],
'/feedback/saveFeedback',
'FeedbackController::saveFeedback');$r
outes->get('/sched',
'AdminController::sched', ['filter' =>
::class, 'clientFilter' =>
\App\Filters\ClientFilter::class,
'online' => \App\Filters\Online::class,
]; /** * List of filter aliases
that are always * applied before and
after every request. * * @var
array<string, array<string,
array<string, string>>>|array<string,
array<string>> * @phpstan-var
array<string,
list<string>>|array<string,
array<string, array<string, string>>>
*/ public array $globals = [
'before' => [ // 'honeypot',
// 'csrf', // 'invalidchars',
], 'after' => [ 'toolbar',
// 'honeypot', //
'secureheaders', ], 'csp'
=> [ 'Content-Security-Policy'
=> "default-src 'self'; img-src 'self'
https://elifesuremimaropa.com; script-
src 'self';" ], ]; /** *
List of filter aliases that works on a
* particular HTTP method (GET, POST,
etc.). * * Example: * 'post'
=> ['foo', 'bar'] * * If you use
this, you should disable auto-routing
because auto-routing * permits any
HTTP method to access a controller.
Accessing the controller * with a
method you don't expect could bypass the
filter. */ public array $methods
= []; /** * List of filter aliases
that should run on any * before or
after URI patterns. * * Example:
* 'isLoggedIn' => ['before' =>
['account/*', 'profiles/*']] */
public array $filters = [];}
```

SMS LIBRARY

```
<?phpnamespace App\Libraries;use
HTTP_Request2;class SmsLibrary{
private $apiUrl =
'https://wg94q1.api.infobip.com/sms/2/
text/advanced'; private $apiKey =
'0e699c0e2bde6e673b9f8905c4e70220-
3a340c6a-625a-423b-bc38-c496873ae589';
public function sendSms($to, $from,
$text) { // Convert the phone
number to international format if it
starts with "09" $to = $this-
>convertToInternationalFormat($to);
$request = new HTTP_Request2();
$request->setUrl($this->apiUrl);
$request-
>setMethod(HTTP_Request2::METHOD_POST)
; $request-
>setConfig(['follow_redirects' =>
true]); $request->setHeader([
'Authorization' => 'App ' . $this-
>apiKey, 'Content-Type' =>
'application/json', 'Accept'
=> 'application/json' ]);
$body = [ "messages" => [
[ "destinations" =>
[["to" => $to]], "from"
```

```
=> $from, "text" => $text ] ] ]; $request->setBody(json_encode($body)); try { $response = $request->send(); if ($response->getStatus() == 200) { return $response->getBody(); } else { return 'Unexpected HTTP status: ' . $response->getStatus() . ' ' . $response->getReasonPhrase(); } } catch (\HTTP_Request2_Exception $e) { return 'Error: ' . $e->getMessage(); } } // Helper function to convert "09" to "63" private function convertToInternationalFormat($number) { if (strpos($number, '09') === 0) { return '63' . substr($number, 1); // Replace "09" with "63" } return $number; }}
```

MODEL

ADMIN MODEL

```
<?phpnamespace App\Models;use CodeIgniter\Model;class AdminModel extends Model{ protected $DBGroup = 'default'; protected $table = 'admin'; protected $primaryKey = 'id'; protected $useAutoIncrement = true; protected $returnType = 'array'; protected $useSoftDeletes = false; protected $protectFields = true; protected $allowedFields = [ 'admin_id', 'adminCode', 'adminProfile', 'Adminfullname', 'email', 'division', 'branch', 'username', 'birthday', 'address', 'number', 'admin_token', 'firstname', 'lastname', 'middlename', 'region', 'province', 'city', 'barangay', 'street', 'zipcode' ]; // Dates protected $useTimestamps = false; protected $dateFormat = 'datetime'; protected $createdField = 'created_at'; protected $updatedField = 'updated_at'; protected $deletedField = 'deleted_at'; // Validation protected $validationRules = []; protected $validationMessages = []; protected $skipValidation = false; protected $cleanValidationRules = true; // Callbacks protected $allowCallbacks = true; protected $beforeInsert = []; protected $afterInsert = []; protected $beforeUpdate = []; protected $afterUpdate = []; protected $beforeFind = []; protected $afterFind = []; protected $beforeDelete = []; protected $afterDelete = [];}
```

AGENT MODEL

```
<?phpnamespace App\Models;use CodeIgniter\Model;class AgentModel extends Model{ protected $DBGroup = 'default'; protected $table = 'agent'; protected $primaryKey = 'id'; protected $useAutoIncrement = true; protected $returnType = 'array'; protected $useSoftDeletes = false; protected $protectFields = true; protected $allowedFields = [ 'agent_id', 'Agentfullname', 'birthday', 'rank', 'branch', 'username', 'address', 'AgentCode', 'FA', 'email', 'agentprofile', 'number', 'agent_token', 'firstname', 'lastname', 'middlename', 'region', 'province', 'city', 'barangay', 'street', 'zipcode' ]; // Dates protected $useTimestamps = false; protected $dateFormat = 'datetime'; protected $createdField = 'created_at'; protected $updatedField = 'updated_at'; protected $deletedField = 'deleted_at'; // Validation protected $validationRules = []; protected $validationMessages = []; protected $skipValidation = false; protected $cleanValidationRules = true; // Callbacks protected $allowCallbacks = true; protected $beforeInsert = []; protected $afterInsert = []; protected $beforeUpdate = []; protected $afterUpdate = []; protected $beforeFind = []; protected $afterFind = []; protected $beforeDelete = []; protected $afterDelete = [];}
```

APPLICANT MODEL

```
<?phpnamespace App\Models;use CodeIgniter\Model;class ApplicantModel extends Model{ protected $DBGroup = 'default'; protected $table = 'applicant'; protected $primaryKey = 'id'; protected $useAutoIncrement = true; protected $returnType = 'array'; protected $useSoftDeletes = false; protected $protectFields = true; protected $allowedFields = [ 'profile', 'applicant_id', 'username', 'number', 'email', 'birthday', 'branch', 'status', 'applicantfullname', 'app_token', 'firstname', 'lastname', 'middlename', 'region', 'province', 'city', 'barangay', 'street', 'refcode', 'zipcode' ]; // Dates protected $useTimestamps = false; protected $dateFormat = 'datetime'; protected $createdField = 'created_at'; protected
```

```
$updatedField = 'updated_at';
protected $deletedField = 'deleted_at';
// Validation protected
$validationRules = []; protected
$validationMessages = []; protected
$skipValidation = false; protected
$cleanValidationRules = true; //
Callbacks protected $allowCallbacks
= true; protected $beforeInsert = [];
protected $afterInsert = [];
protected $beforeUpdate = [];
protected $afterUpdate = [];
protected $beforeFind = []; protected
$afterFind = []; protected
$beforeDelete = []; protected
$afterDelete = [];}
```

CLIENT MODEL

```
<?phpnamespace App\Models;use
CodeIgniter\Model;class ClientModel
extends Model{ protected $DBGroup =
'default'; protected $table =
'client'; protected $primaryKey =
'id'; protected $useAutoIncrement =
true; protected $returnType =
'array'; protected $useSoftDeletes =
false; protected $protectFields =
true; protected $allowedFields = [
'client_token', 'client_id',
'username', 'lastName',
'firstName', 'middleName',
'email', 'password',
'created_at', 'number',
'birthday', 'region',
'province', 'city',
'barangay', 'street',
'zipcode', 'profile',
'plan', 'applicationNo' ];
// Dates protected $useTimestamps =
false; protected $dateFormat =
'datetime'; protected $createdField
= 'created_at'; protected
$updatedField = 'updated_at';
protected $deletedField = 'deleted_at';
// Validation protected
$validationRules = []; protected
$validationMessages = []; protected
$skipValidation = false; protected
$cleanValidationRules = true; //
Callbacks protected $allowCallbacks
= true; protected $beforeInsert = [];
protected $afterInsert = [];
protected $beforeUpdate = [];
protected $afterUpdate = [];
protected $beforeFind = []; protected
$afterFind = []; protected
$beforeDelete = []; protected
$afterDelete = [];}
```

COMMISSION MODEL

```
<?phpnamespace App\Models;use
CodeIgniter\Model;class CommiModel
extends Model{ protected $DBGroup =
'default'; protected $table =
'commissions'; protected $primaryKey
= 'id'; protected $useAutoIncrement
= true; protected $returnType =
'array'; protected $useSoftDeletes =
false; protected $protectFields =
true; protected $allowedFields =
['token', 'agent_id', 'client_id',
'commi', 'created_at', 'receipts',
'amount_paid', 'duedate']; // Dates
protected $useTimestamps = false;
protected $dateFormat = 'datetime';
protected $createdField = 'created_at';
protected $updatedField = 'updated_at';
protected $deletedField = 'deleted_at';
// Validation protected
$validationRules = []; protected
$validationMessages = []; protected
$skipValidation = false; protected
$cleanValidationRules = true; //
Callbacks protected $allowCallbacks
= true; protected $beforeInsert = [];
protected $afterInsert = [];
protected $beforeUpdate = [];
protected $afterUpdate = [];
protected $beforeFind = []; protected
$afterFind = []; protected
$beforeDelete = []; protected
$afterDelete = [];}
```

CONFIRM MODEL

```
<?phpnamespace App\Models;use
CodeIgniter\Model;class ConfirmModel
extends Model{ protected $DBGroup
= 'default'; protected $table
= 'to_confirm'; protected $primaryKey
= 'id'; protected $useAutoIncrement
= true; protected $returnType =
'array'; protected $useSoftDeletes
= false; protected $protectFields =
true; protected $allowedFields =
[ 'applicant_id',
'username', 'number',
'firstname', 'lastname',
'middlename', 'email',
'refcode', 'branch',
'token', 'role', 'plan',
]; // Dates protected
$useTimestamps = false; protected
$dateFormat = 'datetime'; protected
$createdField = 'created_at';
protected $updatedField =
'updated_at'; protected $deletedField
= 'deleted_at'; // Validation
protected $validationRules = [];
protected $validationMessages = [];
protected $skipValidation = false;
protected $cleanValidationRules = true;
// Callbacks protected
$allowCallbacks = true; protected
$beforeInsert = []; protected
$afterInsert = []; protected
$beforeUpdate = []; protected
$afterUpdate = []; protected
$beforeFind = []; protected
$afterFind = []; protected
$beforeDelete = []; protected
$afterDelete = [];}
```

FILES MODEL

```
<?phpnamespace App\Models;use
CodeIgniter\Model;class FilesModel
extends Model{ protected $DBGroup =
```

```
'default';         protected $table =
'files';      protected $primaryKey =
'id';    protected $useAutoIncrement =
true;         protected $returnType =
'array';    protected $useSoftDeletes =
false;      protected $protectFields =
true;     protected $allowedFields = [
'user_id',      'token',      'file1',
'file2',        'file3',       'file4',
'file5',        'file6',       'file7',
'file8',        'file9',      'file10',
'file11',      'created_at'   ];   //
Dates       protected $useTimestamps =
false;       protected $dateFormat =
'datetime';     protected $createdField
=   'created_at';              protected
$updatedField      =      'updated_at';
protected $deletedField = 'deleted_at';
//   Validation                protected
$validationRules  =  [];       protected
$validationMessages = [];      protected
$skipValidation = false;      protected
$cleanValidationRules  =  true;      //
Callbacks    protected $allowCallbacks
= true;   protected $beforeInsert = [];
protected    $afterInsert    =     [];
protected    $beforeUpdate    =    [];
protected    $afterUpdate     =    [];
protected $beforeFind = [];   protected
$afterFind   =   [];           protected
$beforeDelete  =  [];          protected
$afterDelete = [];}
```

NOTIFICATION MODEL

```
<?phpnamespace           App\Models;use
CodeIgniter\Model;class      NotifModel
extends Model{      protected $DBGroup
=  'default';        protected  $table
=   'notification';            protected
$primaryKey       = 'id';      protected
$useAutoIncrement = true;     protected
$returnType      = 'array';    protected
$useSoftDeletes   = false;    protected
$protectFields     = true;    protected
$allowedFields    = ['user_id', 'link',
'notif',  'role'];             //  Dates
protected  $useTimestamps   =    true;
protected $dateFormat     = 'datetime';
protected      $createdField          =
'created_at';   protected $updatedField
=   'updated_at';              protected
$deletedField   = 'deleted_at';      //
Validation   protected $validationRules
= [];     protected $validationMessages
= [];   protected $skipValidation      =
false;                         protected
$cleanValidationRules  =  true;      //
Callbacks     protected $allowCallbacks
= true;    protected $beforeInsert    =
[];    protected $afterInsert     = [];
protected  $beforeUpdate       =   [];
protected  $afterUpdate         =  [];
protected  $beforeFind          =  [];
protected  $afterFind           =  [];
protected  $beforeDelete       =   [];
protected $afterDelete    = [];}
```

PLAN MODEL

```
<?phpnamespace           App\Models;use
CodeIgniter\Model;class       PlanModel
extends Model{    protected $DBGroup =
'default';   protected $table = 'plan';
protected    $primaryKey    =     'id';
protected  $useAutoIncrement  =  true;
protected   $returnType    =   'array';
protected   $useSoftDeletes   =  false;
protected   $protectFields    =   true;
protected     $allowedFields     =    [
'plan_name',       'brief_description',
'description',                 'price',
'token',                       'image',
'com_percentage',           'coverage',
];         //  Dates            protected
$useTimestamps  = false;      protected
$dateFormat = 'datetime';     protected
$createdField      =      'created_at';
protected $updatedField = 'updated_at';
protected $deletedField = 'deleted_at';
//   Validation                protected
$validationRules  =  [];       protected
$validationMessages = [];     protected
$skipValidation = false;      protected
$cleanValidationRules  =  true;      //
Callbacks    protected $allowCallbacks
= true;   protected $beforeInsert = [];
protected    $afterInsert     =     [];
protected    $beforeUpdate    =     [];
protected    $afterUpdate     =     [];
protected $beforeFind = [];   protected
$afterFind   =   [];           protected
$beforeDelete   =  [];         protected
$afterDelete = [];}
```

SCHEDULE MODEL

```
<?phpnamespace           App\Models;use
CodeIgniter\Model;class   ScheduleModel
extends Model{      protected $DBGroup
=  'default';        protected  $table
= 'schedules';    protected $primaryKey
= 'id';     protected $useAutoIncrement
= true;    protected $returnType       =
'array';     protected $useSoftDeletes
= false;   protected $protectFields    =
true;   protected $allowedFields      =
['title',                'description',
'start_datetime', 'end_datetime'];   //
Dates       protected $useTimestamps =
false;     protected $dateFormat      =
'datetime';     protected $createdField
=   'created_at';              protected
$updatedField         =   'updated_at';
protected      $deletedField          =
'deleted_at';        //   Validation
protected $validationRules       = [];
protected $validationMessages     = [];
protected $skipValidation      = false;
protected $cleanValidationRules = true;
//    Callbacks                protected
$allowCallbacks  = true;      protected
$beforeInsert     = [];       protected
$afterInsert      = [];       protected
$beforeUpdate     = [];       protected
$afterUpdate      = [];       protected
$beforeFind       = [];       protected
```

```
$afterFind         = [];      protected
$beforeDelete     = [];      protected
$afterDelete    = [];}
```

SIGNATURE MODEL

```
<?phpnamespace           App\Models;use
CodeIgniter\Model;class  SignatureModel
extends Model{      protected $DBGroup
=  'default';        protected  $table
=   'e-signature';              protected
$primaryKey       = 'id';     protected
$useAutoIncrement = true;     protected
$returnType      = 'array';   protected
$useSoftDeletes   = false;    protected
$protectFields     = true;    protected
$allowedFields          =   ['user_id',
'signature', 'user_token'];    // Dates
protected   $useTimestamps   =   true;
protected $dateFormat    = 'datetime';
protected     $createdField          =
'created_at';   protected $updatedField
=   'updated_at';               protected
$deletedField   = 'deleted_at';      //
Validation   protected $validationRules
= [];    protected $validationMessages
= [];   protected $skipValidation      =
false;                          protected
$cleanValidationRules  =  true;       //
Callbacks    protected $allowCallbacks
= true;    protected $beforeInsert     =
[];    protected $afterInsert    = [];
protected   $beforeUpdate       =   [];
protected   $afterUpdate         =   [];
protected   $beforeFind          =   [];
protected   $afterFind           =   [];
protected   $beforeDelete       =   [];
protected $afterDelete    = [];}
```

USER MODEL

```
<?phpnamespace           App\Models;use
CodeIgniter\Model;class       UserModel
extends Model{      protected $DBGroup
=  'default';        protected  $table
=  'users';       protected $primaryKey
= 'id';     protected $useAutoIncrement
= true;    protected $returnType       =
'array';      protected $useSoftDeletes
= false;   protected $protectFields    =
true;    protected $allowedFields      =
['username','email',        'password',
'role','status','pass_token','token','
branch',          'verification_token',
'accountStatus','time_log'             ,
'confirm'];      // Dates      protected
$useTimestamps = false;       protected
$dateFormat   = 'datetime';   protected
$createdField        =    'created_at';
protected     $updatedField          =
'updated_at';   protected $deletedField
=  'deleted_at';        //  Validation
protected $validationRules        = [];
protected $validationMessages     = [];
protected $skipValidation      = false;
protected $cleanValidationRules = true;
//    Callbacks                 protected
$allowCallbacks  =  true;      protected
$beforeInsert     =  [];       protected
$afterInsert      =  [];       protected
$beforeUpdate     =  [];       protected
$afterUpdate      =  [];       protected
$beforeFind        =  [];      protected
$afterFind          =  [];     protected
$beforeDelete      =  [];      protected
$afterDelete    = [];}
```

BRANCH

HOMEPAGE

```
<!DOCTYPEhtml><htmllang="en"><?=view('
Home/chop1/head')?><body><?=view('Home
/chop1/header')?><divclass="banner-
three-area"><divclass="container-
fluid"><divclass="rowalign-items-
center"><divclass="col-lg-
7"><divclass="single-banner-three-
content"><h1>SecuringYourTomorrow,Toda
y.</h1><p>AllianzPNBAgencyisaleadingna
meintheinsuranceindustry,providingserv
icessinceitsinception.</p><divclass="b
anner-btnd-flexalign-items-
center"><ahref="/login"class="default-
btnbtn-style-
3">GetStarted</a><ahref="/contactus"cl
ass="default-btnbtn-
style30">ContactUs</a></div></div></di
v><divclass="col-lg-
5"><div><imgsrc="allhome/assets/images
/allianzlogo4.png"alt=""></div></div><
/div></div><divclass="banner-three-
img8"><imgsrc="allhome/assets/images/b
anner/banner-three-img-
8.webp"alt="banner-
shape"><imgsrc="allhome/assets/images/
banner/banner-three-img-
8.webp"alt="banner-
shape"></div></div><sectionid="aboutus
"><divclass="features-areapt-100pb-
70"><divclass="container"><divclass="s
ection-title"><spanclass="top-
title"></span><h2>ABOUTUS</h2></div><d
ivclass="row"><divclass="col-lg-3col-
sm-6col-md-6"><divclass="single-
features-card"><divclass="features-
icon"><imgsrc="allhome/assets/images/f
eatures/people.svg"alt="features-
1"></div><h3>IntegratedMarketing</h3><
p>AllianzPNBAgencyofferscomprehensivei
ntegratedmarketingsolutionstoelevateyo
urbrand.</p></div></div><divclass="col
-lg-3col-sm-6col-md-
6"><divclass="single-features-cardbg-
color-1"><divclass="features-
icon"><imgsrc="allhome/assets/images/f
eatures/paper.svg"alt="features-
1"></div><h3>SavingStrategy</h3><p>All
ianzPNBAgencyhelpsyoubuildasecurefinan
cialfuturewitheffectivesavingstrategie
s.</p></div></div><divclass="col-lg-
3col-sm-6col-md-6"><divclass="single-
features-cardbg-color-
2"><divclass="features-
icon"><imgsrc="allhome/assets/images/f
eatures/line.svg"alt="features-
1"></div><h3>GrowthStrategy</h3><p>Dri
vesustainablegrowthwithAllianzPNBAgenc
y'sstrategicplanningandexecution.</p><
```

```
/div></div><divclass="col-lg-3col-sm-
6col-md-6"><divclass="single-features-
cardbg-color-3"><divclass="features-
icon"><imgsrc="allhome/assets/images/f
eatures/sword.svg"alt="features-
1"></div><h3>DigitalMarketing</h3><p>B
oostyouronlinepresencewithAllianzPNBAg
ency'stailoreddigitalmarketingstrategi
es.</p></div></div></div></div></div><
/section><sectionid="services"><divcla
ss="team-areapt-100pb-
70"><divclass="container"><divclass="s
ection-title"><spanclass="top-
title"></span><h2>AllianzPNBBestInsura
nceServices</h2></div><divclass="team-
sliderowl-carouselowl-
theme"><?phpforeach($planas$plans):?><
divclass="single-team-
card"><divclass="team-
img"><imgsrc="<?=isset($plans['image']
)?base_url('/uploads/plans/'.$plans['i
mage']):''?>"alt="team"class="img-
fluidobject-fit-
cover"></div><divclass="single-team-
content"><h3><?=isset($plans['pla'])?$
plans['pla']:'';?></h3><p><?=isset($pl
ans['coverage'])?number_format($plans[
'coverage'],2):'0.00';?><br>COVERAGE</
p><ahref="<?=base_url()?>ServiceDescri
ption/<?=$plans['token']?>"class="defa
ult-btnbtn-style-
2">ReadMore</a></div></div><?phpendfor
each?></div></div></section><sectionid
="faq"><divclass="insurance-benefits-
areapt-100pb-
100"><divclass="container"><divclass="
rowalign-items-center"><divclass="col-
lg-6"><divclass="insurance-benefits-
img"><imgsrc="allhome/assets/images/in
surance-benefits.webp"alt="insurance-
benefits"><divclass="insurance-
benefits-shape-
1"><imgsrc="allhome/assets/images/insu
rance-benefits-shape-
1.webp"alt="insurance-benefits-shape-
1"></div><divclass="insurance-
benefits-shape-
2"><imgsrc="allhome/assets/images/insu
rance-benefits-shape-
2.webp"alt="insurance-benefits-shape-
1"></div></div></div><divclass="col-
lg-6"><divclass="single-insurance-
benefits-content"><divclass="section-
titleleft-title"><spanclass="top-
title">WhyChooseAllianzPNB?</span><h2>
DiscovertheBenefits</h2><p>AllianzPNBi
scommittedtoprovidingexceptionalservic
esandbenefitstoourcustomers.Hearesom
ereasonswhyyoushouldchooseus:</p></div
><divclass="row"><divclass="col-lg-
6col-sm-6col-md-
6"><divclass="insurance-benefits-
card"><divclass="insurance-benefits-
textd-flexalign-items-
center"><span>01</span><h3>100%SecureS
ervices</h3></div><p>Restassuredwithou
r100%secureservices.Weprioritizethesaf
etyandsecurityofyourinvestmentsandpers
onalinformation.</p></div></div><divcl
ass="col-lg-6col-sm-6col-md-
6"><divclass="insurance-benefits-
card"><divclass="insurance-benefits-
textd-flexalign-items-
center"><span>02</span><h3>AnytimeMone
yBack</h3></div><p>WithAllianzPNB,enjo
ytheflexibilityofanytimemoneybackguara
ntee.Weunderstandyourchangingneedsando
fferhassle-
freerefunds.</p><divclass="insurance-
shape"><imgsrc="allhome/assets/images/
insurance-benefits-shape-
3.webp"alt="insurance"></div></div></d
iv></div></div></div></div></div></div
></section><sectionid="testimonials"><
divclass="testimonials-
areatestimonials-two-areapt-100pb-
70"><divclass="container"><divclass="s
ection-titleleft-
title"><spanclass="top-
title">AllianzPNBTestimonials</span><h
2>WhatOurCustomersSay...</h2></div><di
vclass="testimonials-three-sliderowl-
carouselowl-
theme"><?phpforeach($feedas$feedback):
?><divclass="testimonials-
item"><divclass="testimonials-clientd-
flexalign-items-
center"><imgsrc="allhome/assets/images
/testimonials/def.jpg"alt="testimonial
s"><divclass="testimonials-
text"><h3><?=$feedback['name']?></h3><
/div></div><divclass="testimonials-
cardtestimonials-card-
three"><divclass="quote-
icon"><imgsrc="allhome/assets/images/t
estimonials/qoutes.jpg"class="quote1"a
lt="quote"><divclass="quote-icon-
2"><imgsrc="allhome/assets/images/quot
e-
two.svg"alt="quote"></div></div><p><?=
date('Mj,Y',strtotime($feedback['creat
ed_at']))?></p><br><br><p><?=$feedback
['content']?></p></div></div><?phpendf
oreach;?></div></div><divclass="testim
onials-3-shape"data-cue="zoomIn"data-
duration="2000"><imgsrc="allhome/asset
s/images/testimonials/testimonials-3-
shape.webp"alt="testimonials"></div></
div></section><divclass="team-page-
areapt-50pb-
100"><divclass="container"><divclass="
section-
title"><h2>MeetOurGreatAgent</h2></div
><divclass="row"><?phpforeach($agentsa
s$agent):?><divclass="col-lg-3col-sm-
6col-md-6"><divclass="single-team-
cardteam-page-card"><divclass="team-
img"><imgsrc="<?=isset($agent['agentpr
ofile'])&&!empty($agent['agentprofile'
])?base_url('/uploads/'.$agent['agentp
rofile']):base_url('/uploads/def.png')
?>"alt="team"></div><divclass="single-
team-
content"><h5><?=$agent['lastname']?>,<
?=$agent['firstname']?><?=$agent['midd
lename']?>.</h5><p><?=$agent['email']?
```

```
></p><ulclass="d-flexalign-items-
centerjustify-content-
center"><liclass="list-
inline"><ahref="https://www.facebook.c
om/"target="_blank"><iclass="bxbxl-
facebook"></i></a></li><liclass="list-
inline"><ahref="https://twitter.com/"t
arget="_blank"><iclass="bxbxl-
twitter"></i></a></li><liclass="list-
inline"><ahref="https://www.linkedin.c
om/"target="_blank"><iclass="bxbxl-
linkedin"></i></a></li><liclass="list-
inline"><ahref="https://www.google.com
/"target="_blank"><iclass="bxbxl-
google"></i></a></li></ul></div></div>
</div><?phpendforeach;?></div><?=$page
r-
>links('agent','clientpage')?></div></
div><divclass="free-quote-2-areapt-
100pb-
100"><divclass="container"><divclass="
row"><divclass="col-lg-
6"><divclass="single-free-quote-
formquote-form-
three"><divclass="section-titleleft-
title"><spanclass="top-
title">Hello!</span><h2>Feedback/Comme
nt</h2></div><formmethod="post"action=
"<?=base_url('feedback/saveFeedback')?
>"><?=csrf_field()?><divclass="row"><d
ivclass="col-lg-12"><divclass="form-
group"><inputtype="text"name="name"cla
ss="form-
control"placeholder="Name"value="<?=ol
d('name')?>"required><?phpif(session()
->has('errors')):?><divclass="text-
danger"><?=session('errors.name')?></d
iv><?phpendif?></div></div><divclass="
col-lg-12"><divclass="form-
group"><inputtype="email"name="email"c
lass="form-
control"placeholder="Email"value="<?=o
ld('email')?>"required><?phpif(session
()->has('errors')):?><divclass="text-
danger"><?=session('errors.email')?></
div><?phpendif?></div></div><divclass=
"col-lg-12"><divclass="form-
group"><textareaname="content"class="f
orm-
control"placeholder="Feedback/Comment.
.."required><?=old('content')?></texta
rea><?phpif(session()-
>has('errors')):?><divclass="text-
danger"><?=session('errors.content')?>
</div><?phpendif?></div></div></div><b
uttontype="submit"class="default-
btn">Submit</button></form></div></div
><divclass="col-lg-6"><divclass="free-
quote-image-
three"><imgsrc="allhome/assets/images/
free-quote-three.webp"alt="free-
quote"><divclass="free-quote-three-
shape-
1"><imgsrc="allhome/assets/images/free
-quote-three-shape-1.webp"alt="free-
quote"></div><divclass="free-quote-
three-shape-
2"><imgsrc="allhome/assets/images/free
-quote-three-shape-2.webp"alt="free-
quote"></div><divclass="free-quote-
video-
play"><ahref="https://youtube.com/watc
h?v=9weZzMRejvo"class="popup-
youtube"><iclass="bxbx-
play"></i></a></div></div></div></div>
</div></div><?=view('Home/chop1/footer
')?><?=view('Home/chop1/js')?></body><
/html>
```

AGENT

```
<!doctypehtml><htmllang="en"><?=view('
head')?><body><?=view('Agent/chop/head
er')?><divclass="container-
fluid"><divclass="row"><navid="sidebar
Menu"class="col-md-3col-lg-3d-md-
blocksidebarcollapse"><divclass="posit
ion-stickypy-4px-3sidebar-
sticky"><ulclass="navflex-columnh-
100"><liclass="nav-item"><aclass="nav-
linkactive"aria-
current="page"href="/AgDash"><iclass="
bi-house-fillme-
2"></i>Overview</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/AgForms"><iclass=
"bibi-file-earmark-slidesme-
2"></i>Forms</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/Agsignature"><icl
ass="bibi-penme-
2"></i>Signature</a></li><liclass="nav
-item"><aclass="nav-
link"href="/subagent"><iclass="bi-
personme-
2"></i>SubAgents</a></li><liclass="nav
-item"><aclass="nav-
link"href="/clients"><iclass="bi-
personme-
2"></i>Clients</a></li><liclass="nav-
item"><aclass="nav-
link"href="/agentsched"><iclass="bibi-
check-lgme-
2"></i>Schedule</a></li><liclass="nav-
item"><aclass="nav-
link"href="/cliSched"><iclass="bibi-
check-lgme-
2"></i>Transactions</a></li><liclass="
nav-item"><aclass="nav-
link"href="/mycommi"><iclass="bibi-
wallet2me-
2"></i>MyCommission</a></li><liclass="
nav-itemborder-topmt-autopt-
2"><aclass="nav-
link"href="/logout"><iclass="bi-box-
arrow-leftme-
2"></i>Logout</a></li></ul></div></nav
><mainclass="main-wrappercol-md-9ms-
sm-autopy-4col-lg-9px-md-4border-
start"><divclass="title-groupmb-
3"><h1class="h2mb-
0">Overview</h1><smallclass="text-
muted">Hello<?=isset($agent['username'
])?$agent['username']:''?>,welcomeback
!</small></div><divclass="row"><!--
leftsidecolumns--><divclass="col-lg-
```

```
8"><divclass="custom-blockbg-
white"><divid="barChart"></div></div><
divclass="custom-blockbg-
white"><divid="yearlyComm"></div></div
></div><!--endofleftside--><!--
rightsidecolumns--><divclass="col-lg-
4"><divclass="col-lg-12col-sm-
12"><divclass="row"><divclass="col-lg-
12col-12"><divclass="cardmb-
2"><divclass="card-bodyd-flexflex-
columnalign-items-center"><!--Setmax-
widthandmax-
heighttoconstraintheimagewithinthecard
body--><       -center"><iclass="fasfa-
usersfa-2x"></i><smallclass="d-
blockmt-
2">Clients</small><divclass="fs-4mt-
2"><?=$clients?></div></div></div></di
v><divclass="col-lg-6col-
6"><divclass="cardmb-
2"><divclass="card-bodytext-
center"><iclass="bibi-cash-coinfa-
2x"></i><smallclass="d-blockmt-
1">Commissions</small><divclass="fs-
7mt-
2">₱<?=number_format($totalcommi,2,'.'
,',')?></div></div></div></div></div><
/div></div></div></main></div></div><?
=view('js')?><?=view('Charts/visuals')
?><script>functioncopyToClipboard(){va
rinput=document.getElementById('myInpu
t');input.style.display='block';//Make
inputvisibletemporarilyinput.select();
document.execCommand('copy');input.sty
le.display='none';//Hideinputagainafte
rcopyingalert('Textcopiedtoclipboard:'
+input.value);}</script></body></html>
```

ADMIN

```
<!doctypehtml><htmllang="en"><?=view('
head')?><body><?=view('Admin/chop/head
er')?><divclass="container-
fluid"><divclass="row"><navid="sidebar
Menu"class="col-md-3col-lg-3d-md-
blocksidebarcollapse"><divclass="posit
ion-stickypy-4px-3sidebar-
sticky"><ulclass="navflex-columnh-
100"><liclass="nav-item"><aclass="nav-
linkactive"aria-
current="page"href="/AdDash"><iclass="
bi-house-fillme-
2"></i>Overview</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/usermanagement"><
iclass="fafa-userme-
2"></i>UserManagement</a></li><liclass
="nav-item"><aclass="nav-link"aria-
current="page"href="/Forms"><iclass="b
ibi-file-earmark-slidesme-
2"></i>Forms</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/promotion"><iclas
s="fafa-userme-
2"></i>Promotion</a></li><liclass="nav
-item"><aclass="nav-link"aria-
current="page"href="/confirmation"><ic
lass="bibi-check-lgme-
2"></i>Confirmation</a></li><liclass="
nav-item"><aclass="nav-link"aria-
current="page"href="/sched"><iclass="b
ibi-check-lgme-
2"></i>Schedule</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/ManageAgent"><icl
ass="fasfa-user-tieme-
2"></i>Agents</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/ManageApplicant">
<iclass="fafa-usersme-
2"></i>Applicants</a></li><liclass="na
v-item"><aclass="nav-link"aria-
current="page"href="/map"><iclass="bib
i-mapme-
2"></i>Maps</a></li><hr><liclass="nav-
item"><aclass="nav-
link"href="/plans"><iclass="bibi-
hospitalme-
2"></i>Plans</a></li><liclass="nav-
itemborder-topmt-autopt-2"><
3"><divclass="card-bodytext-
center"><iclass="fasfa-user-tiefa-
2x"></i><smallclass="d-blockmt-
2">Applicants</small><h3class="card-
titlemt-
2"><?=$pendingApplicants?></h3></div><
/div></div></div></div><divclass="col-
lg-12col-sm-12"><divclass="cardmb-
3text-center"><h5class="card-titlemt-
3">Top3Recruiters</h5><?phpforeach($to
pas$topagent):?><divclass="card-
body"><ahref="<?=base_url()?>/agentpro
file/<?=$topagent['agent_token'];?>"><
imgsrc="<?=isset($topagent['agentprofi
le'])&&!empty($topagent['agentprofile'
])?base_url('/uploads/'.$topagent['age
ntprofile']):base_url('/uploads/def.pn
g')?>"class="card-img-topimg-
fluidrounded-circlemx-
auto"alt="AgentImage"style="width:80px
;height:80px;"></a><h5class="card-
titlemt-
2small"><?=$topagent['username']?><!--
<?=$topagent['total_fA']?>--></h5><!--
Addotherrelevantinformationasneeded--
></div><?phpendforeach;?></div></div><
divclass="col-lg-
12"><divclass="cardmb-3text-
center"><h6class="card-titlemt-
3">Top3Awardies</><?phpforeach($top_co
mmias$topagent):?><divclass="card-
body"><ahref="<?=base_url()?>/agentpro
file/<?=$topagent['agent_token'];?>"><
imgsrc="<?=isset($topagent['agentprofi
le'])&&!empty($topagent['agentprofile'
])?base_url('/uploads/'.$topagent['age
ntprofile']):base_url('/uploads/def.pn
g')?>"class="card-img-topimg-
fluidrounded-circlemx-
auto"alt="AgentImage"style="width:80px
;height:80px;"></a><h5class="card-
titlemt-
2small"><?=$topagent['username']?><!--
<?=number_format($topagent['total_comm
issions'])?>--></h5><!--
Addotherrelevantinformationasneeded--
```

```
></div><?phpendforeach;?></div></div><
/div></div></div></main></div></div><?
=view('js');?><?=view('Charts/visuals'
)?></body></html>
```

APPLICANT

```
<!doctypehtml><htmllang="en"><?=view('
head');?><style>.file-
upload{position:relative;display:inlin
e-block;width:100%;max-
width:600px;margin:20pxauto;padding:20
px;border:2pxdashed#ddd;border-
radius:10px;text-
align:center;transition:background-
color0.3s;}.file-
upload.dragover{background-
color:#f0f0f0;}.file-
uploadinput[type="file"]{position:abso
lute;top:0;left:0;width:100%;height:10
0%;opacity:0;cursor:pointer;}</style><
body><?=view('Applicant/chop/header');
?><divclass="container-
fluid"><divclass="row"><navid="sidebar
Menu"class="col-md-3col-lg-3d-md-
blocksidebarcollapse"><divclass="posit
ion-stickypy-4px-3sidebar-
sticky"><ulclass="navflex-columnh-
100"><liclass="nav-item"><aclass="nav-
link"aria-
current="page"href="/AppDash"><iclass=
"bi-house-fillme-
2"></i>Overview</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/AppForms"><iclass
="bibi-file-earmark-slidesme-
2"></i>Forms</a></li><liclass="nav-
item"><aclass="nav-link"aria-
current="page"href="/signature"><iclas
s="bibi-penme-
2"></i>Signature</a></li><liclass="nav
-item"><aclass="nav-linkactive"aria-
current="page"href="/appfiles"><iclass
="bibi-filesme-
2"></i></i>Files</a></li><liclass="nav
-item"><aclass="nav-
link"href="/FA"><iclass="bi-personme-
2"></i>Agents</a></li><liclass="nav-
itemborder-topmt-autopt-
2"><aclass="nav-
link"href="/logout"><iclass="bi-box-
arrow-leftme-
2"></i>Logout</a></li></ul></div></nav
><mainclass="main-wrappercol-md-10ms-
sm-autopy-4col-lg-9px-md-4border-
start"><!--<divclass="title-groupmb-
3"><h1class="h2mb-
0">ApplicationForms</h1></div>--
><divclass="container"><divclass="rowj
ustify-content-centerm-
2"><divclass="col-lg-
12"><divclass="card"><?phpif(session()
-
>getFlashdata('error')):?><divclass="a
lertalert-dangermt-3text-
center"role="alert"><?=session()-
>getFlashdata('error')?></div><?phpend
if;?><?phpif(session()-
>getFlashdata('success')):?><divclass=
"alertalert-successmt-3text-
center"role="alert"><?=session()-
>getFlashdata('success')?></div><?phpe
ndif;?><divclass="card-
body"><?=$userIdExists?'<buttontype="b
utton"class="btnbtn-info"data-bs-
toggle="modal"data-bs-
target="#myModal"><iclass="bibi-
eye"></i></button>':''?><formaction="f
ileuploads"method="post"enctype="multi
part/form-
data"><divclass="row"><divclass="col-
6"><divclass="file-
upload"id="fileUpload1"><pid="uploadTe
xt1"><?=isset($files['file1'])&&$files
['file1']?$files['file1']:'Dragfile'?>
</p><inputtype="file"id="fileInput1"na
me="file1"multiple><buttonclass="btnbt
n-primarymt-
2"type="button"onclick="document.getEl
ementById('fileInput1').click()">Uploa
d</button></div></div><divclass="col-
6"><divclass="file-
upload"id="fileUpload2"><pid="uploadTe
xt2"><?=isset($files['file2'])&&$files
['file2']?$files['file2']:'Dragfile'?>
</p><inputtype="file"id="fileInput2"na
me="file2"multiple><buttonclass="btnbt
n-primarymt-
2"type="button"onclick="document.getEl
ementById('fileInput2').click()">Uploa
d</button></div></div></div><divclass=
"row"><divclass="col-
6"><divclass="file-
upload"id="fileUpload3"><pid="uploadTe
xt3"><?=isset($files['file3'])&&$files
['file3']?$files['file3']:'Dragfile'?>
</p><inputtype="file"id="fileInput3"na
me="file3"multiple><buttonclass="btnbt
n-primarymt-
2"type="button"onclick="document.getEl
ementById('fileInput3').click()">Uploa
d</button></div></div><divclass="col-
6"><divclass="file-
upload"id="fileUpload4"><pid="uploadTe
xt4"><?=isset($files['file4'])&&$files
['file4']?$files['file4']:'Dragfile'?>
</p><inputtype="file"id="fileInput4"na
me="file4"multiple><buttonclass="btnbt
n-primarymt-
2"type="button"onclick="document.getEl
ementById('fileInput4').click()">Uploa
d</button></div></div></div><divclass=
"row"><divclass="col-
6"><divclass="file-
upload"id="fileUpload5"><pid="uploadTe
xt5"><?=isset($files['file5'])&&$files
['file5']?$files['file5']:'Dragfile'?>
</p><inputtype="file"id="fileInput5"na
me="file5"multiple><buttonclass="btnbt
n-primarymt-
2"type="button"onclick="document.getEl
ementById('fileInput5').click()">Uploa
d</button></div></div><divclass="col-
6"><divclass="file-
upload"id="fileUpload6"><pid="uploadTe
xt6"><?=isset($files['file6'])&&$files
['file6']?$files['file6']:'Dragfile'?>
```

```
</p><inputtype="file"id="fileInput6"na
me="file6"multiple><buttonclass="btnbt
n-primarymt-
2"type="button"onclick="document.getEl
ementById('fileInput6').click()">Uploa
d</button></div></div></div><divclass=
"row"><divclass="col-
12"><buttontype="submit"class="btnbtn-
successmt-
3"><?=$userIdExists?'Update':'Save'?><
/button></div></div></form></div></div
></div></div><divclass="modalfade"id="
myModal"tabindex="-1"aria-
labelledby="exampleModalLabel"aria-
hidden="true"><divclass="modal-
dialogmodal-md"><divclass="modal-
content"><divclass="modal-
header"><h5class="modal-
title"id="exampleModalLabel">Files</h5
><buttontype="button"class="btn-
close"data-bs-dismiss="modal"aria-
label="Close"></button></div><divclass
="modal-
body"><divclass="container"><divclass=
"row"><?phpforeach(range(1,6)as$i):?><
?phpif(isset($files["file$i"])&&$files
["file$i"]):?><?php//Determinethefilet
ypeforicon$filePath=base_url('uploads/
files/'.$username.'/'.$files["file$i"]
);$fileExt=pathinfo($files["file$i"],P
ATHINFO_EXTENSION);$iconClass='fa-
file';//Defaulticon//Seticonclassbased
onfileextensionif(in_array($fileExt,['
jpg','jpeg','png','gif'])){$iconClass=
'fa-file-
image';}elseif($fileExt==='pdf'){$icon
Class='fa-file-
pdf';}elseif(in_array($fileExt,['doc',
'docx'])){$iconClass='fa-file-
word';}elseif(in_array($fileExt,['ppt'
,'pptx'])){$iconClass='fa-file-
powerpoint';}?><divclass="col-
6"><divclass="card"><divclass="card-
bodytext-center"><h5class="card-
title">File<?=$i?></h5><pclass="card-
text"><ahref="<?=$filePath?>"target="_
blank"><iclass="fas<?=$iconClass?>fa-
3x"></i></a></p><ahref="<?=$filePath?>
"target="_blank"class="btnbtn-
link"><spanstyle="font-
size:9px;"><?=$files["file$i"]?></span
></a></div></div></div><?phpendif;?><?
phpendforeach;?></div></div></div></di
v></div></div></div></main></div></div
><?=view('js');?><scriptsrc="https://c
ode.jquery.com/jquery-
3.5.1.slim.min.js"></script><scriptsrc
="https://cdn.jsdelivr.net/npm/@popper
js/core@2.5.4/dist/umd/popper.min.js">
</script><scriptsrc="https://stackpath
.bootstrapcdn.com/bootstrap/4.5.2/js/b
ootstrap.min.js"></script><script>docu
ment.querySelectorAll('.file-
upload').forEach((fileUpload,index)=>{
constfileInput=document.getElementById
('fileInput'+(index+1));constuploadTex
t=document.getElementById('uploadText'
+(index+1));fileUpload.addEventListene
r('dragover',(event)=>{event.preventDe
fault();fileUpload.classList.add('drag
over');});fileUpload.addEventListener(
'dragleave',()=>{fileUpload.classList.
remove('dragover');});fileUpload.addEv
entListener('drop',(event)=>{event.pre
ventDefault();fileUpload.classList.rem
ove('dragover');constfiles=event.dataT
ransfer.files;handleFiles(files,upload
Text);});fileInput.addEventListener('c
hange',()=>{constfiles=fileInput.files
;handleFiles(files,uploadText);});func
tionhandleFiles(files,uploadTextElemen
t){constfileNames=Array.from(files).ma
p(file=>file.name).join(',');uploadTex
tElement.textContent=fileNames||'Dragf
iletoupload';}});</script></body></htm
l>
```

CLIENT

```
<!DOCTYPEhtml><html><metaname="viewpor
t"content="width=device-width,initial-
scale=1.0"><linkhref="https://fonts.go
ogleapis.com/css?family=Raleway"rel="s
tylesheet"><?=view('Home/chop1/head')?
><style>*{box-sizing:border-
box;}body{background-
color:#f1f1f1;}#avail{background-
color:#ffffff;margin:100pxauto;font-
family:Raleway;padding:40px;width:70%;
min-width:300px;}h1{text-
align:center;}input{padding:10px;width
:100%;font-size:17px;font-
family:Raleway;border:1pxsolid#aaaaaa;
}/*Markinputboxesthatgetsanerroronvali
dation:*/input.invalid{background-
color:#ffdddd;}/*Hideallstepsbydefault
:*/.tab{display:none;}button{backgroun
d-
color:#04AA6D;color:#ffffff;border:non
e;padding:10px20px;font-
size:17px;font-
family:Raleway;cursor:pointer;}button:
hover{opacity:0.8;}#prevBtn{background
-
color:#bbbbbb;}/*Makecirclesthatindica
tethestepsoftheform:*/.step{height:15p
x;width:15px;margin:02px;background-
color:#bbbbbb;border:none;border-
radius:50%;display:inline-
block;opacity:0.5;}.step.active{opacit
y:1;}/*Markthestepsthatarefinishedandv
alid:*/.step.finish{background-
color:#04AA6D;}</style><body><?=view('
Home/chop1/header')?><formid="avail"ac
tion="<?=base_url()?>avail"method="pos
t"><h1>Register</h1><?phpif(session()-
>has('errors')):?><divclass="alertaler
t-dangermt-3text-
center"role="alert"><?phpforeach(sessi
on('errors')as$error):?><?=$error?><br
><?phpendforeach;?></div><?phpendif;?>
<!--One"tab"foreachstepintheform:--
><divclass="tab">Name:<br><p><inputtyp
e="text"placeholder="Firstname..."onin
put="this.className=''"name="firstname
"></p><br><p><inputtype="text"placehol
der="Middlename..."oninput="this.class
```

```
Name=''"name="middlename"></p><br><p><
inputtype="text"placeholder="Lastname.
.."oninput="this.className=''"name="la
stname"></p></div><divclass="tab">Cont
actInfo:<br><p><inputtype="email"place
holder="E-
mail..."oninput="this.className=''"nam
e="email"></p><br><p><inputtype="text"
placeholder="Phone..."oninput="this.cl
assName=''"name="phone"></p></div><div
class="tab">LoginInfo:<br><p><inputtyp
e="text"placeholder="Username..."oninp
ut="this.className=''"name="username">
</p><br><p><inputtype="password"placeh
older="Password..."oninput="this.class
Name=''"name="password"></p><br><p><in
puttype="password"placeholder="Confirm
Password..."oninput="this.className=''
"name="confirmpassword"></p></div><br>
<divstyle="overflow:auto;"><divstyle="
float:right;"><buttontype="button"id="
prevBtn"onclick="nextPrev(-
1)">Previous</button><buttontype="butt
on"id="nextBtn"onclick="nextPrev(1)">N
ext</button></div></div><!--
Circleswhichindicatesthestepsoftheform
:--><divstyle="text-
align:center;margin-
top:40px;"><spanclass="step"></span><s
panclass="step"></span><spanclass="ste
p"></span></div></form><script>varcurr
entTab=0;//Currenttabissettobethefirst
tab(0)showTab(currentTab);//Displaythe
currenttabfunctionshowTab(n){//Thisfun
ctionwilldisplaythespecifiedtabofthefo
rm...varx=document.getElementsByClassN
ame("tab");x[n].style.display="block";
//...andfixthePrevious/Nextbuttons:if(
n==0){document.getElementById("prevBtn
").style.display="none";}else{document
.getElementById("prevBtn").style.displ
ay="inline";}if(n==(x.length-
1)){document.getElementById("nextBtn")
.innerHTML="Submit";}else{document.get
ElementById("nextBtn").innerHTML="Next
";}//...andrunafunctionthatwilldisplay
thecorrectstepindicator:fixStepIndicat
or(n)}functionnextPrev(n){//Thisfuncti
onwillfigureoutwhichtabtodisplayvarx=d
ocument.getElementsByClassName("tab");
//Exitthefunctionifanyfieldinthecurren
ttabisinvalid:if(n==1&&!validateForm()
)returnfalse;//Hidethecurrenttab:x[cur
rentTab].style.display="none";//Increa
seordecreasethecurrenttabby1:currentTa
b=currentTab+n;//ifyouhavereachedtheen
doftheform...if(currentTab>=x.length){
//...theformgetssubmitted
"active"classofallsteps...vari,x=docum
ent.getElementsByClassName("step");for
(i=0;i<x.length;i++){x[i].className=x[
i].className.replace("active","");}//.
..andaddsthe"active"classonthecurrents
tep:x[n].className+="active";}</script
></body></html><?=view('Home/chop1/js'
)?>
```

ALERT CONTROLLER

```
<?phpnamespace App\Controllers;use
App\Controllers\BaseController;use
App\Models\NotifModel;class
AlertController extends BaseController{
private $notif; public function
__construct() { $this->notif = new
NotifModel(); } public function
getAlerts() { $session = session();
$userId = $session->get('id'); $role =
$session->get('role'); $alerts = $this-
>notif->where('role', $role)-
>where('user_id', $userId)-
>orderBy('created_at', 'DESC')-
>findAll(); return $this->response-
>setJSON($alerts); } public function
markAsRead($id) { $this->notif-
>delete($id); return $this->response-
>setJSON(['status' => 'success']); }
public function clearAll() { $session =
session(); $userId = $session-
>get('id'); $role = $session-
>get('role'); $this->notif-
>where('user_id', $userId)-
>where('role', $role)->delete(); return
$this->response->setJSON(['status' =>
'success']); }}
```

AUTH CONTROLLER

```
<?phpnamespace App\Controllers;use
App\Controllers\BaseController;use
App\Models\UserModel;use
App\Models\AdminModel;use
App\Models\AgentModel;use
App\Models\ApplicantModel;use
App\Models\ClientModel;class
AuthController extends BaseController{
private $user; private $admin; private
$agent; private $applicant; private
$client; public function __construct()
{ $this->user = new UserModel(); $this-
>admin = new AdminModel(); $this->agent
= new AgentModel(); $this->applicant =
new ApplicantModel(); $this->client =
new ClientModel(); } public function
login() { return view('login'); } public
function authlog() { $username = $this-
>request->getPost('username');
$password = $this->request-
>getPost('password'); $user = $this-
>user->where('username', $username)-
>first(); if (!$user ||
!password_verify($password,
$user['password'])) { return
redirect()->back()->with('error',
'Invalid credentials'); } $session =
session(); $session->set([ 'id' =>
$user['id'], 'username' =>
$user['username'], 'role' =>
$user['role'], 'token' =>
$user['token'] ]); switch
($user['role']) { case 'admin': return
redirect()->to('/AdDash'); case
'agent': return redirect()-
>to('/AgDash'); case 'applicant':
return redirect()->to('/AppDash'); case
'client': return redirect()-
```

```
>to('/ClientPage'); default: return
redirect()->to('/'); } } public
function logout() { session()-
>destroy(); return redirect()-
>to('/login'); }}
```

DASHBOARD CONTROLLER

```
<?phpnamespace App\Controllers;use
App\Controllers\BaseController;use
App\Models\AdminModel;use
App\Models\AgentModel;use
App\Models\ApplicantModel;use
App\Models\ClientModel;use
App\Models\CommiModel;use
App\Controllers\NotifController;use
App\Controllers\ReportsController;clas
s DashboardController extends
BaseController{ private $admin; private
$agent; private $applicant; private
$client; private $commi; private
$notifcont; private $reportscont;
public function __construct() { $this-
>admin = new AdminModel(); $this->agent
= new AgentModel(); $this->applicant =
new ApplicantModel(); $this->client =
new ClientModel(); $this->commi = new
CommiModel(); $this->notifcont = new
NotifController(); $this->reportscont =
new ReportsController(); } public
function index() { $data = array_merge(
$this->notifcont->notification(),
$this->reportscont->usersreportdata()
); $data['totalAgents'] = count($this-
>agent->findAll());
$data['totalApplicants'] = count($this-
>applicant->findAll());
$data['pendingApplicants'] = $this-
>applicant->where('status', 'pending')-
>countAllResults();
$data['totalClients'] = $this->client-
>countAllResults(); return
view('Admin/AdDash', $data); }}
```

E-COM MANAGEMENT CONTROLLER

```
<?phpnamespace App\Controllers;use
App\Controllers\BaseController;use
App\Models\PlanModel;use
App\Models\ClientPlanModel;use
App\Models\ClientModel;use
App\Controllers\NotifController;class
EComManagementController extends
BaseController{ private $plan; private
$client_plan; private $client; private
$notifcont; public function
__construct() { $this->plan = new
PlanModel(); $this->client_plan = new
ClientPlanModel(); $this->client = new
ClientModel(); $this->notifcont = new
NotifController(); } public function
plans() { $data = $this->notifcont-
>notification(); $data['plans'] =
$this->plan->findAll(); return
view('Admin/plans', $data); } public
function newplan() { $data = [
'plan_name' => $this->request-
>getPost('plan_name'),
'brief_description' => $this->request-
>getPost('brief_description'),
'description' => $this->request-
>getPost('description'), 'price' =>
$this->request->getPost('price'),
'com_percentage' => $this->request-
>getPost('com_percentage'), 'coverage'
=> $this->request->getPost('coverage'),
'token' => bin2hex(random_bytes(25)),
]; if ($imageFile = $this->request-
>getFile('image')) { if ($imageFile-
>isValid()) { $imageName = $imageFile-
>getRandomName(); $imageFile-
>move(FCPATH . 'uploads/plans/',
$imageName); $data['image'] =
$imageName; } } $this->plan-
>save($data); return redirect()-
>back()->with('success', 'Plan Added
Successfully!'); } public function
newplanUpdate($token) { $data = [
'plan_name' => $this->request-
>getPost('plan_name'),
'brief_description' => $this->request-
>getPost('brief_description'),
'description' => $this->request-
>getPost('description'), 'price' =>
$this->request->getPost('price'),
'com_percentage' => $this->request-
>getPost('com_percentage'), 'coverage'
=> $this->request->getPost('coverage'),
]; $this->plan->where('token', $token)-
>set($data)->update(); return
redirect()->back()->with('success',
'Plan Updated Successfully!'); }}
```

FORECAST CONTROLLER

```
<?phpnamespace App\Controllers;use
App\Controllers\BaseController;use
App\Models\CommiModel;use
App\Models\AgentModel;use
App\Models\ApplicantModel;class
ForecastController extends
BaseController{ private $commi; private
$agent; private $applicant; public
function __construct() { $this->commi =
new CommiModel(); $this->agent = new
AgentModel(); $this->applicant = new
ApplicantModel(); } public function
predictMonthlyAgents() { $agents =
$this->agent-
>select("MONTH(created_at) as month,
YEAR(created_at) as year, COUNT(*) as
count")->groupBy("YEAR(created_at),
MONTH(created_at)")->orderBy("year,
month")->findAll(); return $this-
>response->setJSON($agents); } public
function predictMonthlyApplicants() {
$applicants = $this->applicant-
>select("MONTH(created_at) as month,
YEAR(created_at) as year, COUNT(*) as
count")->groupBy("YEAR(created_at),
MONTH(created_at)")->orderBy("year,
month")->findAll(); return $this-
>response->setJSON($applicants); }
public function
predictMonthlyCommissions() {
$commissions = $this->commi-
>select("MONTH(created_at) as month,
YEAR(created_at) as year, SUM(commi) as
total")->groupBy("YEAR(created_at),
```

```
MONTH(created_at)")->orderBy("year, month")->findAll(); return $this->response->setJSON($commissions); }}
```

GCASH CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\ClientPlanModel;use App\Models\ClientModel;class GcashController extends BaseController{ private $client_plan; private $client; public function __construct() { $this->client_plan = new ClientPlanModel(); $this->client = new ClientModel(); } public function gcashPayment() { $token = $this->request->getPost('token'); $plan = $this->client_plan->where('token', $token)->first(); if (!$plan) { return redirect()->back()->with('error', 'Plan not found'); } return view('gcash/payment', ['plan' => $plan]); } public function processPayment() { $data = [ 'token' => $this->request->getPost('token'), 'status' => 'paid', 'receipt' => $this->request->getPost('receipt'), ]; $this->client_plan->where('token', $data['token'])->set($data)->update(); return redirect()->back()->with('success', 'Payment processed successfully'); }}
```

INVENTORY CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\PlanModel;use App\Models\ClientPlanModel;class InventoryController extends BaseController{ private $plan; private $client_plan; public function __construct() { $this->plan = new PlanModel(); $this->client_plan = new ClientPlanModel(); } public function index() { $plans = $this->plan->findAll(); $data['plans'] = $plans; foreach ($plans as &$plan) { $plan['sold'] = $this->client_plan->where('plan', $plan['token'])->countAllResults(); } return view('Admin/inventory', $data); } public function planDetails($token) { $data['plan'] = $this->plan->where('token', $token)->first(); $data['clients'] = $this->client_plan->select('client_plan.*, client.username as client_name')->join('client', 'client.client_id = client_plan.client_id')->where('client_plan.plan', $token)->findAll(); return view('Admin/planDetails', $data); }}
```

LOG CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\UserModel;class LogController extends BaseController{ private $user; public function __construct() { $this->user = new UserModel(); } public function getlogs() { $users = $this->user->select('username, role, time_log, accountStatus')->orderBy('time_log', 'DESC')->findAll(); return $this->response->setJSON($users); } public function updateTimeLog($userId) { $timeLog = date('Y-m-d H:i:s'); $this->user->set(['time_log' => $timeLog])->where('id', $userId)->update(); return $this->response->setJSON(['status' => 'success', 'time_log' => $timeLog]); }}
```

MONTHLY SUMMARY CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\CommiModel;use App\Models\AgentModel;use App\Models\ApplicantModel;use App\Models\ClientModel;class MonthlySummaryController extends BaseController{ private $commi; private $agent; private $applicant; private $client; public function __construct() { $this->commi = new CommiModel(); $this->agent = new AgentModel(); $this->applicant = new ApplicantModel(); $this->client = new ClientModel(); } public function monthlySummary() { $month = $this->request->getGet('month') ?? date('m'); $year = $this->request->getGet('year') ?? date('Y'); $data['totalCommissions'] = $this->commi->select('SUM(commi) as total')->where('MONTH(created_at)', $month)->where('YEAR(created_at)', $year)->get()->getRowArray()['total'] ?? 0; $data['newAgents'] = $this->agent->where('MONTH(created_at)', $month)->where('YEAR(created_at)', $year)->countAllResults(); $data['newApplicants'] = $this->applicant->where('MONTH(created_at)', $month)->where('YEAR(created_at)', $year)->countAllResults(); $data['newClients'] = $this->client->where('MONTH(created_at)', $month)->where('YEAR(created_at)', $year)->countAllResults(); return $this->response->setJSON($data); }}
```

ORDER CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\ClientPlanModel;use App\Models\ClientModel;use App\Models\PlanModel;use App\Models\CommiModel;class OrderController extends BaseController{ private $client_plan; private $client; private $plan; private $commi; public function __construct() { $this->client_plan = new ClientPlanModel(); $this->client = new ClientModel(); $this->plan = new PlanModel(); $this->commi = new CommiModel(); } public function getOrders() { $orders = $this-
```

```
>client_plan->select('client_plan.*, client.username as client_name, plan.plan_name')->join('client', 'client.client_id = client_plan.client_id')->join('plan', 'plan.token = client_plan.plan')->orderBy('client_plan.created_at', 'DESC')->findAll(); return $this->response->setJSON($orders); } public function getOrdersByAgent($agentId) { $orders = $this->client_plan->select('client_plan.*, client.username as client_name, plan.plan_name')->join('client', 'client.client_id = client_plan.client_id')->join('plan', 'plan.token = client_plan.plan')->where('client_plan.agent', $agentId)->orderBy('client_plan.created_at', 'DESC')->findAll(); return $this->response->setJSON($orders); }}
```

PRODUCT CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\PlanModel;class ProductController extends BaseController{ private $plan; public function __construct() { $this->plan = new PlanModel(); } public function getProducts() { $plans = $this->plan->findAll(); return $this->response->setJSON($plans); } public function getProduct($token) { $plan = $this->plan->where('token', $token)->first(); if (!$plan) { return $this->response->setJSON(['error' => 'Product not found'], 404); } return $this->response->setJSON($plan); } public function ServiceDescription($token) { $data['plan'] = $this->plan->where('token', $token)->first(); $data['plans'] = $this->plan->findAll(); return view('Home/ServiceDescription', $data); }}
```

REORDER CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\ClientPlanModel;use App\Models\PlanModel;class ReorderController extends BaseController{ private $client_plan; private $plan; public function __construct() { $this->client_plan = new ClientPlanModel(); $this->plan = new PlanModel(); } public function renewPlan($token) { $existingPlan = $this->client_plan->where('token', $token)->first(); if (!$existingPlan) { return redirect()->back()->with('error', 'Plan not found'); } $planDetails = $this->plan->where('token', $existingPlan['plan'])->first(); $addDuration = '+1 year'; $currentDate = new \DateTime(); $currentDate->modify($addDuration); $newDueDate = $currentDate->format('Y-m-d'); $this->client_plan->set(['duedate' => $newDueDate, 'status' => 'paid'])->where('token', $token)->update(); return redirect()->back()->with('success', 'Plan renewed successfully'); }}
```

REPORTS CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\AgentModel;use App\Models\ApplicantModel;use App\Models\ClientModel;use App\Models\CommiModel;class ReportsController extends BaseController{ private $agent; private $applicant; private $client; private $commi; public function __construct() { $this->agent = new AgentModel(); $this->applicant = new ApplicantModel(); $this->client = new ClientModel(); $this->commi = new CommiModel(); } public function reports() { $data = $this->usersreportdata(); return view('Admin/reports', $data); } public function usersreportdata() { $data['reportAgents'] = $this->agent->countAllResults(); $data['reportApplicants'] = $this->applicant->countAllResults(); $data['reportClients'] = $this->client->countAllResults(); $data['totalRevenue'] = $this->commi->select('SUM(commi) as total')->get()->getRowArray()['total'] ?? 0; return $data; } public function generateReport($month, $year) { $data['commissions'] = $this->commi->select('commissions.*, agent.username as agent_name')->join('agent', 'agent.agent_id = commissions.agent_id')->where('MONTH(commissions.created_at)', $month)->where('YEAR(commissions.created_at)', $year)->findAll(); $data['month'] = $month; $data['year'] = $year; return view('Admin/reportPdf', $data); }}
```

SALES CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\CommiModel;use App\Models\ClientPlanModel;class SalesController extends BaseController{ private $commi; private $client_plan; public function __construct() { $this->commi = new CommiModel(); $this->client_plan = new ClientPlanModel(); } public function getoverallMonthlysales() { $sales = $this->client_plan->select("MONTH(created_at) as month, YEAR(created_at) as year, SUM(commission) as total")->groupBy("YEAR(created_at), MONTH(created_at)")->orderBy("year, month")->findAll(); return $this-
```

```
>response->setJSON($sales); } public function getoverallYearlysales() { $sales = $this->client_plan->select("YEAR(created_at) as year, SUM(commission) as total")->groupBy("YEAR(created_at)")->orderBy("year")->findAll(); return $this->response->setJSON($sales); } public function predictNext12MonthsSales() { $sales = $this->client_plan->select("MONTH(created_at) as month, YEAR(created_at) as year, SUM(commission) as total")->groupBy("YEAR(created_at), MONTH(created_at)")->orderBy("year DESC, month DESC")->limit(24)->findAll(); return $this->response->setJSON(array_reverse($sales)); }}
```

STOCK OVERVIEW CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\PlanModel;use App\Models\ClientPlanModel;class StockOverviewController extends BaseController{ private $plan; private $client_plan; public function __construct() { $this->plan = new PlanModel(); $this->client_plan = new ClientPlanModel(); } public function stockOverview() { $plans = $this->plan->findAll(); $data = []; foreach ($plans as $plan) { $sold = $this->client_plan->where('plan', $plan['token'])->countAllResults(); $active = $this->client_plan->where('plan', $plan['token'])->where('status', 'paid')->countAllResults(); $data[] = [ 'plan_name' => $plan['plan_name'], 'token' => $plan['token'], 'price' => $plan['price'], 'total_sold' => $sold, 'active_subscriptions' => $active, ]; } return $this->response->setJSON($data); }}
```

SUPPLIER CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\AgentModel;use App\Models\UserModel;class SupplierController extends BaseController{ private $agent; private $user; public function __construct() { $this->agent = new AgentModel(); $this->user = new UserModel(); } public function getSuppliers() { $agents = $this->agent->select('agent_id, username, email, number, AgentCode, FA')->findAll(); return $this->response->setJSON($agents); } public function getSupplier($agentId) { $agent = $this->agent->where('agent_id', $agentId)->first(); if (!$agent) { return $this->response->setJSON(['error' => 'Agent not found'], 404); } return $this->response->setJSON($agent); } public function updateSupplier($agentId) { $data = [ 'username' => $this->request->getPost('username'), 'email' => $this->request->getPost('email'), 'number' => $this->request->getPost('number'), ]; $this->agent->set($data)->where('agent_id', $agentId)->update(); return redirect()->back()->with('success', 'Agent updated successfully'); }}
```

TEMPLATE CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;class TemplateController extends BaseController{ public function head() { return view('head'); } public function js() { return view('js'); } public function adminHeader() { return view('Admin/chop/header'); } public function agentHeader() { return view('Agent/chop/header'); } public function applicantHeader() { return view('Applicant/chop/header'); } public function homeHeader() { return view('Home/chop1/header'); } public function homeFooter() { return view('Home/chop1/footer'); } public function homeJs() { return view('Home/chop1/js'); }}
```

TREND CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\CommiModel;use App\Models\AgentModel;use App\Models\ApplicantModel;class TrendController extends BaseController{ private $commi; private $agent; private $applicant; public function __construct() { $this->commi = new CommiModel(); $this->agent = new AgentModel(); $this->applicant = new ApplicantModel(); } public function getMonthlyCommissions() { $data = $this->commi->select("MONTH(created_at) as month, YEAR(created_at) as year, SUM(commi) as total, agent_id")->where('YEAR(created_at)', date('Y'))->groupBy("agent_id, YEAR(created_at), MONTH(created_at)")->orderBy("year, month")->findAll(); return $this->response->setJSON($data); } public function getYearlyCommissions() { $data = $this->commi->select("YEAR(created_at) as year, SUM(commi) as total, agent_id")->groupBy("agent_id, YEAR(created_at)")->orderBy("year")->findAll(); return $this->response->setJSON($data); } public function getoverallMonthlyCommissions() { $data = $this->commi->select("MONTH(created_at) as month, YEAR(created_at) as year, SUM(commi) as total")->groupBy("YEAR(created_at), MONTH(created_at)")->orderBy("year,
```

```
month")->findAll(); return $this->response->setJSON($data); } public function getoverallYearlyCommissions() { $data = $this->commi->select("YEAR(created_at) as year, SUM(commi) as total")->groupBy("YEAR(created_at)")->orderBy("year")->findAll(); return $this->response->setJSON($data); }}
```

USER CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\UserModel;use App\Models\AdminModel;use App\Models\AgentModel;use App\Models\ApplicantModel;use App\Models\ClientModel;class UserController extends BaseController{ private $user; private $admin; private $agent; private $applicant; private $client; public function __construct() { $this->user = new UserModel(); $this->admin = new AdminModel(); $this->agent = new AgentModel(); $this->applicant = new ApplicantModel(); $this->client = new ClientModel(); } public function usermanagement() { $session = session(); $userId = $session->get('id'); $users = $this->user->where(['role !=' => 'admin'])->paginate(10, 'group1'); $data['users'] = $users; $data['pager'] = $this->user->pager; $data['admin'] = $this->admin->where('admin_id', $userId)->first(); return view('Admin/usermanagement', $data); } public function newuser() { $data = [ 'username' => $this->request->getPost('username'), 'email' => $this->request->getPost('email'), 'password' => password_hash($this->request->getPost('password'), PASSWORD_DEFAULT), 'role' => $this->request->getPost('role'), 'token' => bin2hex(random_bytes(25)), 'status' => 'active', ]; $this->user->save($data); return redirect()->back()->with('success', 'User created successfully'); } public function upuser($id) { $data = [ 'username' => $this->request->getPost('username'), 'email' => $this->request->getPost('email'), 'role' => $this->request->getPost('role'), ]; $this->user->set($data)->where('id', $id)->update(); return redirect()->back()->with('success', 'User updated successfully'); }}
```

VOUCHER CONTROLLER

```
<?phpnamespace App\Controllers;use App\Controllers\BaseController;use App\Models\ClientPlanModel;use App\Models\ClientModel;class VoucherController extends BaseController{ private $client_plan; private $client; public function __construct() { $this->client_plan = new ClientPlanModel(); $this->client = new ClientModel(); } public function applyVoucher() { $voucherCode = $this->request->getPost('voucher_code'); $planToken = $this->request->getPost('plan_token'); // Validate voucher code $validVouchers = ['PROMO10', 'SAVE20', 'DISCOUNT15']; if (!in_array($voucherCode, $validVouchers)) { return $this->response->setJSON(['status' => 'error', 'message' => 'Invalid voucher code']); } $discounts = ['PROMO10' => 10, 'SAVE20' => 20, 'DISCOUNT15' => 15]; $discount = $discounts[$voucherCode]; return $this->response->setJSON(['status' => 'success', 'discount' => $discount, 'message' => "Voucher applied! {$discount}% discount"]); } public function getVouchers() { $vouchers = [ ['code' => 'PROMO10', 'discount' => 10, 'description' => '10% off on any plan'], ['code' => 'SAVE20', 'discount' => 20, 'description' => '20% off on annual plans'], ['code' => 'DISCOUNT15', 'discount' => 15, 'description' => '15% off for new clients'], ]; return $this->response->setJSON($vouchers); }}
```

APPENDIX H

Data Elements/Dictionary

Field for Users				
Field	**Data Type**	**Size**	**Default**	**Description**
id	Int		NOT NULL	Primary key, auto-increment
username	Varchar	50	NULL	Index for username
password_hash	Text		NOT NULL	User password (hashed)
role	Text		NOT NULL	User role in the system
email	Varchar	100	NOT NULL	User email address
profile_image	Text		NULL	Profile image file path
status	Text		NULL	Account status
token	Varchar	50	NULL	Index for authentication
time_log	Timestamp		CURRENT_TIMESTAMP	Time log entry
created_at	Timestamp		CURRENT_TIMESTAMP	Record creation timestamp
updated_at	Timestamp		CURRENT_TIMESTAMP	Record last updated timestamp

Field for Products				
Field	**Data Type**	**Size**	**Default**	**Description**
product_id	Int		NOT NULL	Primary key, auto-increment
product_name	Text		NOT NULL	Name of the product
product_sku	Varchar	255	NULL	Stock Keeping Unit identifier
product_brand	Varchar	255	NULL	Brand name of the product
product_category	Text		NULL	Product category classification
total_bundle_qty	Double	12,2	NOT NULL	Total quantity available per bundle
quantity_per_bundle	Int		NOT NULL	Number of units per bundle
wholesale_price	Double	12,2	NOT NULL	Wholesale price per unit
unit_price	Double	12,2	NOT NULL	Retail price per unit
unit_amount	Double		NOT NULL	Amount per unit of measurement
unit	Varchar	255	NULL	Unit of measurement (e.g., pcs, box, case)
packaging_type	Varchar	255	NULL	Type of packaging

				for the product
availability	Varchar	255	NULL	Availability status of the product
image_path	Text		NULL	File path to the product image

Field for Inventory List				
Field	**Data Type**	**Size**	**Default**	**Description**
id	Int		NOT NULL	Primary key, auto-increment
product_id	Int		NOT NULL	Foreign key referencing rm_products
supplier_id	Int		NOT NULL	Foreign key referencing rm_suppliers
bundle_qty	Int		NOT NULL	Number of bundles currently in stock
unit_qty	Int		NOT NULL	Number of individual units in stock
batch_number	Int		NULL	Batch number for inventory lot tracking
expiry_date	Date		NULL	Expiry date of the inventory batch
storage_location	Varchar	100	NULL	Physical storage

				location in warehouse
source	Enum		NOT NULL	Source: purchase_order or received
created_at	Timestamp		CURRENT_TIMESTAMP	Record creation timestamp
updated_at	Timestamp		CURRENT_TIMESTAMP	Record last updated timestamp
status	Enum		NOT NULL	Inventory status: active or archived

Field for Suppliers				
Field	**Data Type**	**Size**	**Default**	**Description**
id	Int		NOT NULL	Primary key, auto-increment
supplier_name	Varchar	100	NOT NULL	Full name of the supplier
contact_person	Varchar	255	NULL	Name of the contact person
email	Text		NULL	Email address of the supplier
phone_number	Bigint		NOT NULL	Phone number of the supplier
address	Varchar	255	NULL	Physical address of the supplier

created_at	Timestamp		CURRENT_TIMESTAMP	Record creation timestamp
updated_at	Timestamp		CURRENT_TIMESTAMP	Record last updated timestamp

Field for Orders				
Field	**Data Type**	**Size**	**Default**	**Description**
id	Int		NOT NULL	Primary key, auto-increment
user_id	Int		NOT NULL	Foreign key referencing rm_users
address_id	Int		NOT NULL	Foreign key referencing billing address
shipping_option	Enum		NOT NULL	Shipping: door-to-door or self-pickup
payment_method	Enum		NOT NULL	Payment: cod, gcash, maya, card, xendit
seller_message	Text		NULL	Optional message from the seller
subtotal	Decimal	10,2	NOT NULL	Order subtotal before fees and discounts
shipping_fee	Decimal	10,2	NOT NULL	Shipping fee applied

				to the order
discount	Decimal	10,2	NOT NULL	Discount amount applied to the order
total_amount	Decimal	10,2	NOT NULL	Final total amount of the order
voucher_code	Varchar	50	NULL	Voucher code applied to the order
status	Enum		NOT NULL	Status: pending, processing, shipped, delivered, cancelled, completed
updated_at	Datetime		NOT NULL	Record last updated timestamp

APPENDIX I

Capstone Evaluation Form

RM'S: AN AI-INTEGRATED INVENTORY AND E-COMMERCE MANAGEMENT SYSTEM WITH REAL-TIME DEMAND FORECASTING AND ENHANCED DISTRIBUTION FOR REMHART

Direction: Test the performance and acceptability of the Project based on the rating below using ISO 25010 Software Metrics Evaluation and Unified Theory of Acceptance and Use of Technology (UTAUT) Model.

Name: ___________________________________ Date: _______

Stakeholder Type:
___ Admin
___ Employee
___ Customer
___ IT Expert
___ Others (specify):

Age:
___ 18-25 yrs old
___ 26-35 yrs old
___ 36-45 yrs old
___ 46-55 yrs old
___ 56 and above

Gender:
___ Female ___ Male

Rate the following question by checking the number that correspond your answer. Use the legend below.

Rating
4-Strongly Agree
3-Agree
2-Disagree
1-Strongly Disagree

1. **ISO 25010 Software Metrics**

1 Functional Suitability		**4**	**3**	**2**	**1**
1.1	The functions of system covers all the specified task and user objectives.				
1.2	The functions of system provides the correct results with the needed degree of precision.				
1.3	The functions of system facilitate the accomplishment of specified tasks and objectives.				

2	**Performance Efficiency**	**4**	**3**	**2**	**1**
2.1	The functions of system response and process the output on time to meet the user requirements.				
2.2	The resources used by the system, when performing its functions, meet requirements.				
2.3	The maximum limits of the product or system, parameter meet requirements.				

3	**Usability**	**4**	**3**	**2**	**1**
3.1	The system is appropriate for my needs.				
3.2	The use of system is effective and efficient in emergency situations.				
3.3	The system is easy to operate, control and appropriate to use.				
3.4	The system protects users against making errors.				
3.5	The user interface of the system enables pleasing and satisfying interaction for the user.				
3.6	The system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.				

4	**Reliability**	**4**	**3**	**2**	**1**
4.1	The system meets needs for reliability under normal operation.				
4.2	The system is operational and accessible when required for use.				
4.3	The system operates as intended despite the presence of hardware or software faults.				

4.4	When an interruption or a failure happened, the system can recover the data on the directly affected and re-establish the desired state of the system.				

2. **Unified Theory of Acceptance and Use of Technology (UTAUT) Model**

1 Performance Expectancy		**4**	**3**	**2**	**1**
1.1	Using the system, my job would increase my productivity.				
1.2	Using the system would enhance my effectiveness on the job.				
1.3	Using the system would make it easier to do my job.				
1.4	I would find the system useful in my job.				
1.5	Using the system enables me to accomplish tasks more quickly.				
1.6	Using the system improves the quality of work I do.				
1.7	Using the system makes it easier to do my job.				
1.8	Using the system enhances my effectiveness on the job.				
1.9	If I will use the system I will increase my effectiveness on the job.				
1.10	If I will use the system I will spend less time on routing job tasks.				

2 Effort Expectancy		**4**	**3**	**2**	**1**
2.1	Learning to operate the system would be easy for me.				
2.2	I would find it easy to get the system to do what I want it to do.				

2.3	My interaction with the system is clear and understandable.				
2.4	My interaction with the system would be clear and understandable.				
2.5	I would find the system to be flexible to interact with.				
2.6	Using the system don't take too much time from my normal duties.				
2.7	Working with the system is so simple, it is not difficult to understand what is going on.				
2.8	Using the system involves lesser time doing mechanical operations (e.g., data input).				
2.9	My interaction with the system is clear and understandable.				
2.10	I believe that it is easy to get the system to do what I want it to do.				

3 Facilitating Conditions		**4**	**3**	**2**	**1**
3.1	I have control over using the system.				
3.2	I have the resources necessary to use the system.				
3.3	I have the knowledge necessary to use the system.				
3.4	Given the resources, opportunities and knowledge it takes to use the system, it would be easy for me to use the system.				
3.5	Guidance was available to me in the selection of the system.				
3.6	Specialized instruction concerning the system was available to me.				
3.7	A specific person (or group) is available for assistance with system difficulties.				

3.8	Using the system is compatible with all aspects of my work.				
3.9	I think that using the system fits well with the way I like to work.				
3.10	Using the system fits into my work style.				

4 Behavioral Intention		**4**	**3**	**2**	**1**
4.1	Using the system is a good idea.				
4.2	Using the system is a wise idea.				
4.3	I like the idea of using the system.				
4.4	I find using the system to be enjoyable.				
4.5	The actual process of using the system is pleasant.				
4.6	I have fun using the system.				
4.7	The system makes work more interesting.				
4.8	Working with the system is fun.				
4.9	I like working with the system.				
4.10	I look forward to those aspects of my job that require me to use the system.				

Comments and Recommendations:

____

____

____

____

____

Signature

APPENDIX J

Pictures During Development

Photos were taken during the development and deployment of RM'S System.

Researchers successfully developed and deployed the system to the client. Demonstrated how the system works and regularly checks them for user feedback.

APPENDIX K

Pictures During Final Defense

Photos were taken during and after the researcher's defense of the system.

APPENDIX L

Research/Capstone Project Waiver

We, **RHAJ S. MADRONA, MARIA HANAH G. MENDOZA and GWEN M. BASCONCILLO** of legal ages, the sole proponent of the Research/Capstone Project entitled **"RM'S: AN AI-INTEGRATED INVENTORY AND E-COMMERCE MANAGEMENT SYSTEM FOR REAL-TIME DEMAND FORECASTING AND ENHANCED DISTRIBUTION AT REMHART"**, hereby give our consent for our project to be submitted for presentation, publication, and potential funding opportunities. We authorize our Adviser, Course Facilitator, and the College of Computer Studies Research Unit to handle this submission on our behalf.

If our project is selected for presentation or publication, we agree to work closely with our Adviser and the Research Unit as needed. If we are unable to attend any event where our work is featured, we authorize our Adviser, Course Facilitator, or Research Unit Head to represent our project and present on our behalf. If revisions are needed to meet conference or journal requirements, we allow our Adviser, Course Facilitator, and/or Research Unit Head to make necessary edits.

If significant changes are made, we agree that these individuals may be credited as co-authors, in addition to us as the original project authors. We understand that these actions and decisions are intended to benefit us, our academic department, and our university community.

RHAJ S. MADRONA
Research Proponent
Date:___________

MARIA HANAH G. MENDOZA
Research Proponent
Date:___________

GWEN M. BASCONCILLO
Research Proponent
Date:___________

APPENDIX M

Consent of Co-Authorship Form

APPENDIX N

IMRAD Format

RM'S: AN AI-INTEGRATED INVENTORY AND E-COMMERCE MANAGEMENT SYSTEM FOR REAL-TIME DEMAND FORECASTING AND ENHANCED DISTRIBUTION AT REMHART

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Abstract -- *This research explains how the researchers made RM'S, an AI-Integrated Inventory and E-Commerce Management System for Real-Time Demand Forecasting and Enhanced Distribution at REMHART: A Unilever Products Distributor. The goal of this system is to help the company manage inventory more accurately and efficiently by replacing manual stock monitoring processes with an AI-powered digital solution. The system includes features for AI-driven demand forecasting using Facebook Prophet, real-time inventory monitoring, automated replenishment suggestions, supplier management, geographic sales analysis, and integrated e-commerce with GCash payment support. The researchers used the Agile Model of the Software Development Life Cycle (SDLC) to create the system. This model helped ensure that the system met the needs of all users including administrators, staff, and customers. The team focused on making the system easy to use, reliable, fast, and secure. To test the system, the researchers used two methods: ISO 25010 evaluation and the Unified Theory of Acceptance and Use of Technology (UTAUT). The ISO evaluation showed positive results, with the system scoring an overall mean of 3.36 out of 4 for overall quality, 3.43 for functional suitability, and 3.35 for usability. The UTAUT results were also good, with an overall mean of 3.32. Users said the system improved their performance, giving performance expectancy a score of 3.38. They also felt it was easy to use, with effort expectancy scoring 3.29. The researchers suggest adding a mobile application to make the system even more accessible. They also recommend continuous user training and integration with Unilever's supplier databases to improve the system in the future.*

Keywords -- RM'S, AI-integrated inventory system, e-commerce management, demand forecasting, Agile SDLC, ISO evaluation, UTAUT

I. INTRODUCTION

The rapid growth of e-commerce has reshaped how businesses operate, pushing companies to adopt smarter, faster, and more customer-centered systems. Online platforms today integrate secure payments, order tracking, and real-time stock visibility to meet the expectations of modern consumers. Difficulties associated with manual stock monitoring and paper-based inventory processes have been pointed out. Researchers mention problems including inaccurate stock counts, delayed order fulfillment, and inefficient supply chain coordination that make inventory management more complicated.

This need is particularly relevant for REMHART, a trusted distributor of Unilever products. REMHART currently encounters several operational challenges: fluctuating demand, manual stock monitoring, and inefficient order processing. These issues often result in excess inventory, shortages, and delays in order fulfillment. Without an intelligent system to connect their e-commerce platform with inventory operations, REMHART risks inefficiencies that hinder both profitability and customer satisfaction.

According to Smith (2024), predictive analytics helps retailers forecast demand with greater accuracy by analyzing historical data and real-time market trends. AI-driven systems reduce stockouts and overstocking, minimizing costs and enhancing operational efficiency. Taylor (2024) provided a comprehensive review of AI applications in inventory control, highlighting how machine learning and automation are reshaping supply chain management. These include real-time stock monitoring, predictive demand forecasting, and automated restocking.

Martinez (2023) presented a case study on AI implementation in inventory management in the FMCG sector, demonstrating that AI-driven demand forecasting and stock optimization significantly improved inventory practices. Zhang and Li (2024) investigated the application of AI and machine learning in demand forecasting for supply chain management, finding that machine learning algorithms outperformed traditional statistical methods. Brown (2023) explored real-time inventory systems and their effect on supply chain efficiency, confirming that AI-powered systems provide detailed insights into product movements and sales trends.

The main concern is that handling manual inventory records is complicated and error-prone, requiring staff to constantly monitor stock levels and order quantities. In inventory management, making the process modern is good for improving productivity and having fewer risks or problems. Choosing an AI-integrated system like RM'S accepts attempts aimed at aligning actual inventory management strategies with current industry requirements. This method increases productivity while satisfying modern needs, thereby enhancing the operational capability of REMHART as well.

There were various factors evaluated before designing and implementing RM'S, which include fluctuating demand patterns, manual stock tracking inefficiencies, lack of real-time inventory visibility, and absence of an integrated e-commerce platform. The RM'S system also offers real-time alerts and notifications to keep staff informed about low stock levels and pending orders. These features make it easier for administrators and warehouse staff to stay updated and ensure that important inventory information is never missed.

II. METHODOLOGY

A. Development Method

In this study, the respondents were surveyed using a questionnaire to provide information on the study in response to several questions. It utilized the Agile Model of the Software Development Life Cycle (SDLC) for the development of the proposed system.

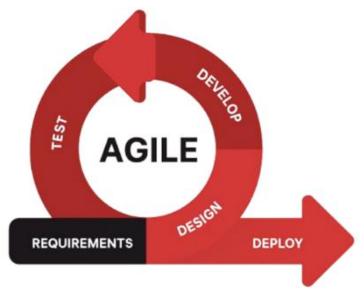

Figure 1. Agile Model

The sequential phases in the Agile model are:

1. **Requirements**

   In this initial phase, the researchers collected detailed information regarding REMHART's operational needs. This includes understanding the objectives of the inventory system and identifying specific features required by the company. Key functionalities include real-time stock monitoring, AI-driven demand forecasting, automated replenishment, supplier management, and integrated e-commerce. The researchers also documented the separate roles and permissions required for Admin, Stock Manager, Sales Personnel, and Customer users.

2. **Design**

   After gathering the requirements, the researchers focused on designing the system. This involves creating layouts, wireframes, and system architecture that illustrate the organization of information, user interaction points, and the relationships between different system components. Special focus was given to usability, ensuring a clean and intuitive interface suitable for both administrative and operational users.

3. **Develop**

   During the development phase, the researchers utilized Node.js for backend processing and MySQL for database management. Core modules including Inventory Management, E-Commerce Ordering, Supplier Management, and AI-Driven Demand Forecasting were coded and integrated. The AI forecasting model utilized Facebook Prophet, trained on historical sales data from REMHART's covered stores. This stage also involves ensuring secure authentication and creating dashboards for administrators to monitor inventory and sales data.

4. **Test**

   Once the system had been developed, testing was conducted to identify any errors or bugs. The researchers evaluated each function to ensure it operates as intended and confirmed that user roles can access only their designated areas. Testing also includes validating the functionality of demand forecasting, inventory updates, GCash payment processing, and file uploads to ensure they work effortlessly.

5. **Deploy**

   After successful testing, the system was deployed into REMHART's operational environment, making it available for user interaction. Deployment includes configuring the system on the client's servers and ensuring that all inventory data and user accounts are correctly integrated. This phase enables admins, stock managers, sales personnel, and customers to log in and execute their respective tasks effectively.

6. **Review**

Following deployment, the system undergoes a continuous review process to monitor performance and address any developing issues. The researchers collect feedback from users, identify areas for improvement, and implement necessary updates. This review phase is key for maintaining the system's effectiveness and ensuring it continues to meet the evolving needs of REMHART.

B. Requirements Specifications

The requirements specifications for the inventory and e-commerce system include the software interface and hardware interface. The specifications highlight the required functions, connection points, and security components that the system needs to be dependable and safe.

Hardware Specifications

Hardware Specifications means technical descriptions of hardware items, their components and functions. Consideration of the hardware elements which will ensure the efficiency and functionality of the project is a must. The table below presents the different hardware components and their recommended specifications:

Table 1. Hardware Specifications

Hardware Specifications	Minimum Specifications	Recommended Specifications
Processor	4 cores, 2.5 GHz	8 cores, 3.0 GHz
Memory	8 GB RAM	16 GB RAM
Storage	256 GB SSD, 500 GB HDD	512 GB SSD, 1 TB HDD
Network	802.11ac 2.4/5 GHz Wireless Adapter	802.11ac 2.4/5 GHz Wireless Adapter

Table 1 shows the hardware requirements needed for the system to work properly. The system needs at least 8 GB of RAM for smooth performance. For storage, a 256 GB SSD is the minimum recommended. The system also requires an 802.11ac wireless adapter for internet connection. An Intel Core i5 processor or higher with at least 8 GB DDR4 RAM are ideal for better performance and faster operation.

Software Specifications

Software specifications are very important for the smooth operation and connectivity of the inventory management system. They help make sure that the system works well and connects properly to different s ervices. By meeting these software requirements, the system will work as expected without any performance or connectivity problems.

Table 2. Software Specifications

Software Components	Minimum Specifications	Recommended Specifications
Operating System	Windows 10 (64-bit)	Windows 11 Pro (64-bit)
Web Server	Apache 2.4 / Nginx 1.14	Apache 2.4 / Nginx (latest stable)
Database	MySQL 5.7 / Firebase Realtime DB	MySQL 8.0 / Firebase Firestore
Web Browser	Google Chrome (latest)	Google Chrome or any web browser
AI/ML Libraries	TensorFlow 2.0	TensorFlow 2.12+ / PyTorch 1.13+
IDE	Visual Studio Code, Node.js 16+	JetBrains PyCharm, Node.js 18+
Frontend Framework	Vue.js 2.6.x / Bootstrap 5	Vue.js 3.x / Tailwind 3.x

As shown in Table 2, it outlines the minimum and recommended specifications for various software components required for the system. By defining these software specifications, the inventory and e-commerce platform is able to work with the necessary software pieces and perform the intended functions

C. System Architecture

RM'S system architecture, shown in Figure 2, explains how the system is organized and how everything works together. To use all the features and functions of RM'S, the user must first register and create an account. After registering, the user can log in to access the system. Once logged in, they can use all the tools and services that RM'S offers. Users access the platform via the frontend built with Vue.js, which sends requests to the Node.js backend for inventory data or demand forecasts. The AI/ML layer processes forecasting using Facebook Prophet and returns results to the dashboard.

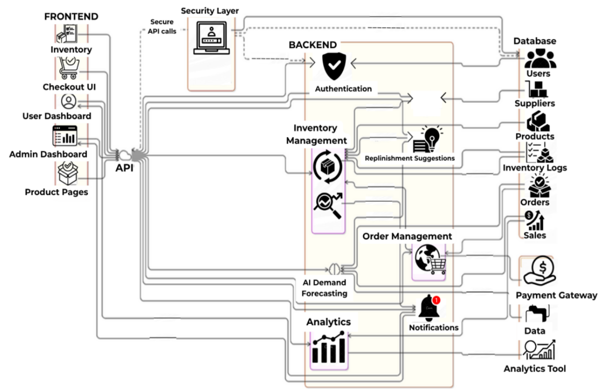

Figure 2. System Architecture

Figure 2 shows that the admin is in charge of managing users, inventory, suppliers, and reports. The Stock Manager handles inventory updates and monitoring. The Sales Personnel process customer orders, while the Customer interacts with the e-commerce module for browsing and purchasing. The system serves as a hub which establishes communication channels between all actors, facilitating data-driven operations and organized inventory management.

D. Use Case Diagram

Users of the developed system must follow the steps shown in Figure 3. This view reveals the perspectives of users regarding the features that the project's proponents have provided.

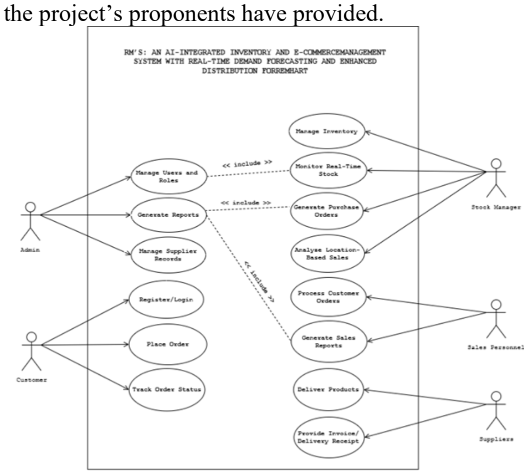

Figure 3. Use Case Diagram

Figure 3 serves as an outline of how these stakeholders all interact with and complement each other in carrying out functions within the system. The Admin has full access to user management, inventory control, and reporting. Stock Managers monitor stock levels and respond to replenishment alerts. Sales Personnel process orders while Customers browse products and complete purchases via the integrated GCash module.

E. Activity Diagram

This section illustrates the project flow through an object-oriented flowchart, depicting the dynamic behavior of the system. Emphasizing execution and system behavior rather than implementation details, the activity diagram provides a complete view of the project's dynamic aspects.

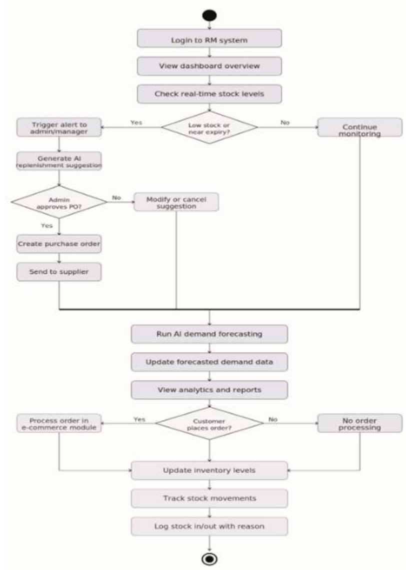

Figure 4. Activity Diagram

Figure 4 shows the activity diagram of the system. The process begins when a user logs in to the platform. Once authenticated, the user accesses the dashboard to view real-time stock levels and analytics. The system monitors stock quantities and automatically triggers alerts when items fall below the reorder threshold. AI-powered replenishment suggestions are generated and the admin may approve purchase orders to suppliers. Customer orders through the e-commerce module automatically update inventory levels upon checkout.

F. Data Flow Diagram

The data flow diagram is like a map that shows how information moves through the system. It helps users understand how different parts of the system work together by showing the flow of data between them. The diagram includes all system processes, showing where the data comes from, how it is used, and where it goes.

Context Diagram

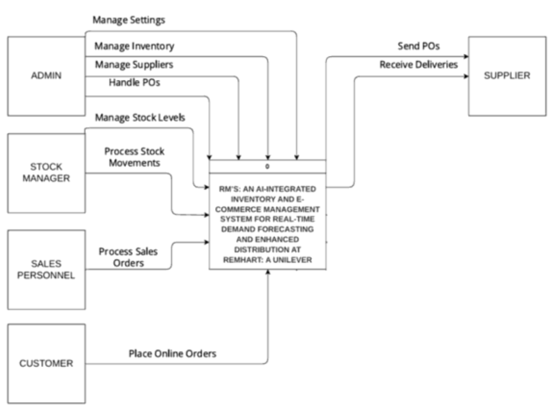

Figure 5. Context Diagram

Figure 5 shows how the system will be developed for different areas and how it will be used by different users. The Admin manages system settings, users, inventory, and purchase orders. The Stock Manager tracks stock movements and monitors inventory levels. The Sales Personnel process customer sales orders, while Customers place orders through the e-commerce module. The system also interfaces with Suppliers (receiving purchase orders) and the AI/ML Engine (supplying demand forecasts).

Diagram 0

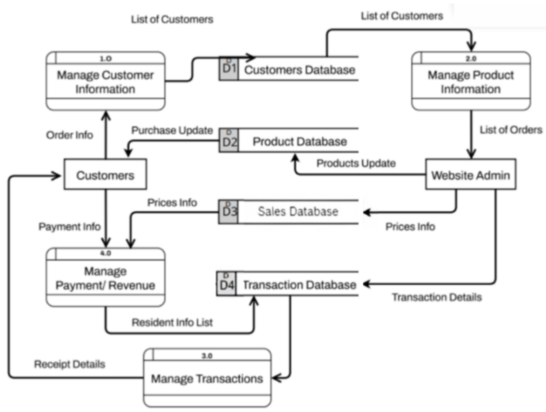

Figure 6. Diagram 0

Figure 6 shows how the Admin, Stock Manager, Sales Personnel, Suppliers, and Customers work together and share information. The admin manages the system and makes important decisions. Stock Managers handle inventory updates while Sales Personnel process orders. Customers place and track orders through the e-commerce interface. Arrows in the diagram show how information moves between all actors, keeping everyone connected and ensuring the right information is shared in real-time.

G. Participants of the Study

The respondents of the study were composed of the Administrator, Warehouse Managers, Sales Representatives, Technical Experts, and End Users/Customers. Their diverse perspectives provided valuable insights into the functionality, usability, and potential improvements of the system.

Table 3. Participants of the Study

Respondents	Number of Respondents	Percentage
Admin	1	2%
Warehouse Managers	7	11%
Sales Representatives	10	15%
Technical Experts	1	2%
End Users/Customers	46	70%
Total	**65**	**100%**

Table 3 shows different groups of people and their percentages out of 65 total respondents. There is 1 Administrator (2%), 7 Warehouse Managers (11%), 10 Sales Representatives (15%), 1 Technical Expert (2%), and 46 End Users/Customers (70%). Together, they make up 100% of the total.

III. RESULTS AND DISCUSSIONS

A. Presentation of System Output

This section shows how the system displays its results, similar to how a good website needs a user-friendly design. To make sure the information is easy to understand and helpful for making decisions, the results are shared using clear summaries and visuals like graphs and figures. These tools help turn complex data into something easier to read, so users can quickly understand the information and make better choices.

Admin/Staff Side

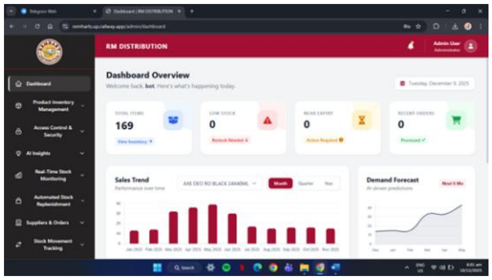
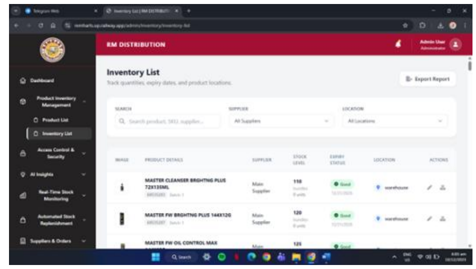

Figure 7. Admin Dashboard, Inventory List, Demand Forecasting, Supplier Management, Activity Logs
(Left-to-Right Row 1 & 2)

Figure 7 shows the user interface designed for administrators and staff, which gives them complete control and management over the system. The admin can monitor inventory levels, view demand forecasts, manage suppliers, process purchase orders, and generate reports. Through the centralized dashboard, administrators can track current stock levels, recent sales, low stock alerts, and AI-powered demand forecasts with organized graphs and charts. They can also manage user accounts by activating or deactivating them.

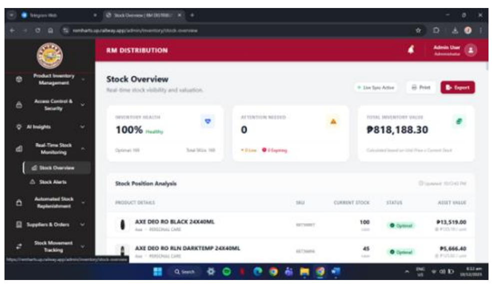
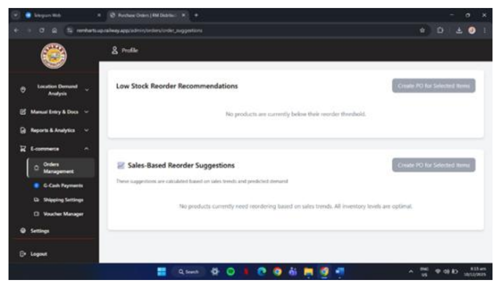

Figure 8. Stock Overview, Reorder Suggestions, Purchase Orders, Sales by Location, Reports Dashboard
(Left-to-Right Row)

Figure 8 shows that the admin can view stock levels and automated reorder suggestions to monitor inventory that needs restocking. They can see a geographic breakdown showing which areas or routes generate the highest sales, helping managers optimize distribution. The admin can also create and manage purchase orders sent to suppliers and view consolidated reports with key business metrics.

Customer Side

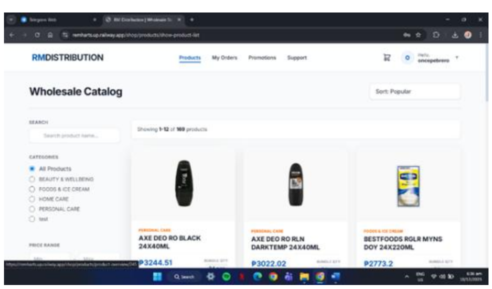
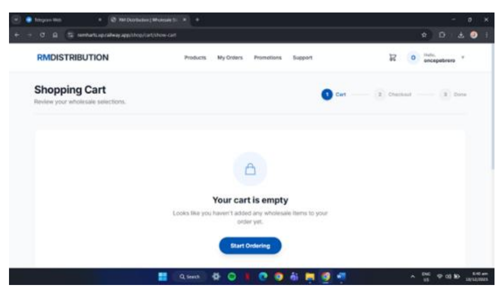

Figure 9. Home Page, All Products, Shopping Cart, My Orders, GCash Payment (Left-to-Right Row)

Figure 9 shows the customer-facing interface which allows users to browse products, manage their shopping cart, place orders, and complete payments through the integrated GCash payment system. Customers have a profile setting where they can update or manage their information. The My Orders module provides users with a convenient overview of their past and ongoing purchases with order details, statuses, and timestamps.

B. Evaluation of the System

After designing and evaluating RM'S: An AI-Integrated Inventory and E-Commerce Management System, the study concludes with a thorough assessment across several key dimensions, including Functional Suitability, Performance

Efficiency, Usability, and Reliability. Feedback was gathered from 65 respondents, including business owners, staff, and customers, through questionnaires. The responses were carefully analyzed and presented in tables providing a detailed interpretation of the system's overall evaluation and performance.

Table 4. Summary Results of the ISO Software Metrics Evaluation

Items	Mean	Rank	Verbal Interpretation
1. Functional Suitability	3.43	1	Agree
2. Performance Efficiency	3.36	2	Agree
3. Usability	3.35	3	Agree
4. Reliability	3.30	4	Agree
Overall Mean	**3.36**		**Agree**

Table 4 shows the results of the ISO 25010 evaluation for the system in four important areas: Functional Suitability, Performance Efficiency, Usability, and Reliability. The system received an overall mean of 3.36, which means most users agree that the system works well and meets their needs. The highest score was 3.43 for Functional Suitability, showing that the system covers all specified tasks and user objectives effectively. Performance Efficiency scored 3.36, Usability scored 3.35, and Reliability scored 3.30. These results indicate that the system is reliable, easy to use, and performs efficiently, making it a good tool for its users.

Table 5. Summary Results of the UTAUT Evaluation

Items	Mean	Rank	Verbal Interpretation
1. Performance Expectancy	3.38	1	Agree
2. Effort Expectancy	3.29	4	Agree
3. Facilitating Conditions	3.31	2	Agree
4. Behavioral Intention	3.30	3	Agree
Overall Mean	**3.32**		**Agree**

Table 5 shows the results of the UTAUT evaluation, which measures four areas: Performance Expectancy, Effort Expectancy, Facilitating Conditions, and Behavioral Intention. The overall mean score is 3.32, meaning most users agree that the system is good and meets their needs. The highest score, 3.38, is for Performance Expectancy, which means users believe the system significantly enhances their performance. Facilitating Conditions scored 3.31, showing users feel they have enough support when using the system. Effort Expectancy scored 3.29, indicating users find the system easy to use, and Behavioral Intention scored 3.30, showing users intend to continue using the system.

IV. CONCLUSIONS AND RECOMMENDATIONS

The researchers have drawn several conclusions during the development of RM'S for REMHART. The system was designed to make inventory management faster and more organized for administrators, stock managers, sales personnel, and customers. The AI-powered demand forecasting feature helped management anticipate product demand, optimize restocking schedules, and reduce wastage through data-driven decisions. The inventory management module enabled staff to manage product information accurately, ensuring real-time stock monitoring across all product categories. Digital forms and the integrated e-commerce module made it easy for customers to browse and purchase products online, with automatic inventory updates after each transaction. The system's dashboard allowed administrators to manage users easily, ensuring secure and controlled access. Geographic sales analysis showing sales performance by location helped management optimize distribution routes and target high-demand areas for better coverage, making the distribution process more efficient. The system also automatically generates reports with filters to sort data and help administrators track inventory and sales details.

For future work, researchers can add a mobile application interface to broaden accessibility and cater to evolving user preferences. They can also include multi-language support, allowing people who speak different languages to use the system more comfortably. Another improvement could be the integration with Unilever's supplier databases and third-party logistics systems to automate procurement, shipment tracking, and supply coordination. Researchers can also improve the system by automating the creation and sending of reports on a schedule, saving time for administrators. Lastly, a predictive resource dashboard could be developed to forecast future inventory needs based on historical data, helping administrators plan better and ensure they have enough stock for future demand.

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APPENDIX O

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